

H^s Direct — Stage 1 Compositional Analysis

Japan EMBER Electricity Generation · 8 Carriers · 26 Years (2000–2025)

$H_s = 1 - H/\ln(D)$ · Shannon (1948) + Aitchison (1986)

Higgins Scale Ring Classification

- Hs-0 ISOTROPIC [0.00 – 0.05]
- Hs-1 TEXTURED [0.05 – 0.15]
- Hs-2 STRUCTURED [0.15 – 0.30]
- Hs-3 POLARISED [0.30 – 0.50]
- Hs-4 DOMINATED [0.50 – 0.75]
- Hs-5 COLLAPSED [0.75 – 0.95]
- Hs-6 SINGULAR [0.95 – 1.00]

Carriers

- Bioenergy
- Coal
- Gas
- Hydro
- Nuclear
- Other Fossil
- Solar
- Wind

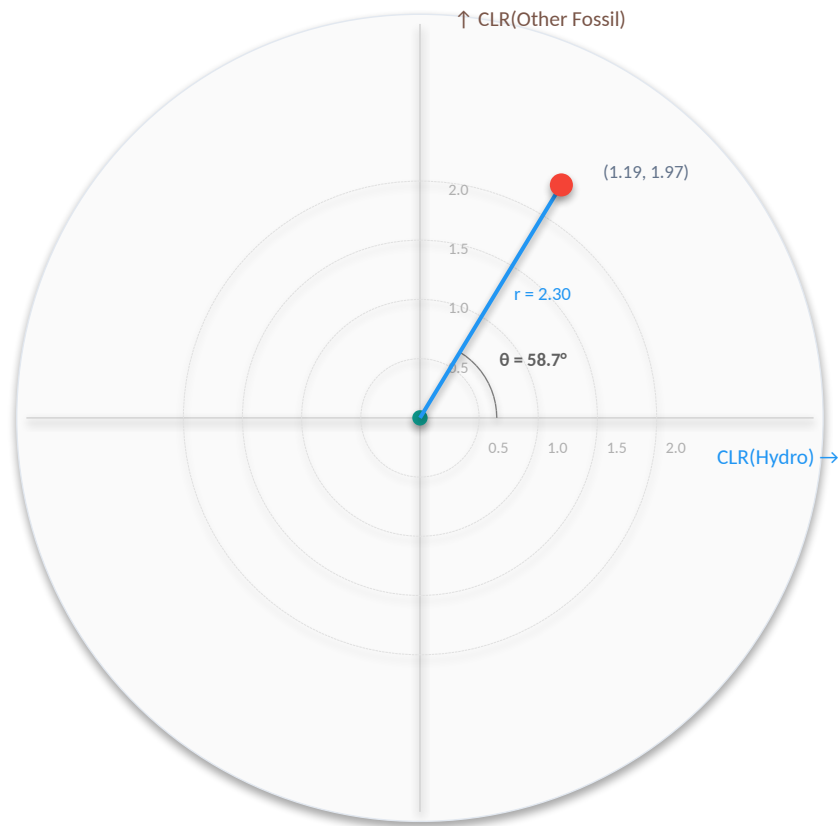
XY Face — Compositional Plane (Plan View)

CLR(Hydro) vs CLR(Other Fossil) · 167° angular spread · Maximum fan-out

Projection selected by largest directional separation across 26 years

Coal-Gas projection rejected: 8.4° spread — informationally locked at 45°

Each composition mapped independently · No point-to-point connections



Polar Decomposition

$\theta = 58.7^\circ$

$r = 2.301$ (projection radius)

$d_A = 8.400$ (full 8D Aitchison)

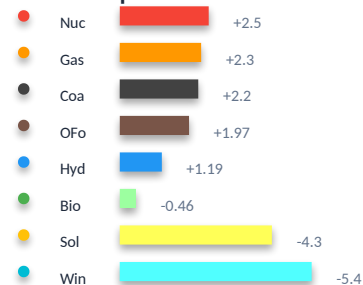
Projection covers 27% of d_A

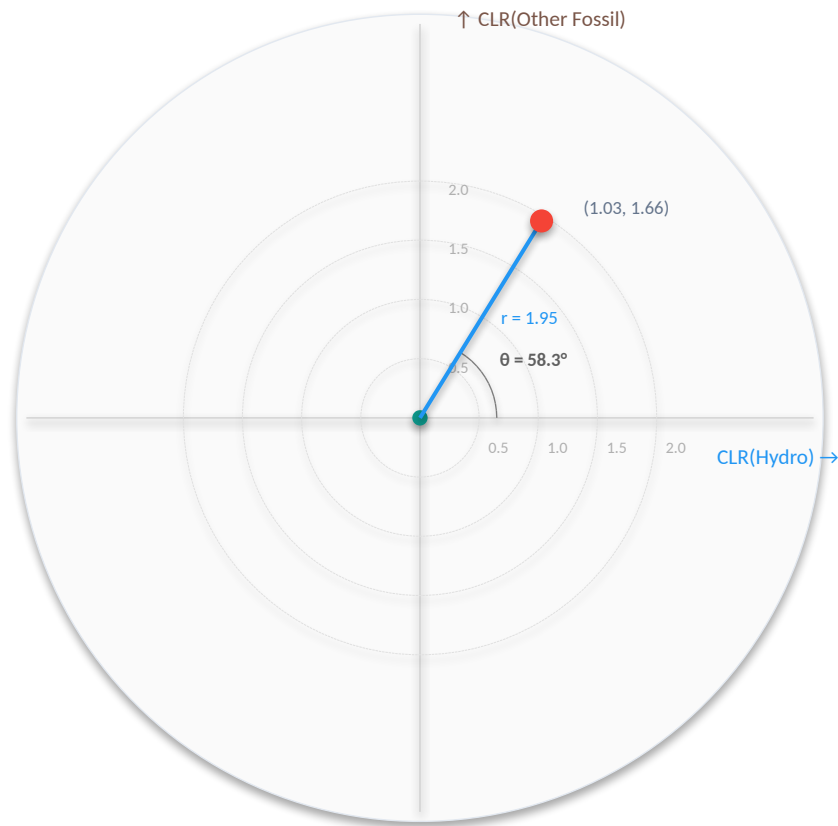
Angular Velocity (8D CLR)

→ 2.8° to 2001

Coal-Gas Lock: $\theta_{CG} = 46.0^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 58.3^\circ$

$r = 1.955$ (projection radius)

$d_A = 7.650$ (full 8D Aitchison)

Projection covers 26% of d_A

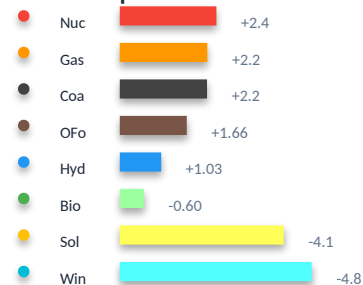
Angular Velocity (8D CLR)

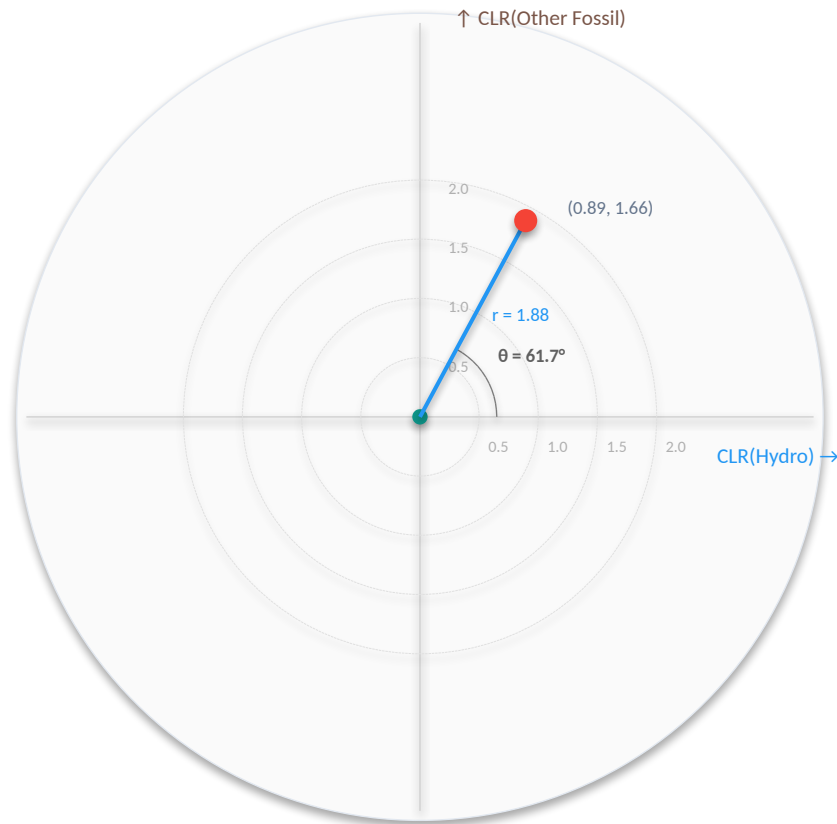
← 2.8° from 2000

→ 1.5° to 2002

Coal-Gas Lock: $\theta_{CG} = 45.1^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 61.7^\circ$

$r = 1.882$ (projection radius)

$d_A = 7.194$ (full 8D Aitchison)

Projection covers 26% of d_A

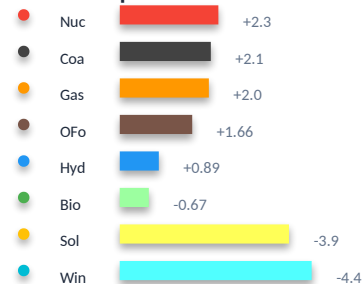
Angular Velocity (8D CLR)

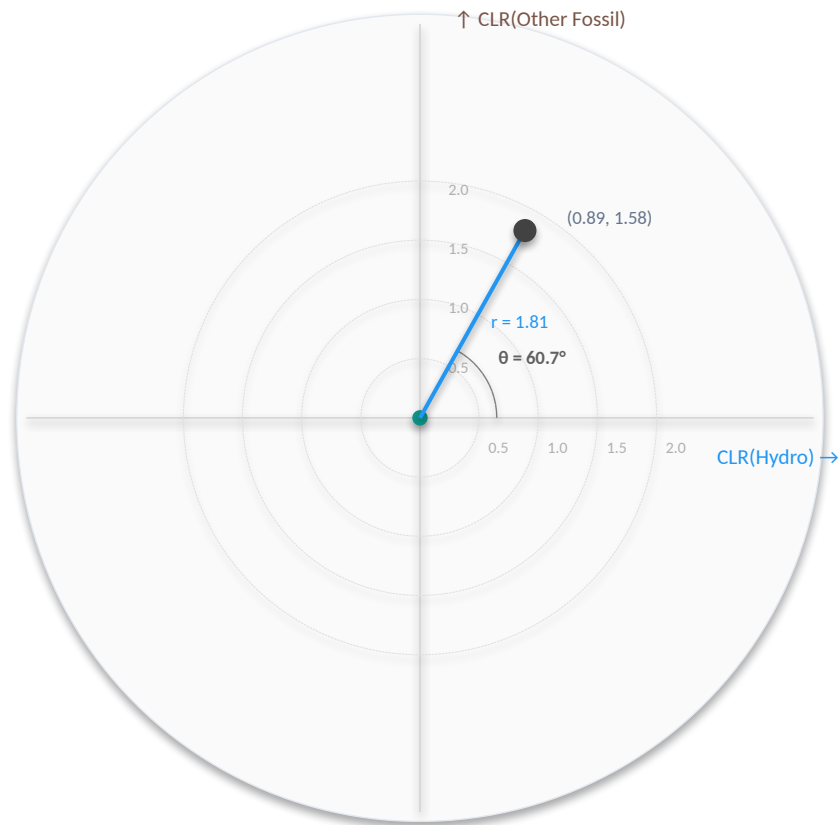
← 1.5° from 2001

→ 3.8° to 2003

Coal-Gas Lock: $\theta_{CG} = 44.4^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$\theta = 60.7^\circ$

$r = 1.813$ (projection radius)

$d_A = 6.587$ (full 8D Aitchison)

Projection covers 28% of d_A

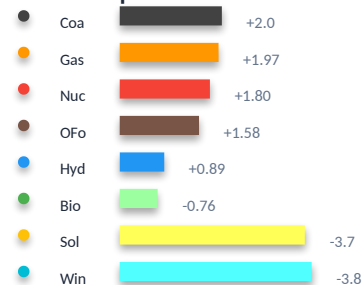
Angular Velocity (8D CLR)

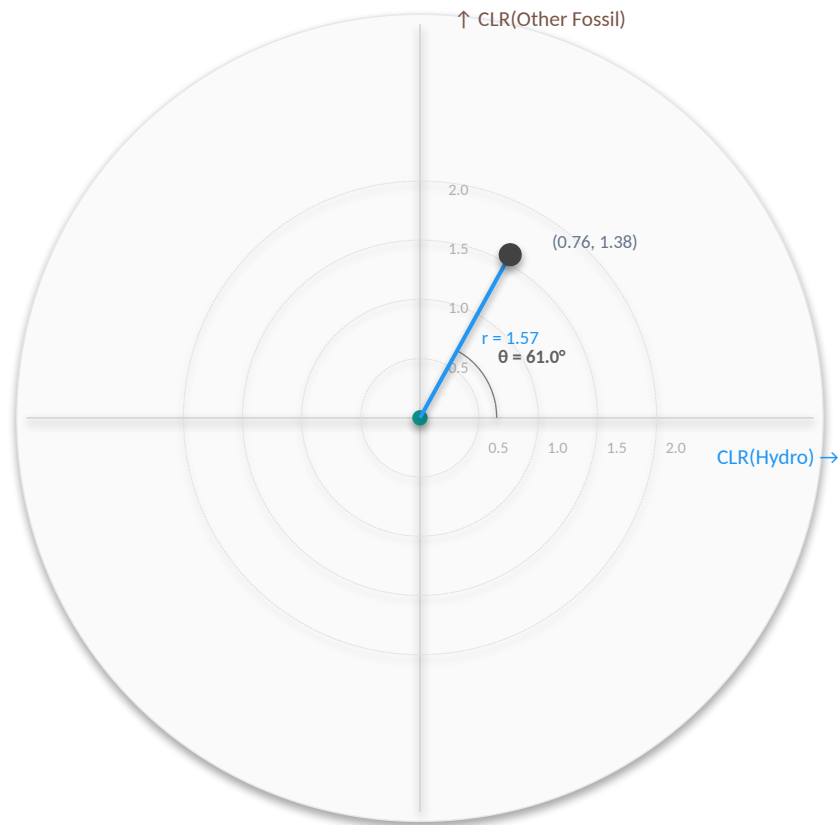
← 3.8° from 2002

→ 3.3° to 2004

Coal-Gas Lock: $\theta_{CG} = 44.0^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$\theta = 61.0^\circ$

$r = 1.575$ (projection radius)

$d_A = 6.137$ (full 8D Aitchison)

Projection covers 26% of d_A

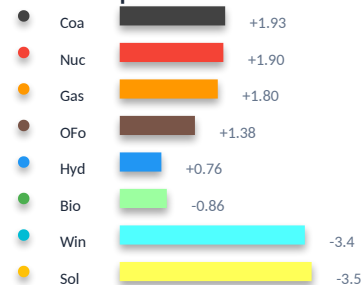
Angular Velocity (8D CLR)

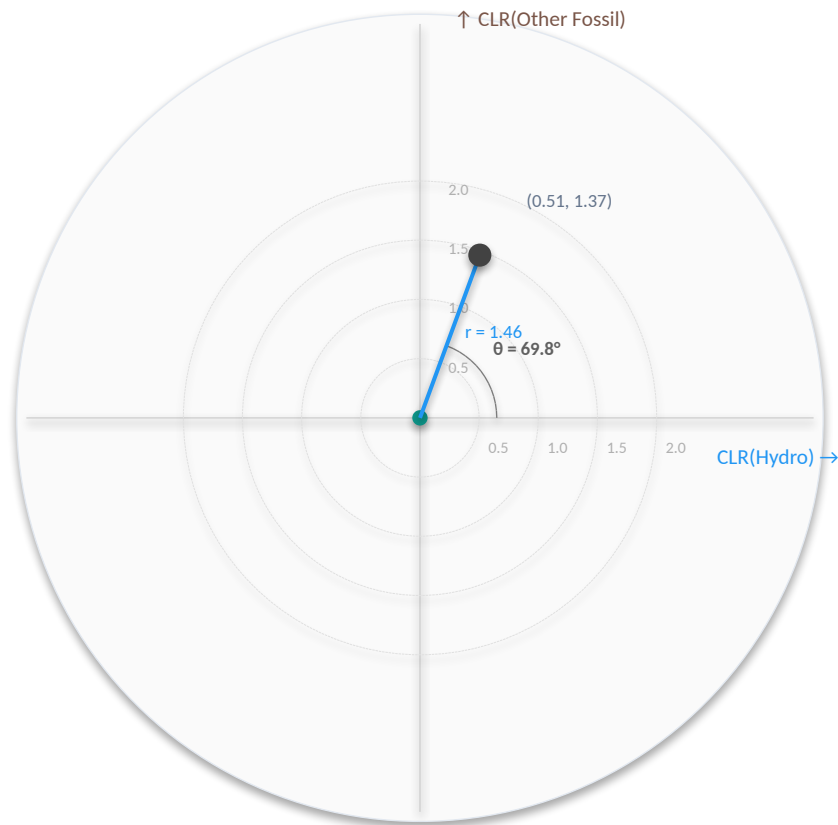
← 3.3° from 2003

→ 2.5° to 2005

Coal-Gas Lock: $\theta_{CG} = 42.9^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$\theta = 69.8^\circ$

$r = 1.465$ (projection radius)

$d_A = 5.843$ (full 8D Aitchison)

Projection covers 25% of d_A

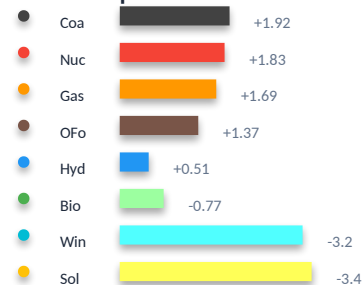
Angular Velocity (8D CLR)

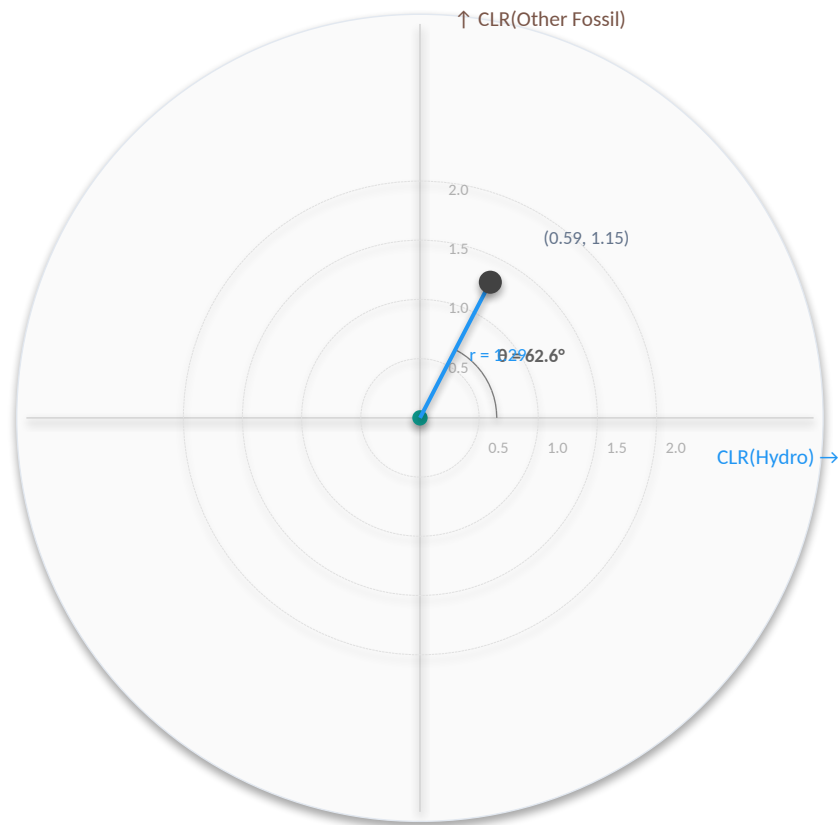
← 2.5° from 2004

→ 2.7° to 2006

Coal-Gas Lock: $\theta_{CG} = 41.3^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$$\theta = 62.6^\circ$$

$$r = 1.291 \text{ (projection radius)}$$

$$d_A = 5.675 \text{ (full 8D Aitchison)}$$

Projection covers 23% of d_A

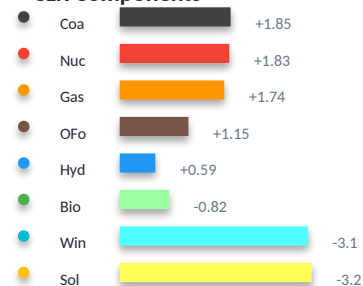
Angular Velocity (8D CLR)

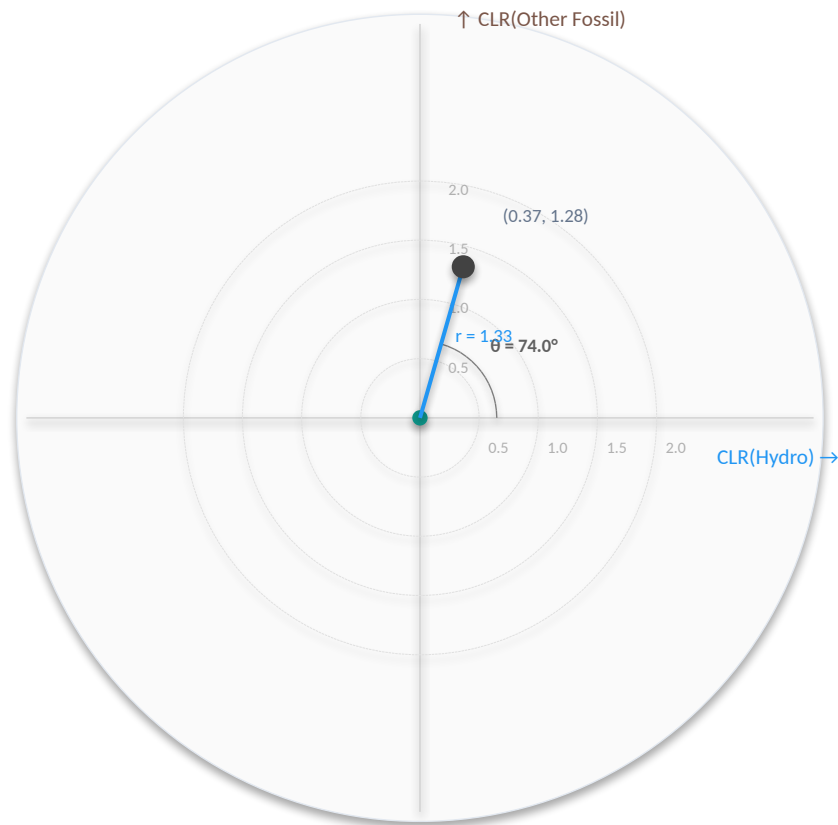
← 2.7° from 2005

→ 3.2° to 2007

Coal-Gas Lock: $\theta_{CG} = 43.3^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$\theta = 74.0^\circ$

$r = 1.327$ (projection radius)

$d_A = 5.473$ (full 8D Aitchison)

Projection covers 24% of d_A

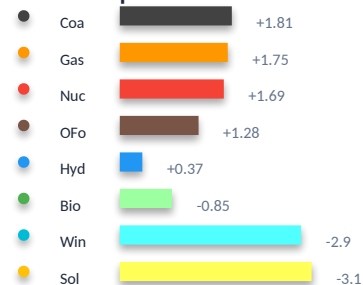
Angular Velocity (8D CLR)

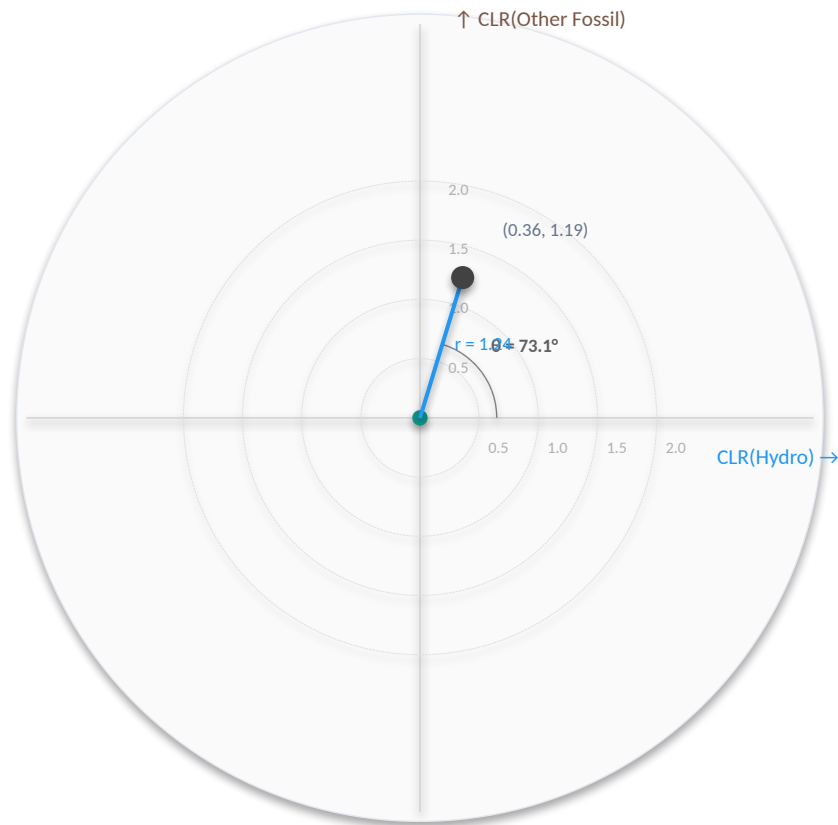
← 3.2° from 2006

→ 2.0° to 2008

Coal-Gas Lock: $\theta_{CG} = 43.9^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$\theta = 73.1^\circ$

$r = 1.239$ (projection radius)

$d_A = 5.376$ (full 8D Aitchison)

Projection covers 23% of d_A

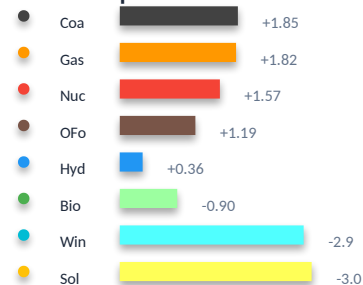
Angular Velocity (8D CLR)

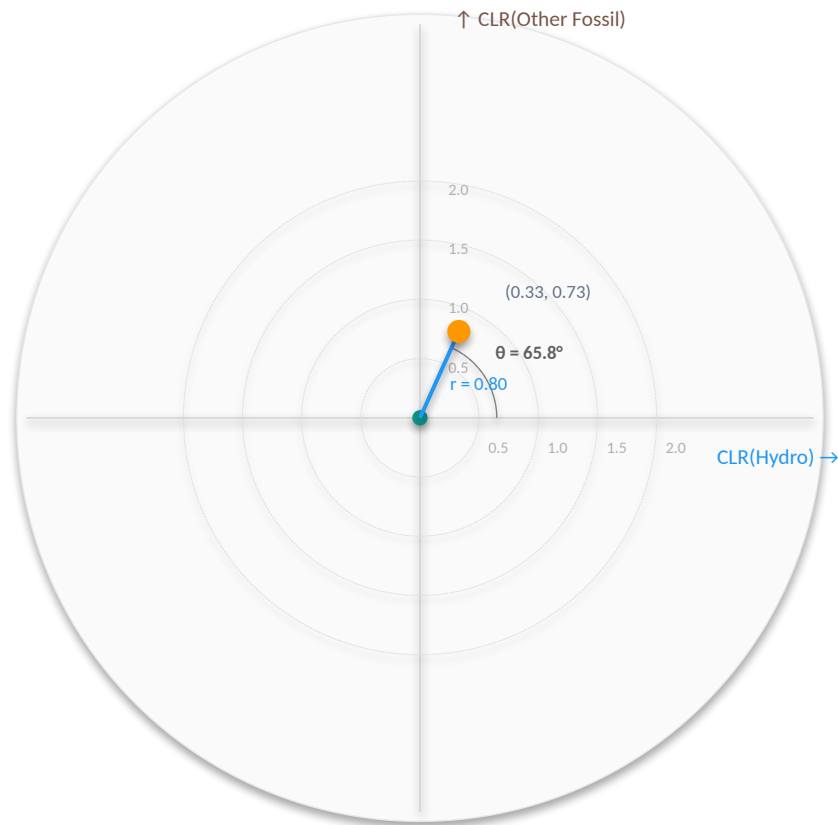
← 2.0° from 2007

→ 5.3° to 2009

Coal-Gas Lock: $\theta_{CG} = 44.6^\circ$ (Coal > Gas)

CLR Components





Polar Decomposition

$\theta = 65.8^\circ$

$r = 0.801$ (projection radius)

$d_A = 5.122$ (full 8D Aitchison)

Projection covers 16% of d_A

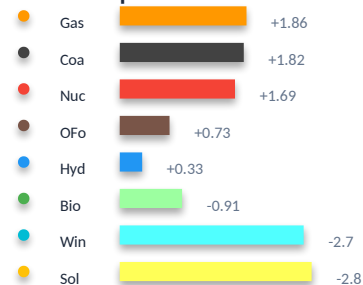
Angular Velocity (8D CLR)

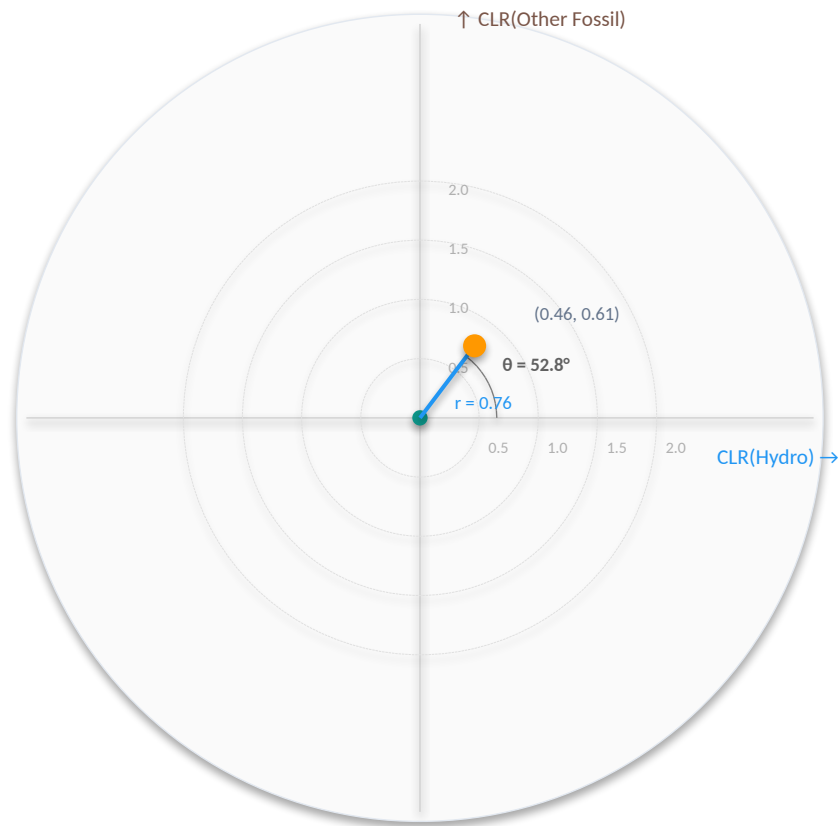
← 5.3° from 2008

→ 2.4° to 2010

Coal-Gas Lock: $\theta_{CG} = 45.6^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 52.8^\circ$

$r = 0.764$ (projection radius)

$d_A = 4.935$ (full 8D Aitchison)

Projection covers 15% of d_A

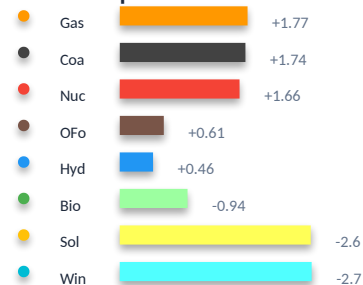
Angular Velocity (8D CLR)

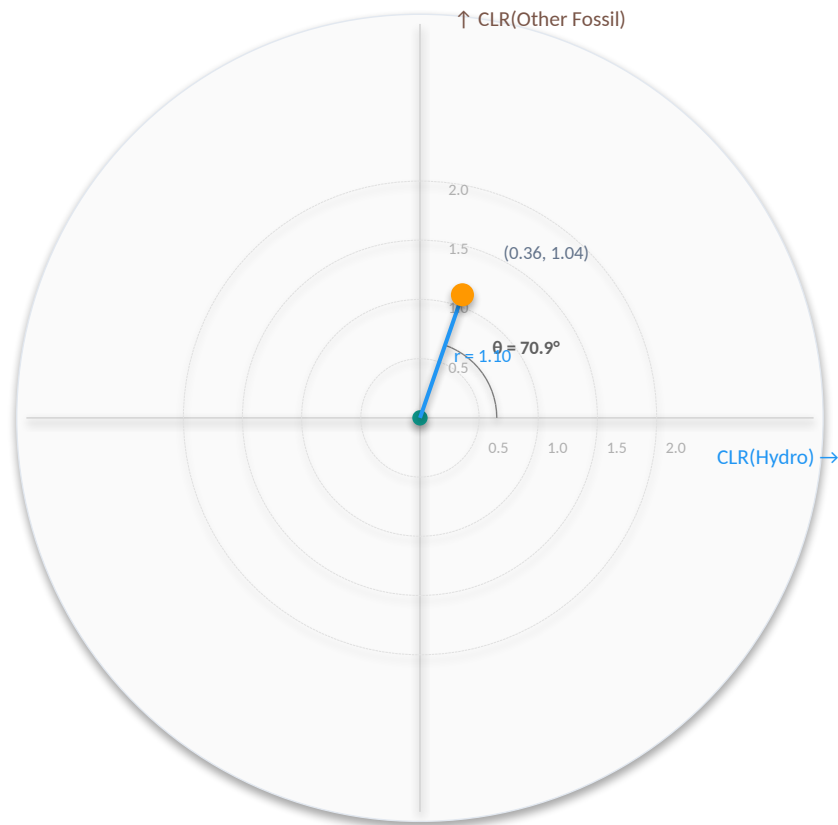
← 2.4° from 2009

→ 9.4° to 2011

Coal-Gas Lock: $\theta_{CG} = 45.5^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 70.9^\circ$

$r = 1.099$ (projection radius)

$d_A = 4.640$ (full 8D Aitchison)

Projection covers 24% of d_A

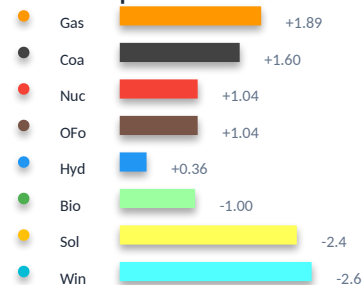
Angular Velocity (8D CLR)

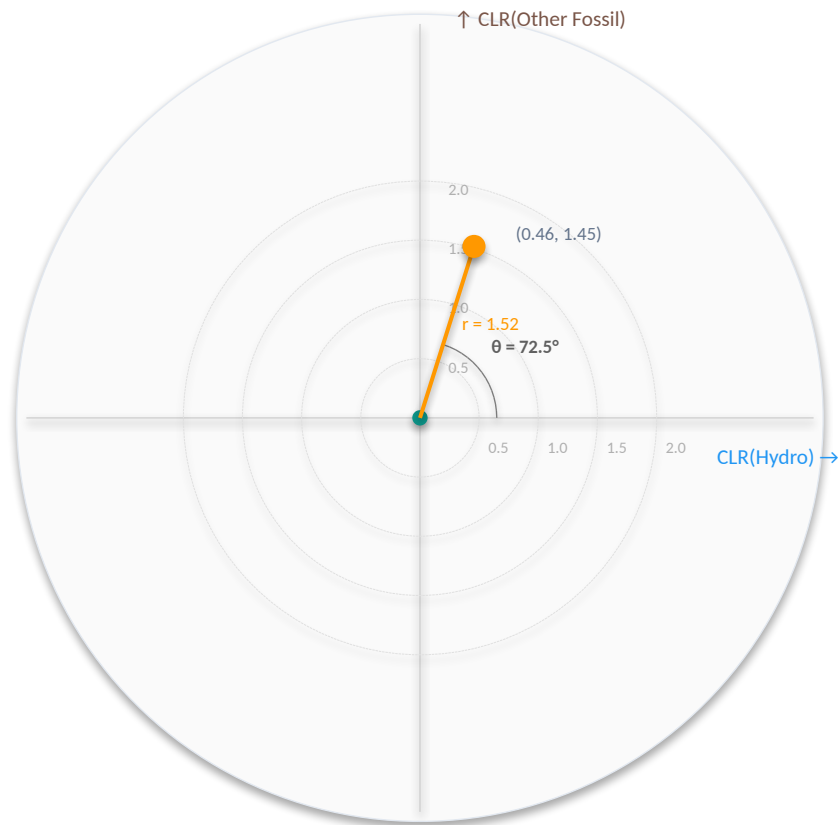
← 9.4° from 2010

→ 27.5° to 2012

Coal-Gas Lock: $\theta_{CG} = 49.7^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 72.5^\circ$

$r = 1.518$ (projection radius)

$d_A = 4.631$ (full 8D Aitchison)

Projection covers 33% of d_A

Angular Velocity (8D CLR)

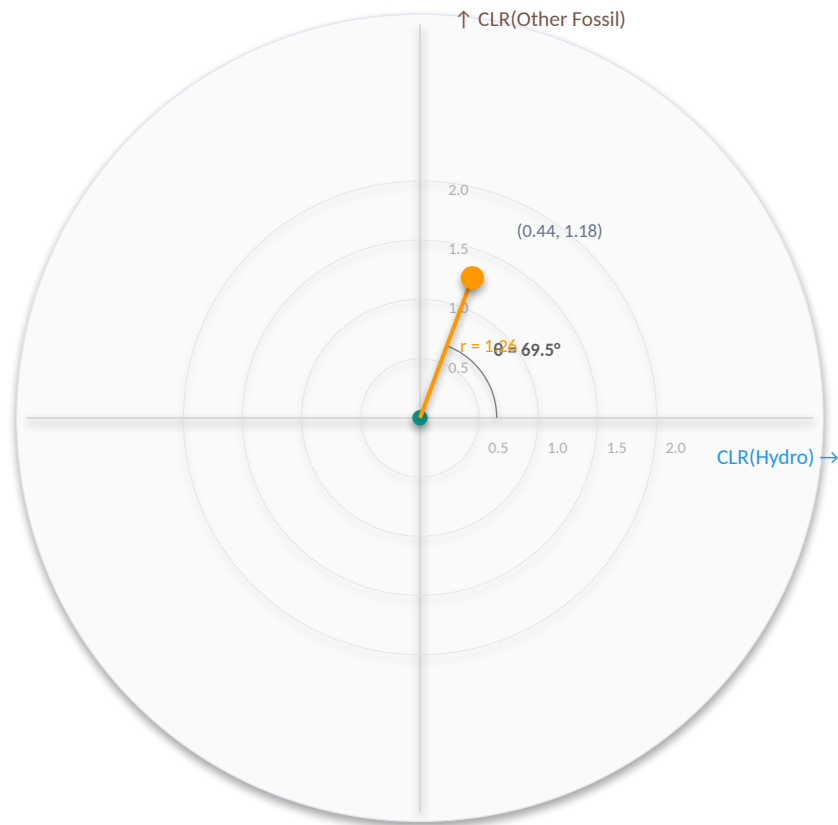
← 27.5° from 2011

→ 7.5° to 2013

Coal-Gas Lock: $\theta_{CG} = 48.7^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 69.5^\circ$

$r = 1.264$ (projection radius)

$d_A = 4.390$ (full 8D Aitchison)

Projection covers 29% of d_A

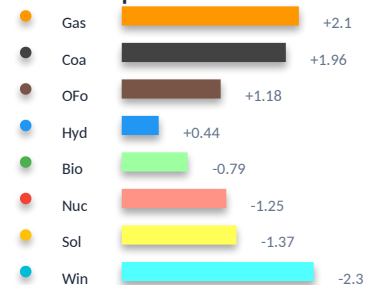
Angular Velocity (8D CLR)

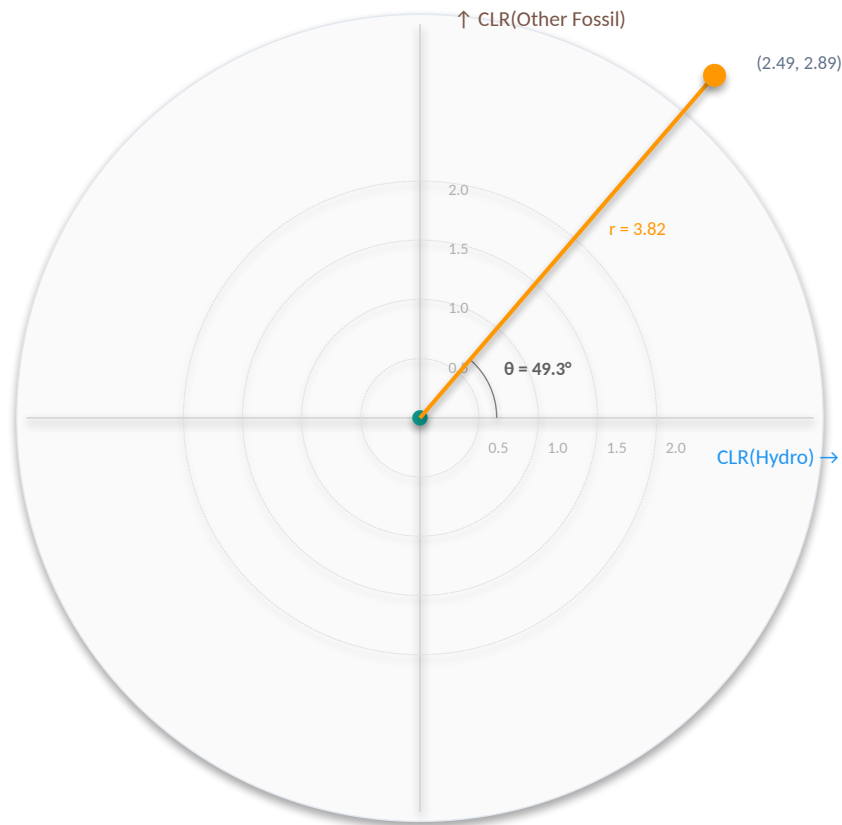
← 7.5° from 2012

→ 59.2° to 2014

Coal-Gas Lock: $\theta_{CG} = 47.2^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 49.3^\circ$

$r = 3.816$ (projection radius)

$d_A = 17.277$ (full 8D Aitchison)

Projection covers 22% of d_A

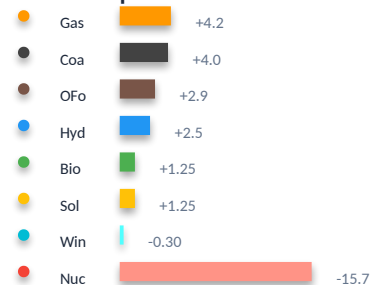
Angular Velocity (8D CLR)

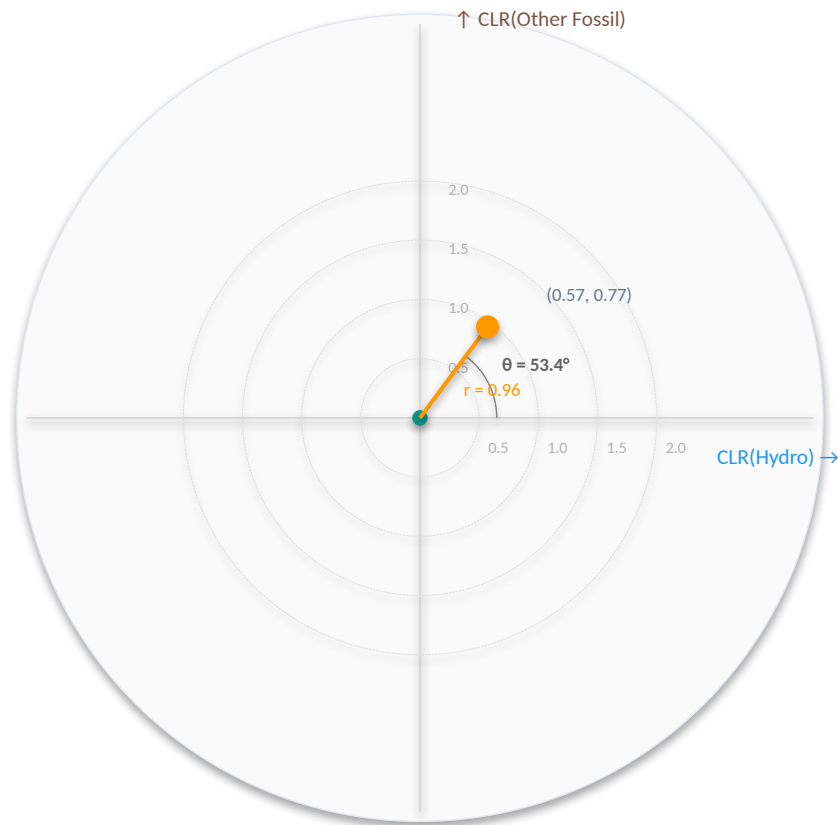
← 59.2° from 2013

→ 42.7° to 2015

Coal-Gas Lock: $\theta_{CG} = 46.7^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 53.4^\circ$

$r = 0.957$ (projection radius)

$d_A = 4.522$ (full 8D Aitchison)

Projection covers 21% of d_A

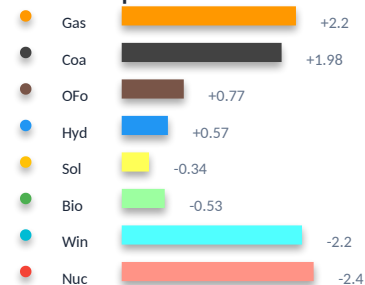
Angular Velocity (8D CLR)

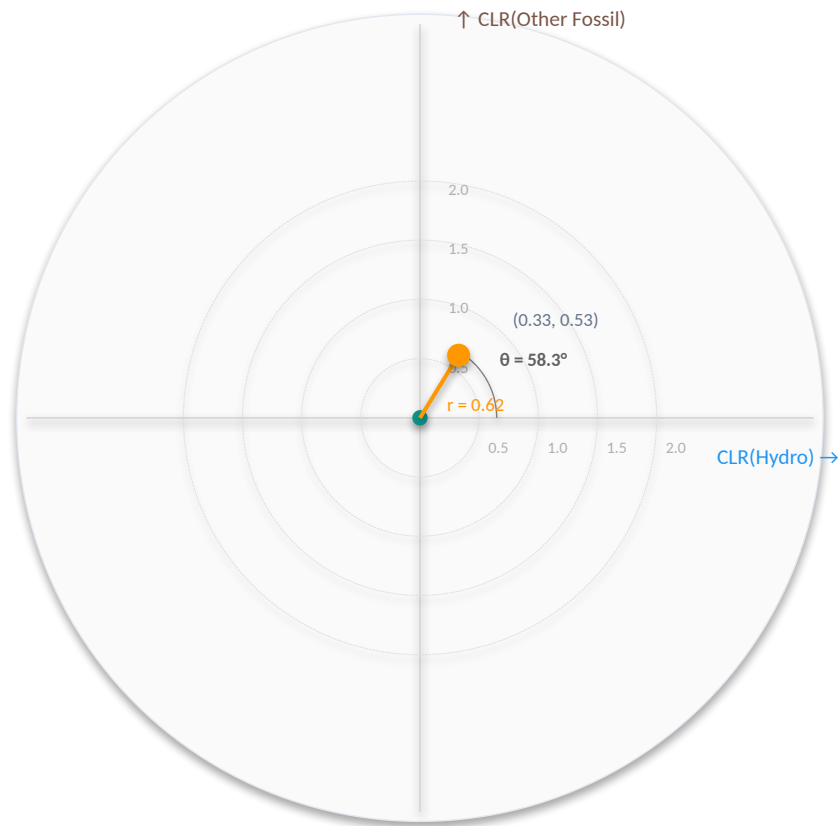
← 42.7° from 2014

→ 16.3° to 2016

Coal-Gas Lock: $\theta_{CG} = 47.5^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = 58.3^\circ$

$r = 0.622$ (projection radius)

$d_A = 3.967$ (full 8D Aitchison)

Projection covers 16% of d_A

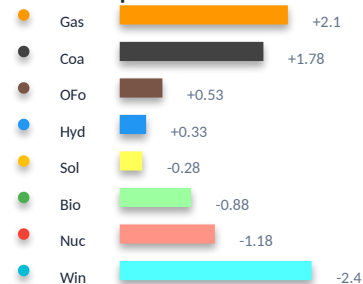
Angular Velocity (8D CLR)

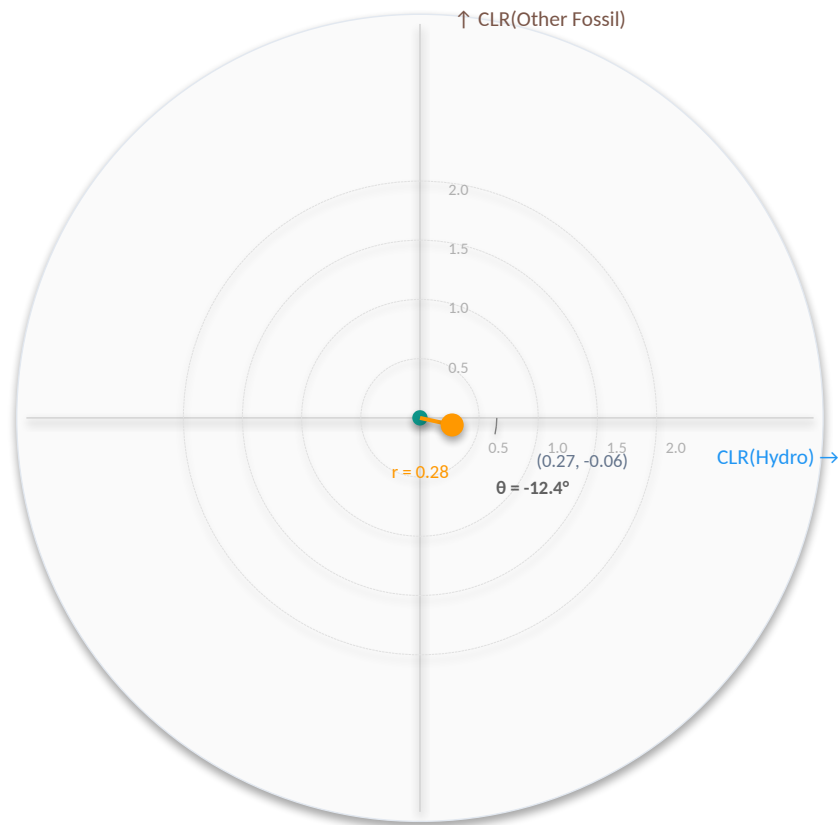
← 16.3° from 2015

→ 10.9° to 2017

Coal-Gas Lock: $\theta_{CG} = 49.5^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -12.4^\circ$

$r = 0.278$ (projection radius)

$d_A = 3.747$ (full 8D Aitchison)

Projection covers 7% of d_A

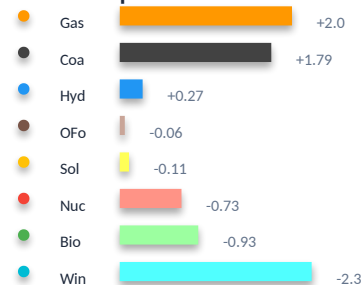
Angular Velocity (8D CLR)

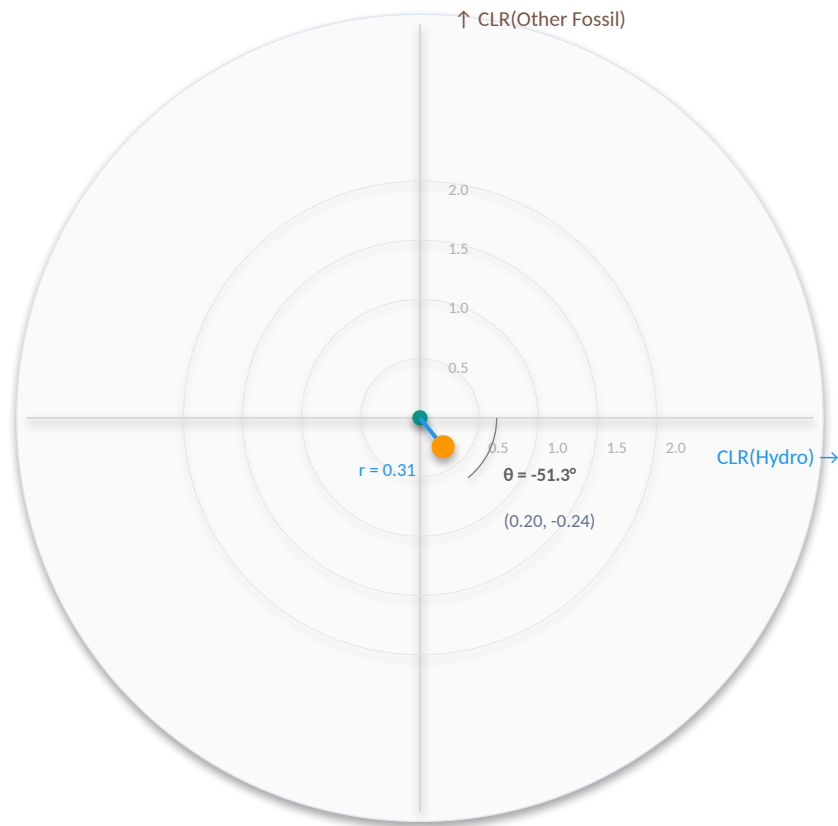
← 10.9° from 2016

→ 7.2° to 2018

Coal-Gas Lock: $\theta_{CG} = 48.6^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -51.3^\circ$

$r = 0.312$ (projection radius)

$d_A = 3.523$ (full 8D Aitchison)

Projection covers 9% of d_A

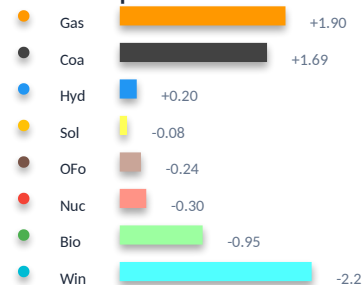
Angular Velocity (8D CLR)

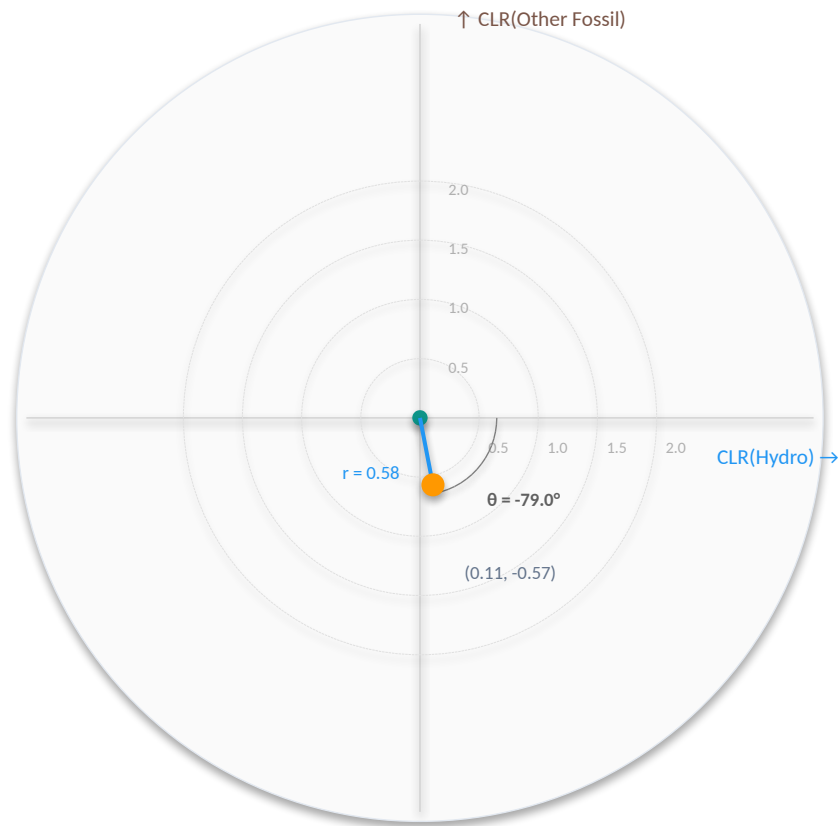
← 7.2° from 2017

→ 7.6° to 2019

Coal-Gas Lock: $\theta_{CG} = 48.4^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -79.0^\circ$

$r = 0.576$ (projection radius)

$d_A = 3.468$ (full 8D Aitchison)

Projection covers 17% of d_A

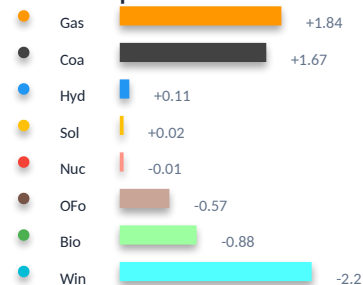
Angular Velocity (8D CLR)

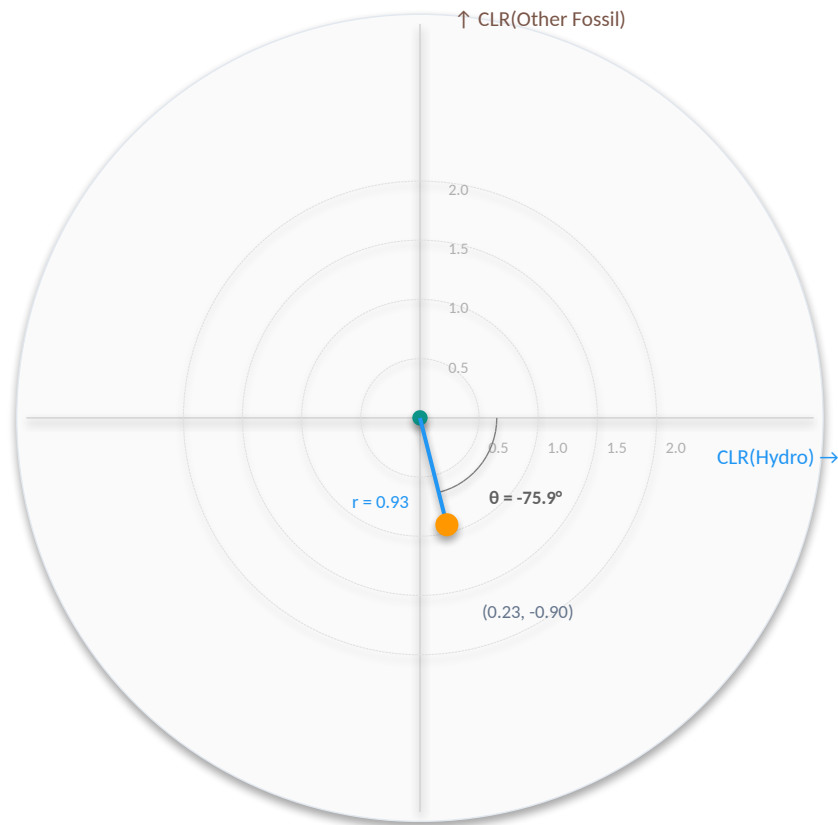
← 7.6° from 2018

→ 9.8° to 2020

Coal-Gas Lock: $\theta_{CG} = 47.8^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -75.9^\circ$

$r = 0.931$ (projection radius)

$d_A = 3.463$ (full 8D Aitchison)

Projection covers 27% of d_A

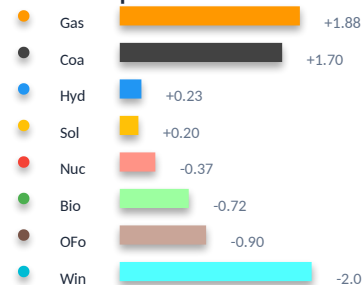
Angular Velocity (8D CLR)

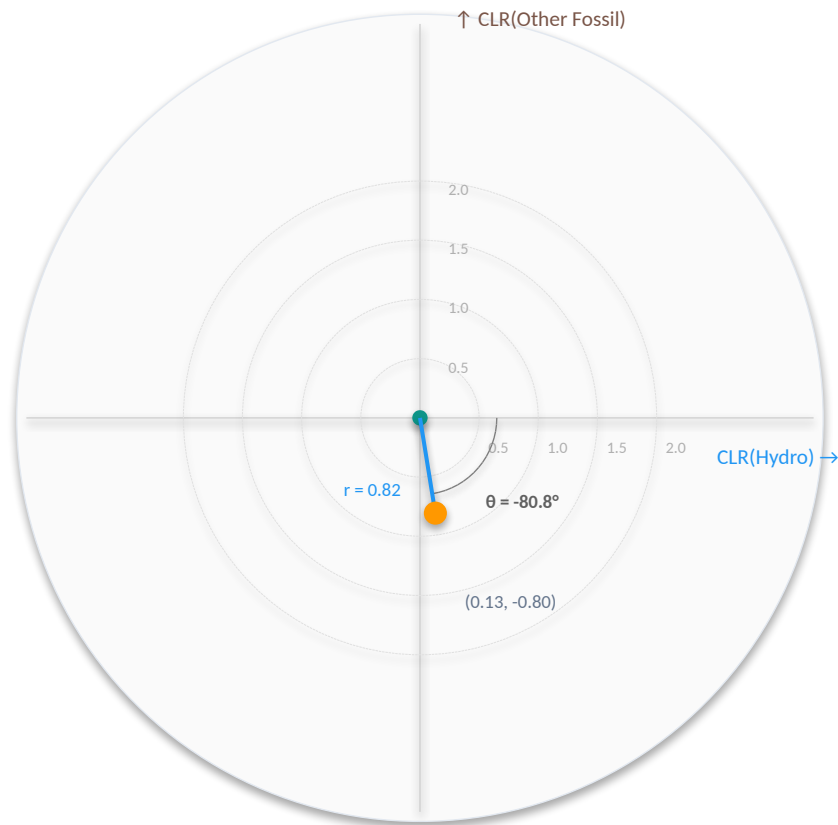
← 9.8° from 2019

→ 4.7° to 2021

Coal-Gas Lock: $\theta_{CG} = 48.0^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -80.8^\circ$

$r = 0.815$ (projection radius)

$d_A = 3.266$ (full 8D Aitchison)

Projection covers 25% of d_A

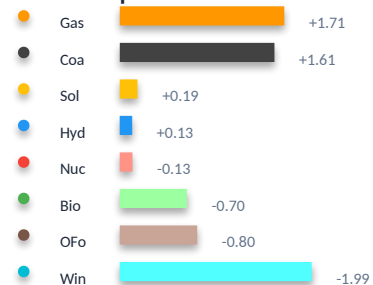
Angular Velocity (8D CLR)

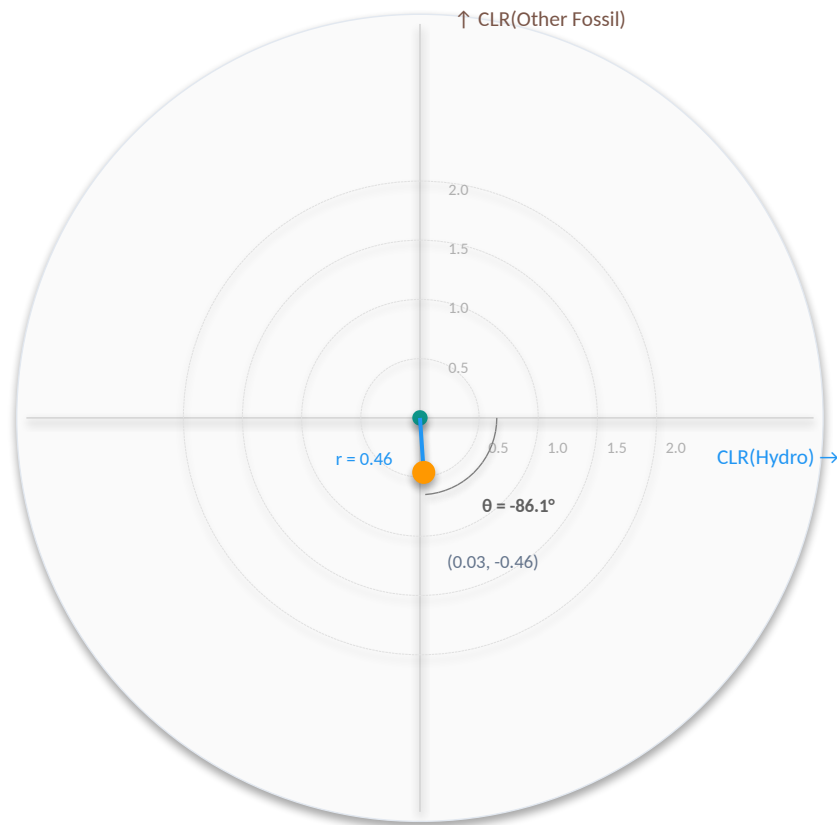
← 4.7° from 2020

→ 7.4° to 2022

Coal-Gas Lock: $\theta_{CG} = 46.8^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -86.1^\circ$

$r = 0.461$ (projection radius)

$d_A = 3.192$ (full 8D Aitchison)

Projection covers 14% of d_A

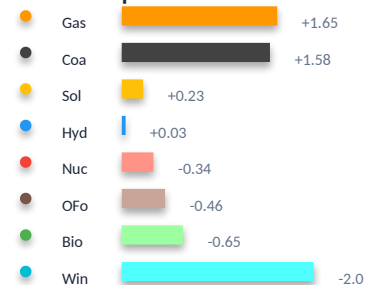
Angular Velocity (8D CLR)

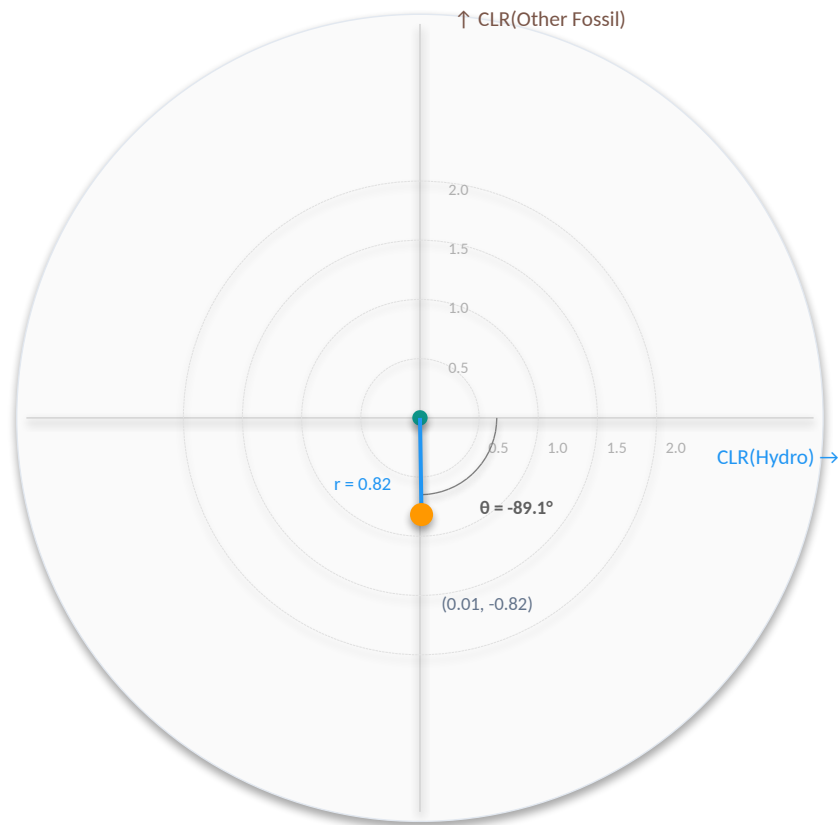
← 7.4° from 2021

→ 10.0° to 2023

Coal-Gas Lock: $\theta_{CG} = 46.4^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

$\theta = -89.1^\circ$

$r = 0.817$ (projection radius)

$d_A = 3.112$ (full 8D Aitchison)

Projection covers 26% of d_A

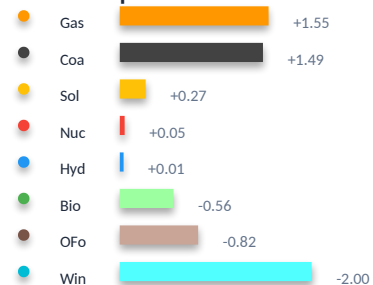
Angular Velocity (8D CLR)

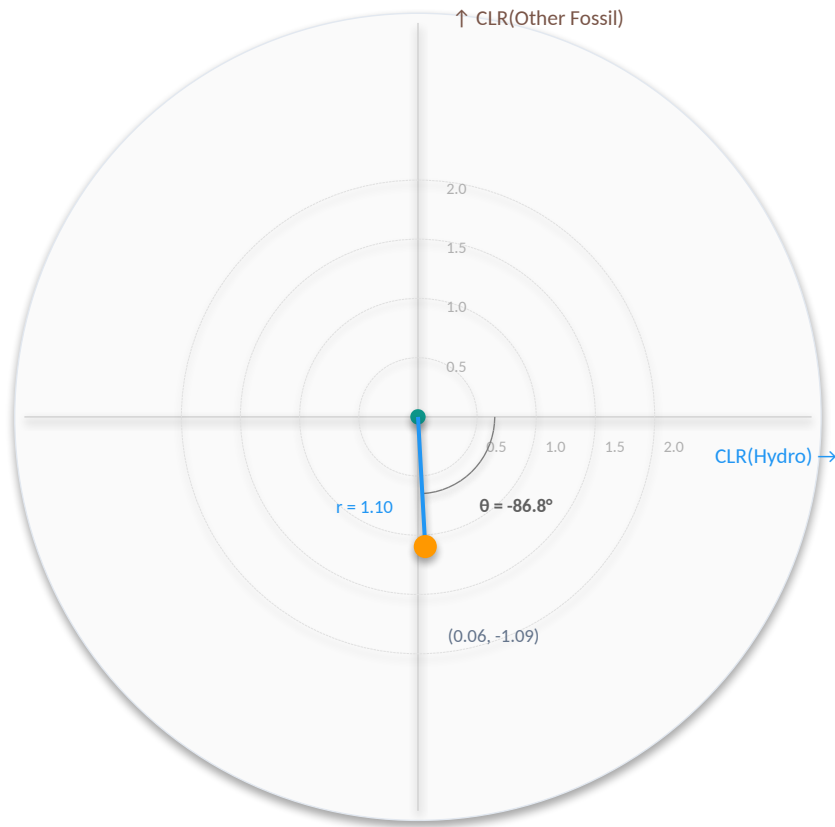
← 10.0° from 2022

→ 6.0° to 2024

Coal-Gas Lock: $\theta_{CG} = 46.1^\circ$ (Gas > Coal)

CLR Components





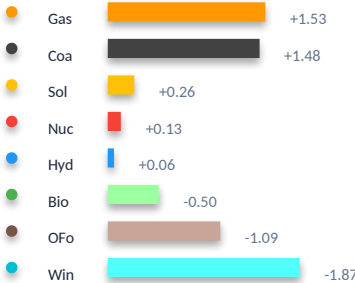
Polar Decomposition

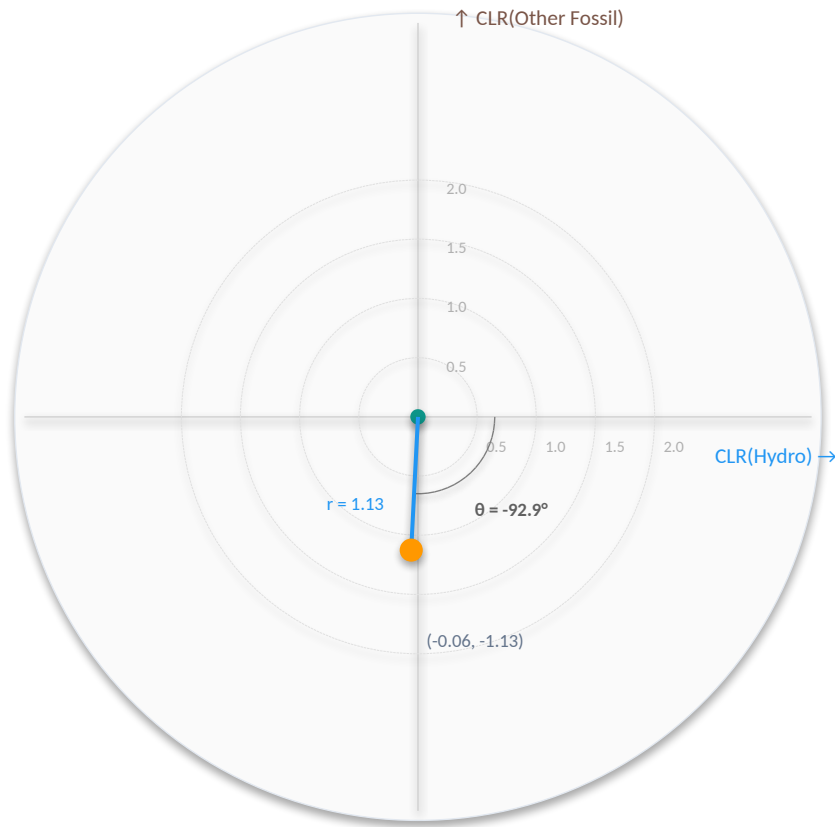
$\theta = -86.8^\circ$
 $r = 1.096$ (projection radius)
 $d_A = 3.091$ (full 8D Aitchison)
Projection covers 35% of d_A

Angular Velocity (8D CLR)

← 6.0° from 2023
→ 3.8° to 2025
Coal-Gas Lock: $\theta_{CG} = 46.1^\circ$ (Gas > Coal)

CLR Components





Polar Decomposition

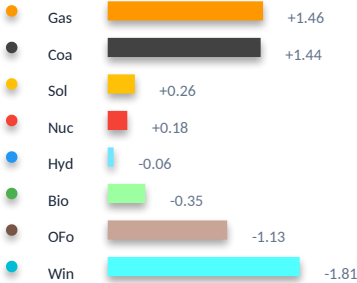
$\theta = -92.9^\circ$
 $r = 1.127$ (projection radius)
 $d_A = 2.995$ (full 8D Aitchison)
Projection covers 38% of d_A

Angular Velocity (8D CLR)

← 3.8° from 2024

Coal-Gas Lock: $\theta_{CG} = 45.4^\circ$ (Gas > Coal)

CLR Components



XZ Face — Front Elevation

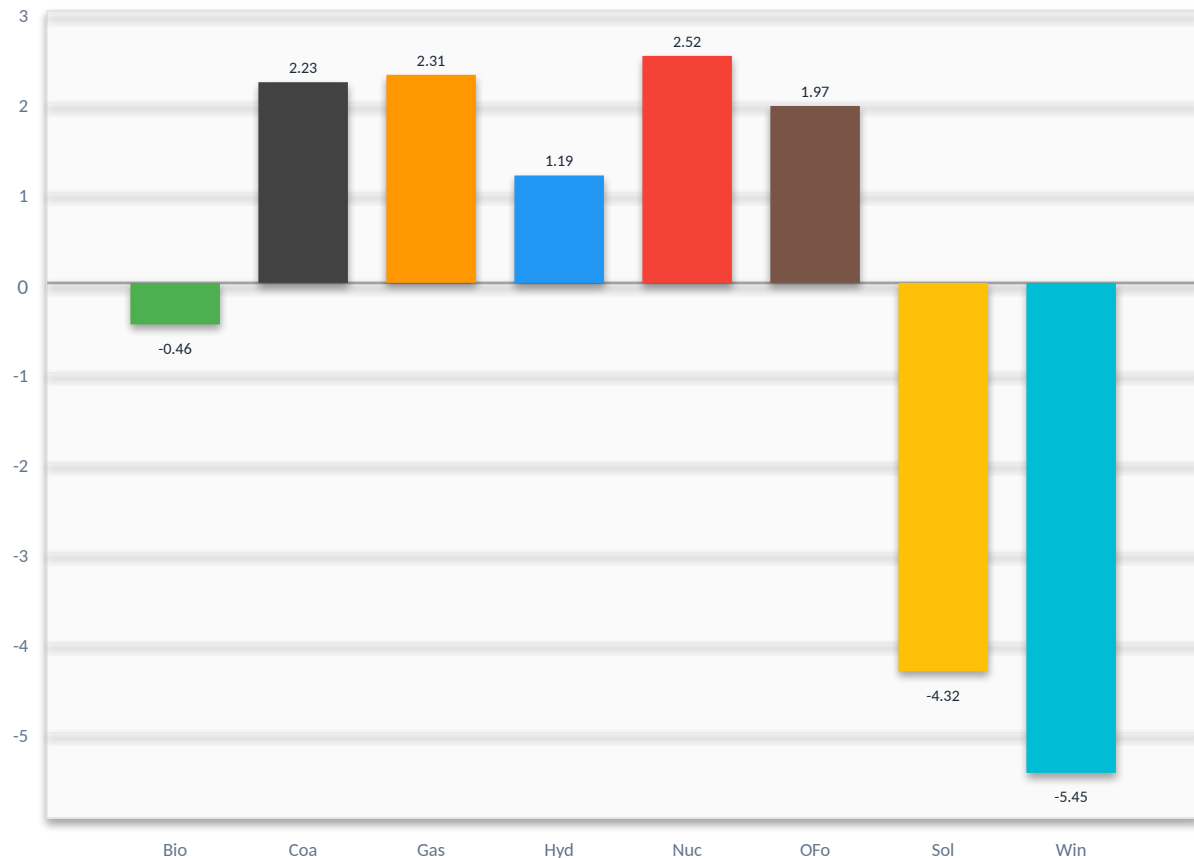
CLR values per carrier · Higgins Scale zone classification

26 annual profiles · Each composition presented as CLR bar chart

XZ Face — 2000

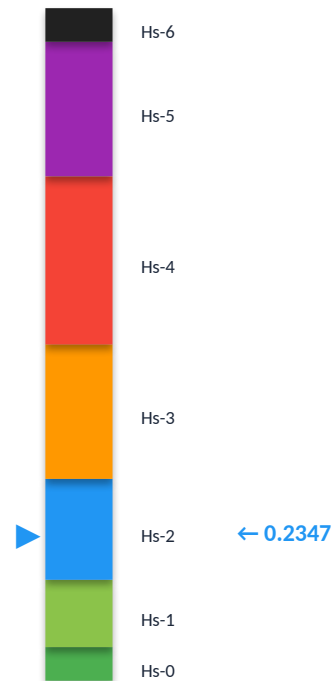
Hs = 0.2347 · Hs-2 STRUCTURED

d_A = 8.40 · Dom: Nuclear



H* Direct · Japan EMBER · 2000 · XZ Face · CLR zero-sum verified

Higgins Scale



d_A = 8.400

N_eff = 4.91 / 8

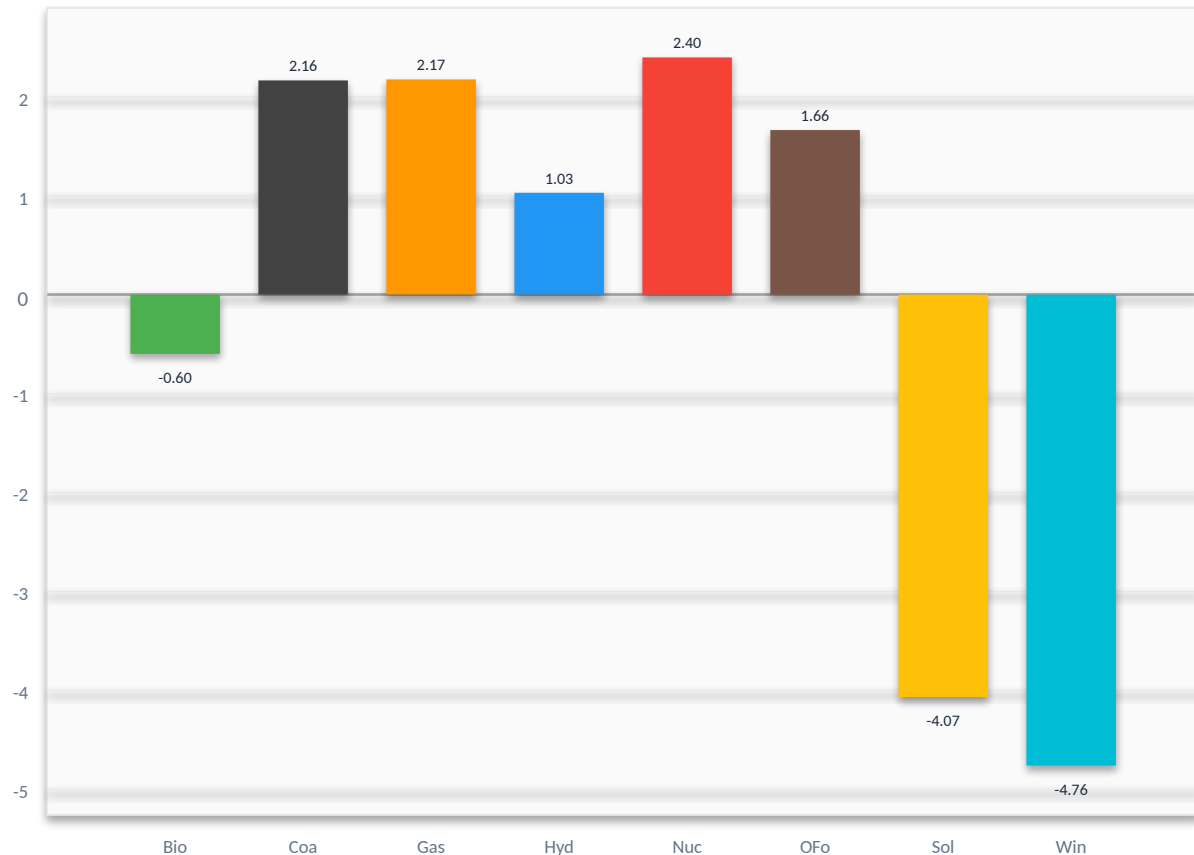
p = 0.0848

Dominant: Nuclear

XZ Face — 2001

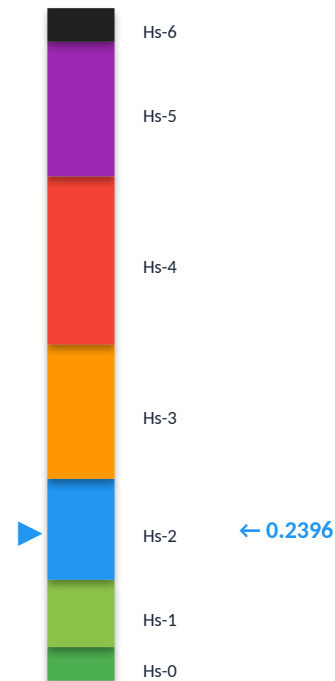
$H_s = 0.2396$ · Hs-2 STRUCTURED

$d_A = 7.65$ · Dom: Nuclear



H* Direct · Japan EMBER · 2001 · XZ Face · CLR zero-sum verified

Higgins Scale



$d_A = 7.650$

$N_{eff} = 4.86 / 8$

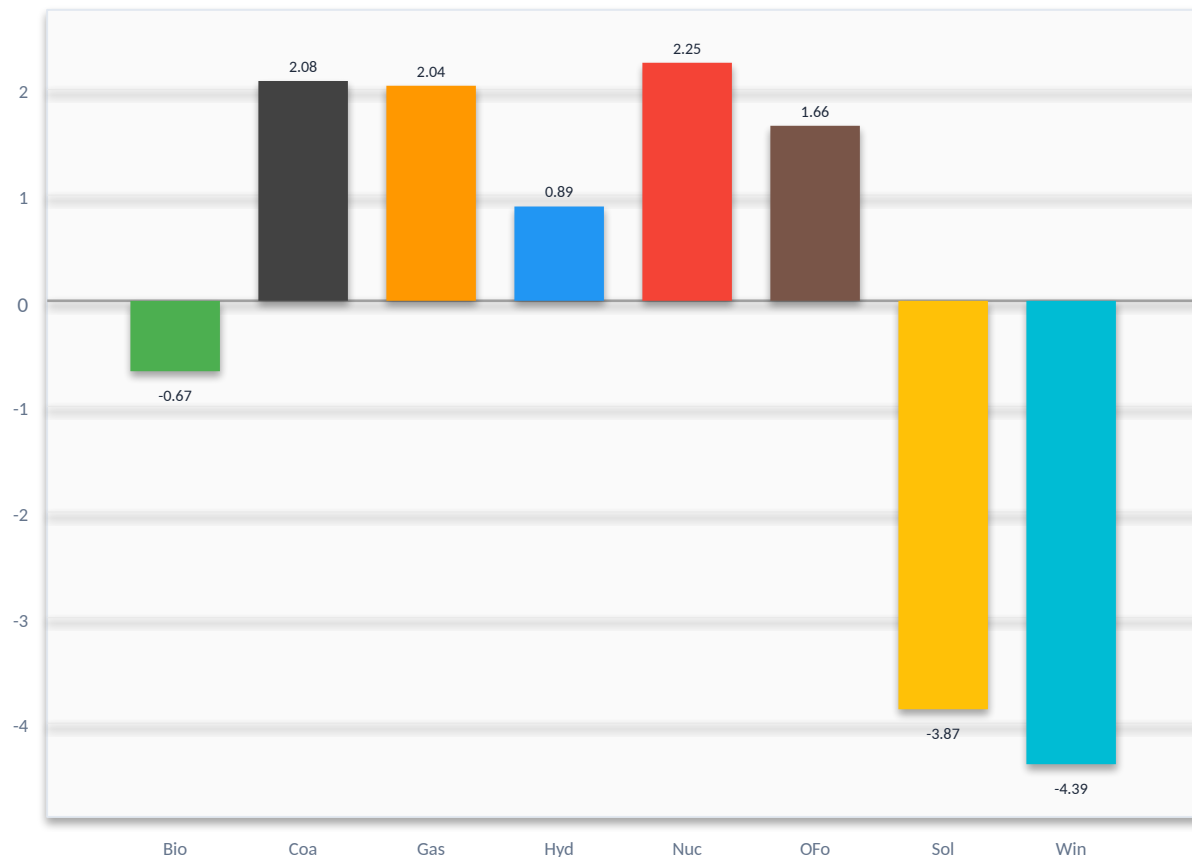
$\rho = 0.1258$

Dominant: Nuclear

XZ Face — 2002

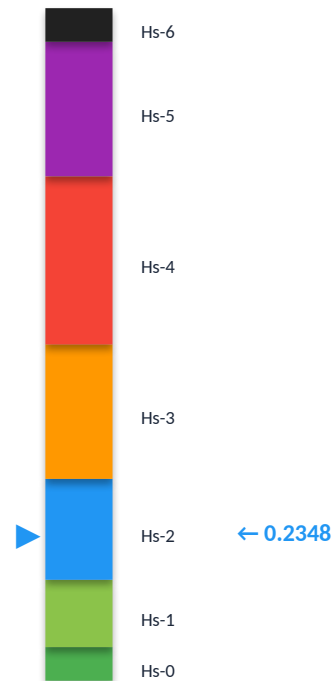
$H_s = 0.2348$ · Hs-2 STRUCTURED

$d_A = 7.19$ · Dom: Nuclear



H* Direct · Japan EMBER · 2002 · XZ Face · CLR zero-sum verified

Higgins Scale



$d_A = 7.194$

$N_{eff} = 4.91 / 8$

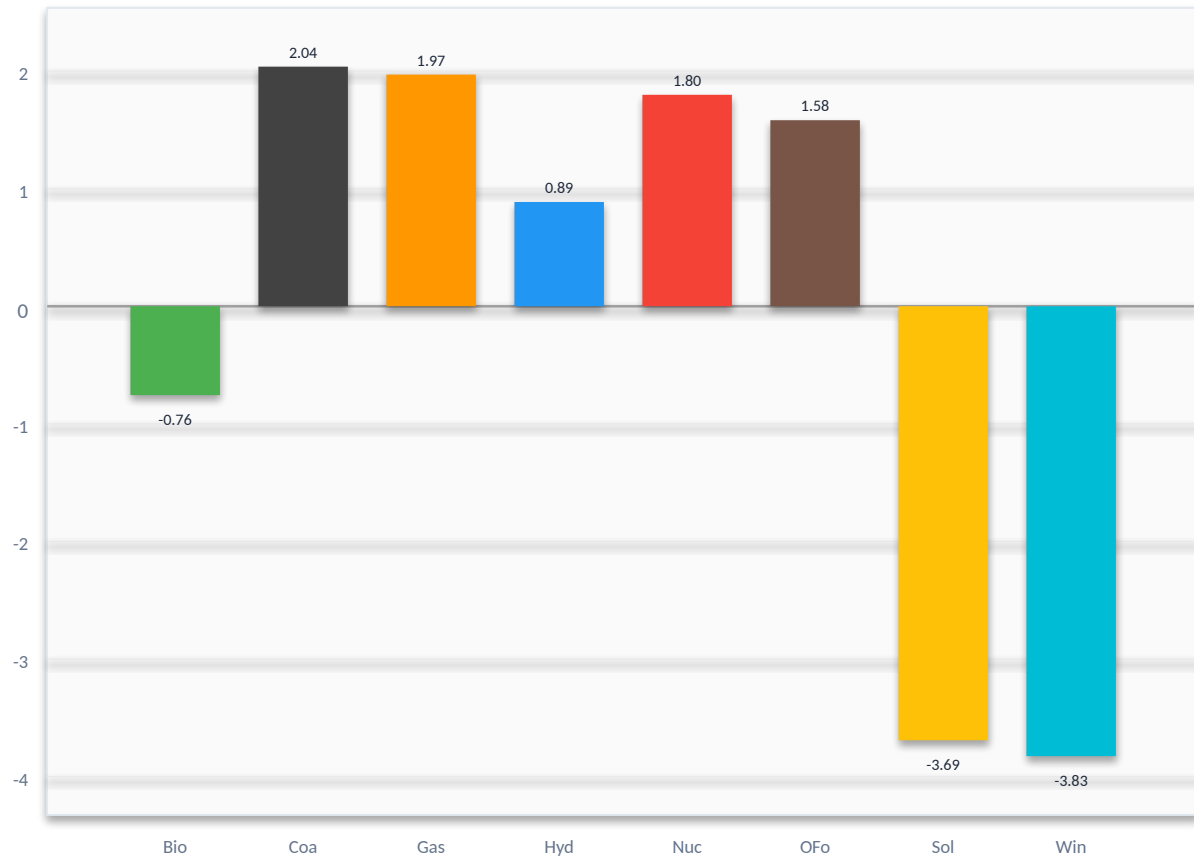
$p = 0.1519$

Dominant: Nuclear

XZ Face — 2003

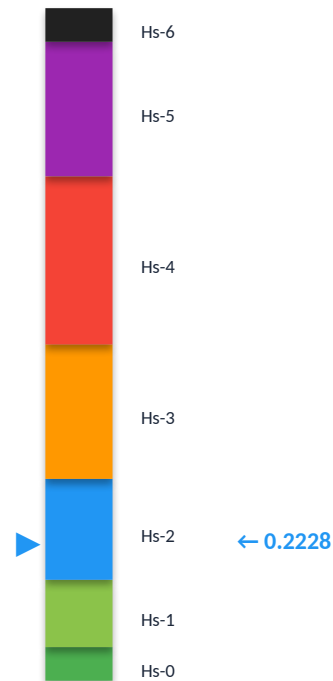
Hs = 0.2228 · Hs-2 STRUCTURED

d_A = 6.59 · Dom: Coal



H* Direct · Japan EMBER · 2003 · XZ Face · CLR zero-sum verified

Higgins Scale



d_A = 6.587

N_eff = 5.03 / 8

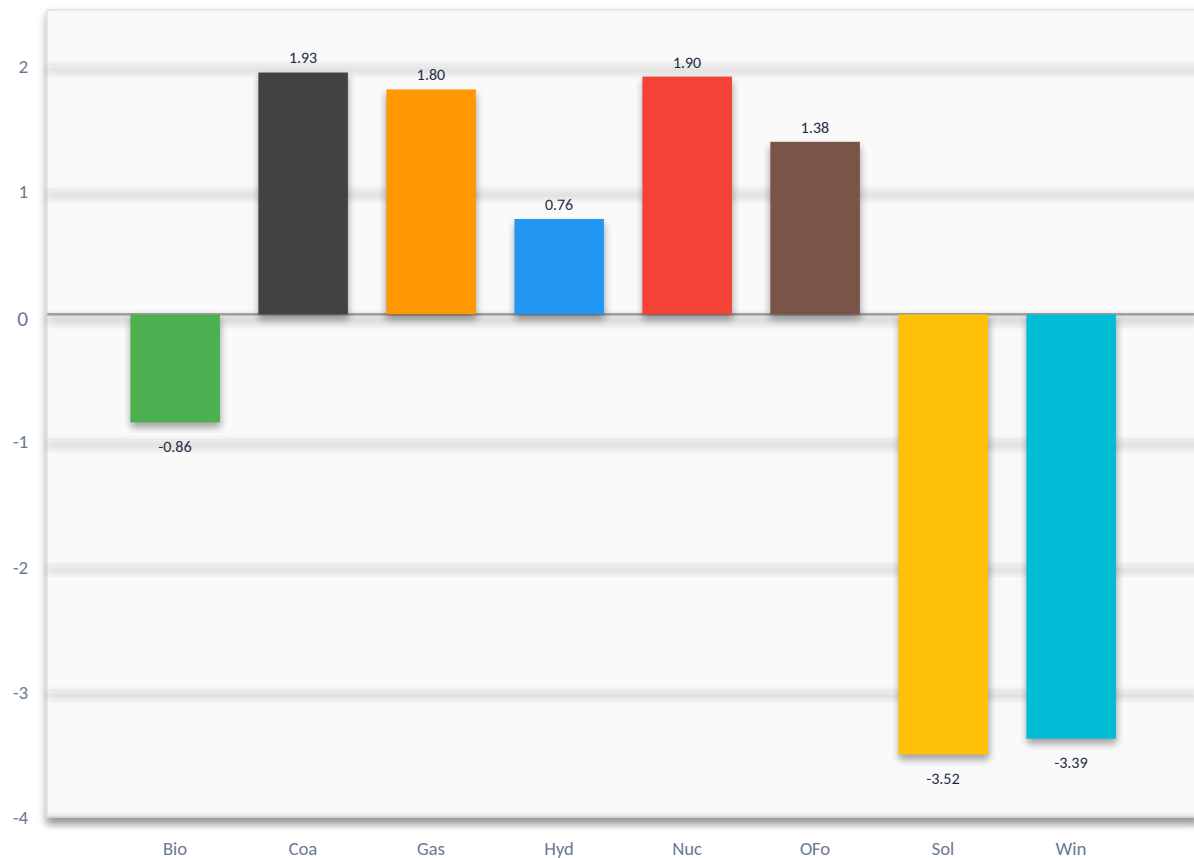
p = 0.1977

Dominant: Coal

XZ Face — 2004

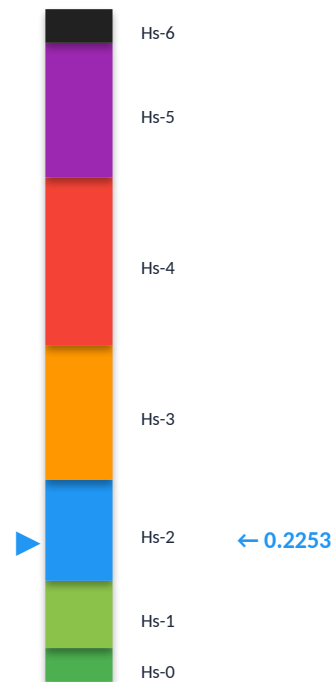
Hs = 0.2253 · Hs-2 STRUCTURED

d_A = 6.14 · Dom: Coal



H* Direct · Japan EMBER · 2004 · XZ Face · CLR zero-sum verified

Higgins Scale



d_A = 6.137

N_eff = 5.01 / 8

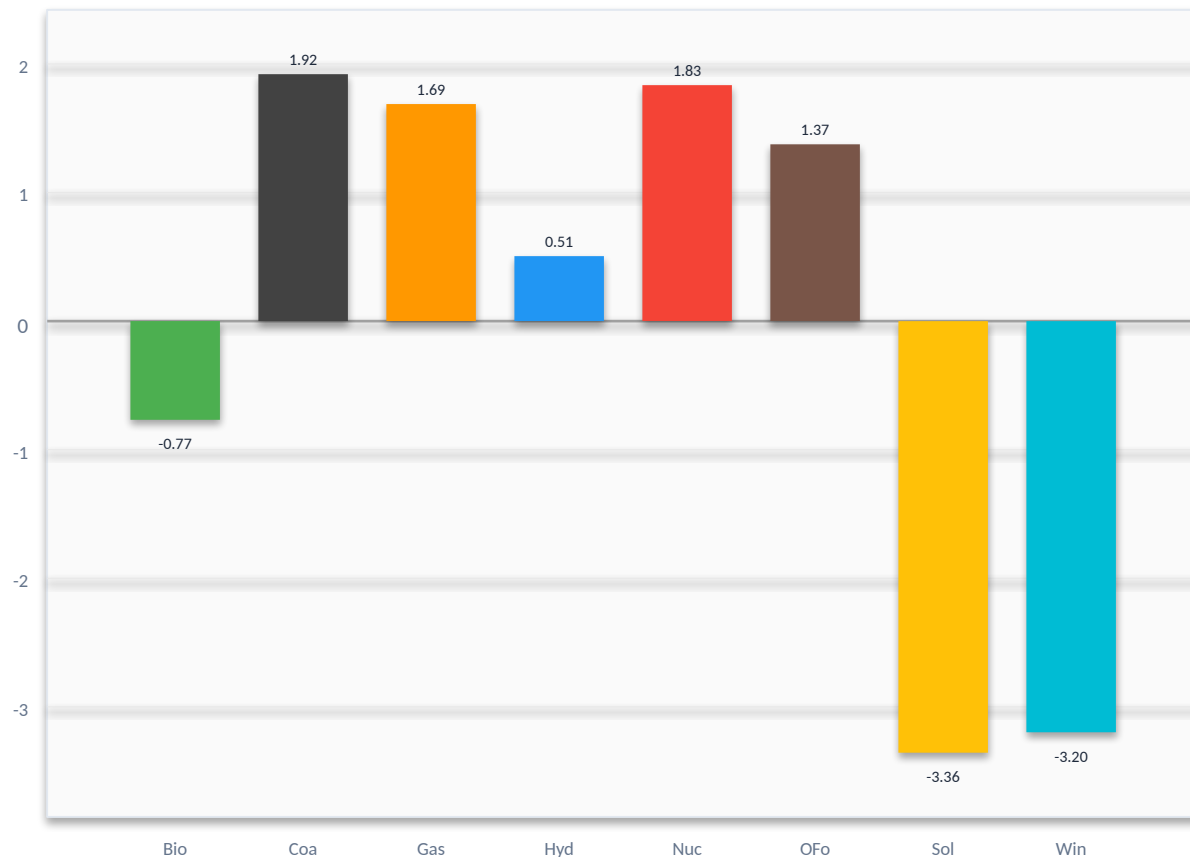
p = 0.2169

Dominant: Coal

XZ Face — 2005

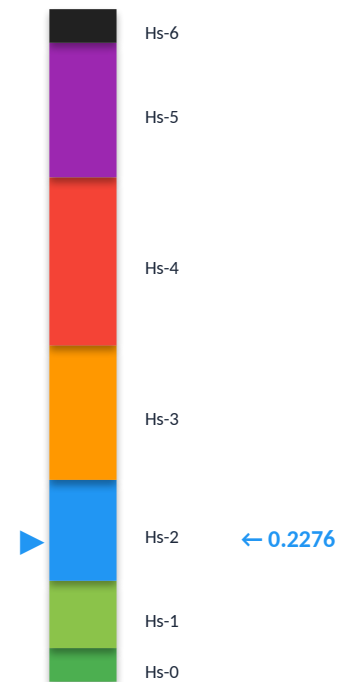
H_s = 0.2276 · H_s-2 STRUCTURED

d_A = 5.84 · Dom: Coal



H* Direct · Japan EMBER · 2005 · XZ Face · CLR zero-sum verified

Higgins Scale



d_A = 5.843

N_{eff} = 4.98 / 8

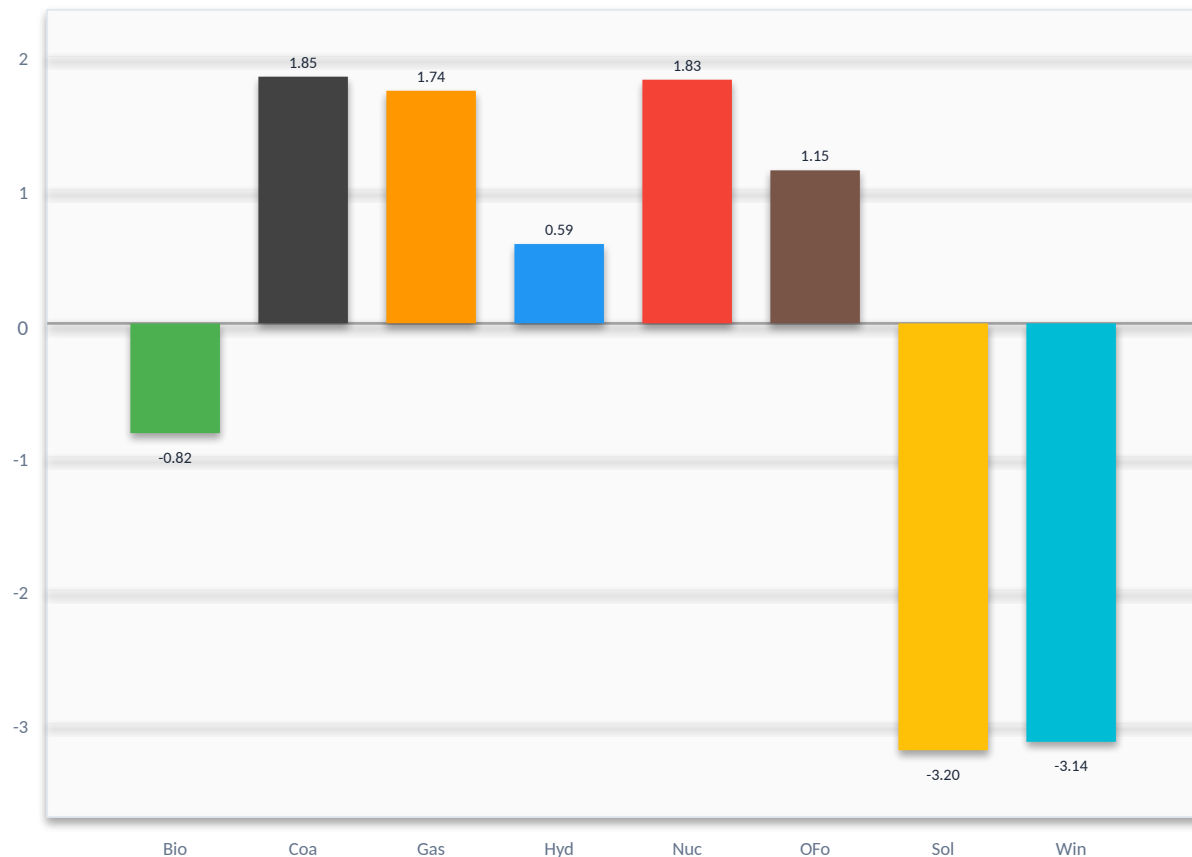
ρ = 0.1506

Dominant: Coal

XZ Face — 2006

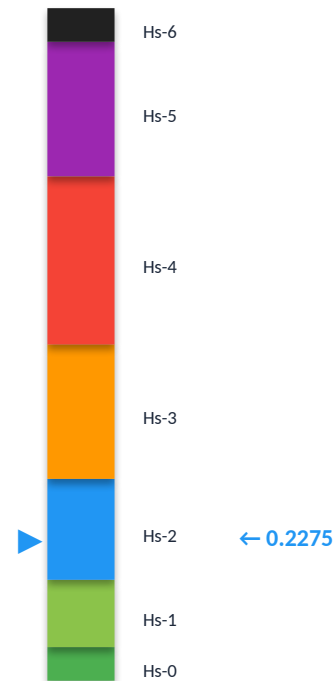
Hs = 0.2275 · Hs-2 STRUCTURED

d_A = 5.67 · Dom: Coal



H* Direct · Japan EMBER · 2006 · XZ Face · CLR zero-sum verified

Higgins Scale

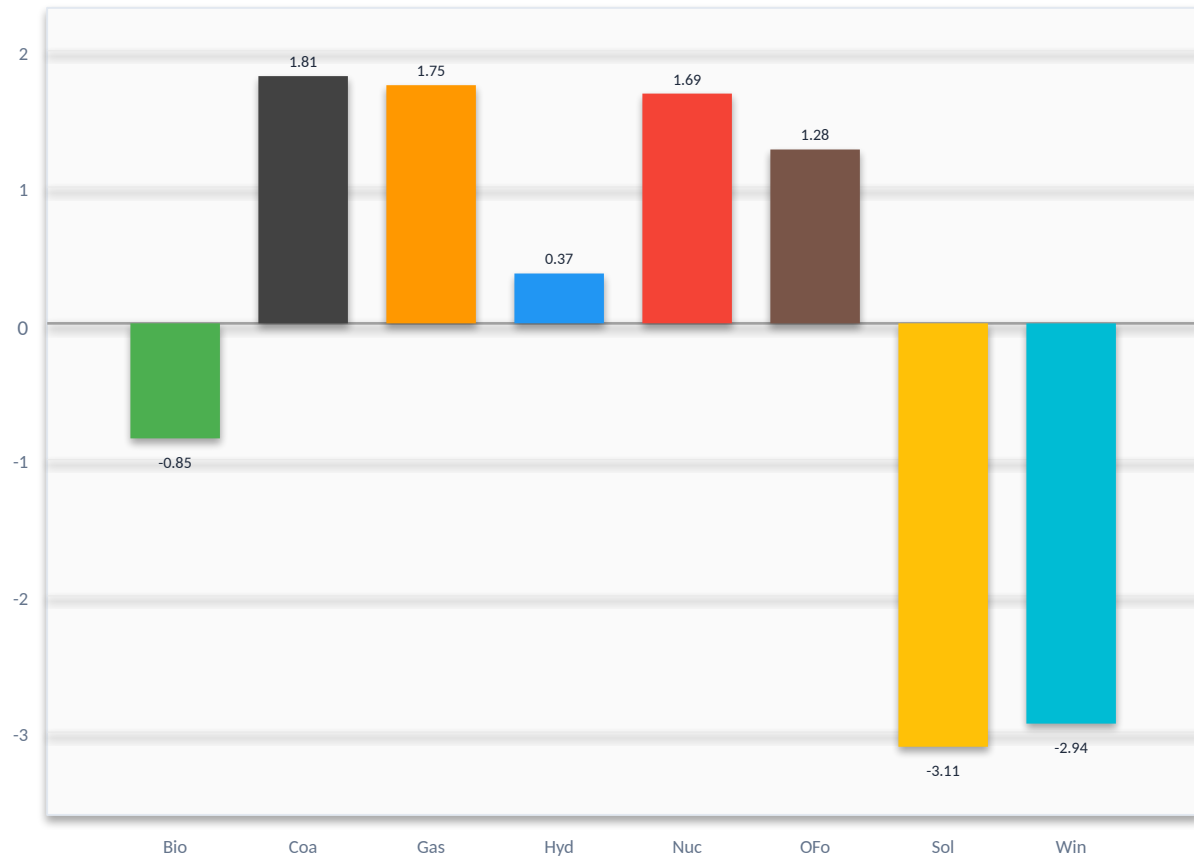


d_A = 5.675

N_eff = 4.98 / 8

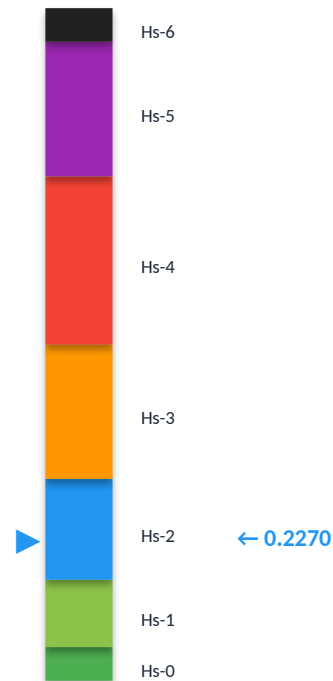
p = 0.1858

Dominant: Coal



H* Direct · Japan EMBER · 2007 · XZ Face · CLR zero-sum verified

Higgins Scale

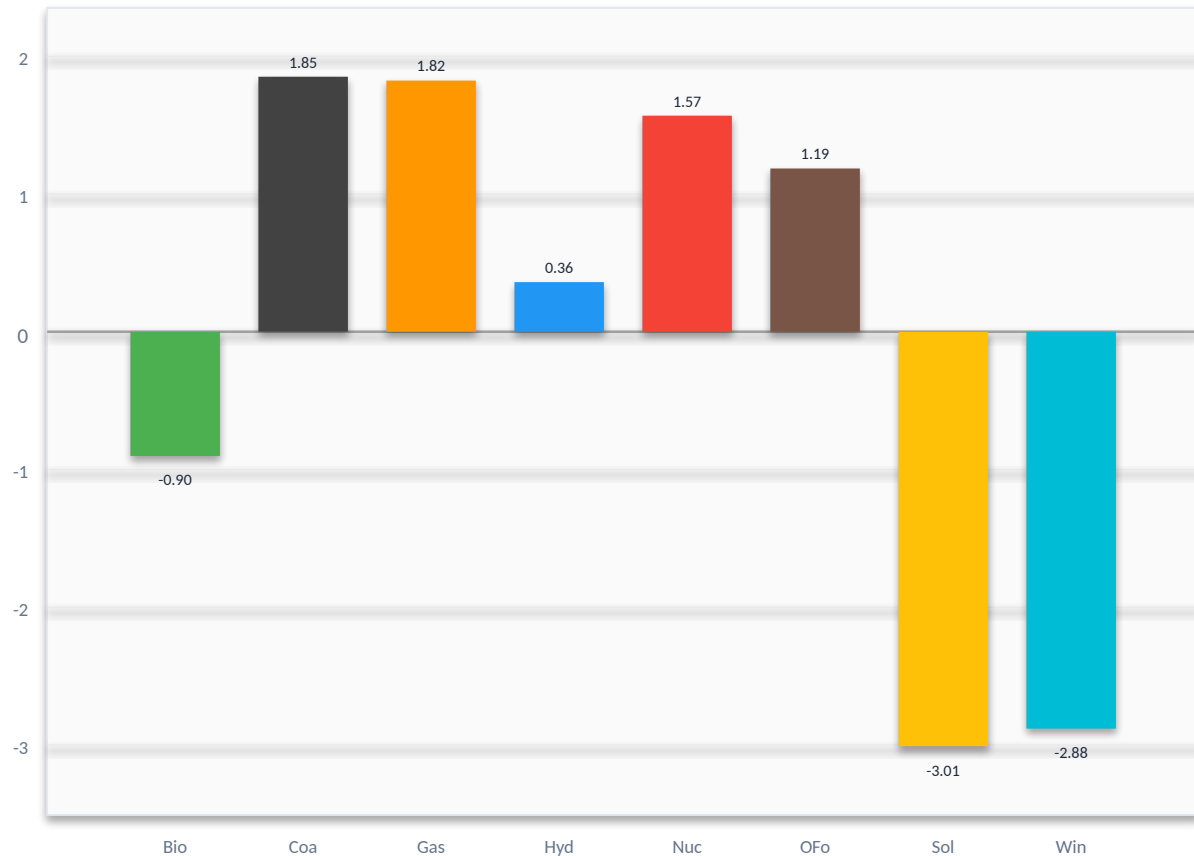


$d_A = 5.473$

$N_{eff} = 4.99 / 8$

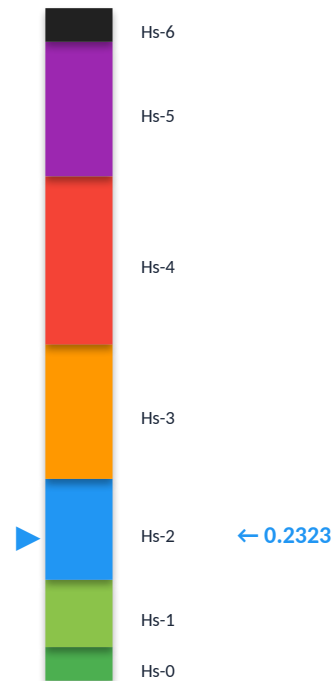
$p = 0.1177$

Dominant: Coal



H* Direct · Japan EMBER · 2008 · XZ Face · CLR zero-sum verified

Higgins Scale



$d_A = 5.376$

$N_{eff} = 4.94 / 8$

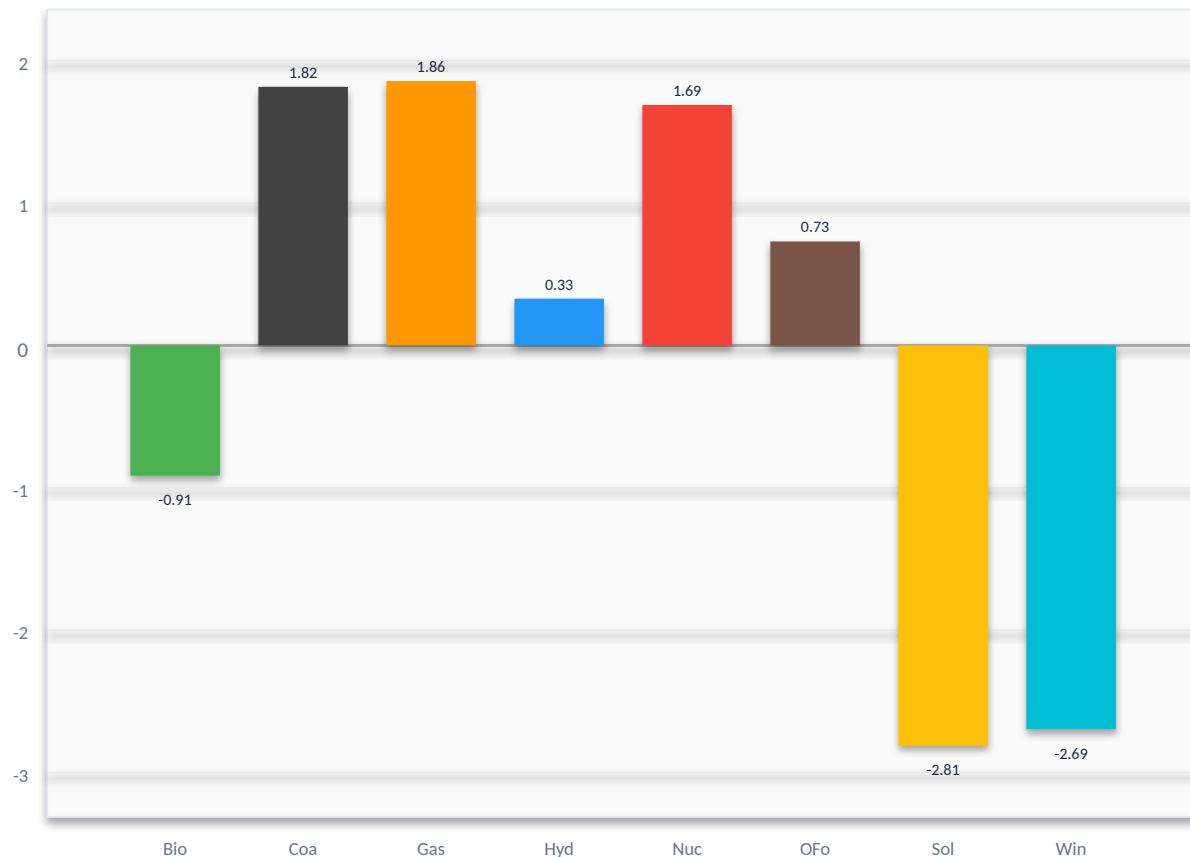
$\rho = 0.1198$

Dominant: Coal

XZ Face — 2009

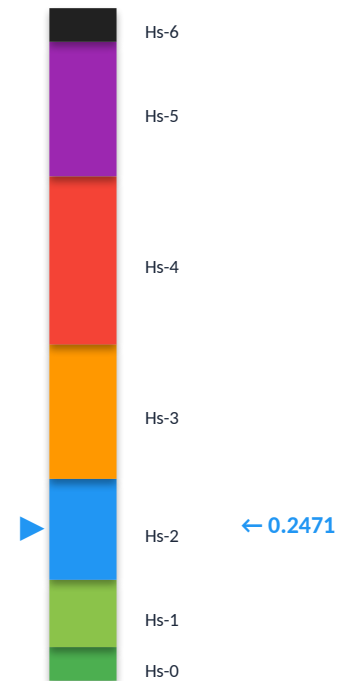
$H_s = 0.2471$ · Hs-2 STRUCTURED

$d_A = 5.12$ · Dom: Gas



H* Direct · Japan EMBER · 2009 · XZ Face · CLR zero-sum verified

Higgins Scale



$d_A = 5.122$

$N_{eff} = 4.79 / 8$

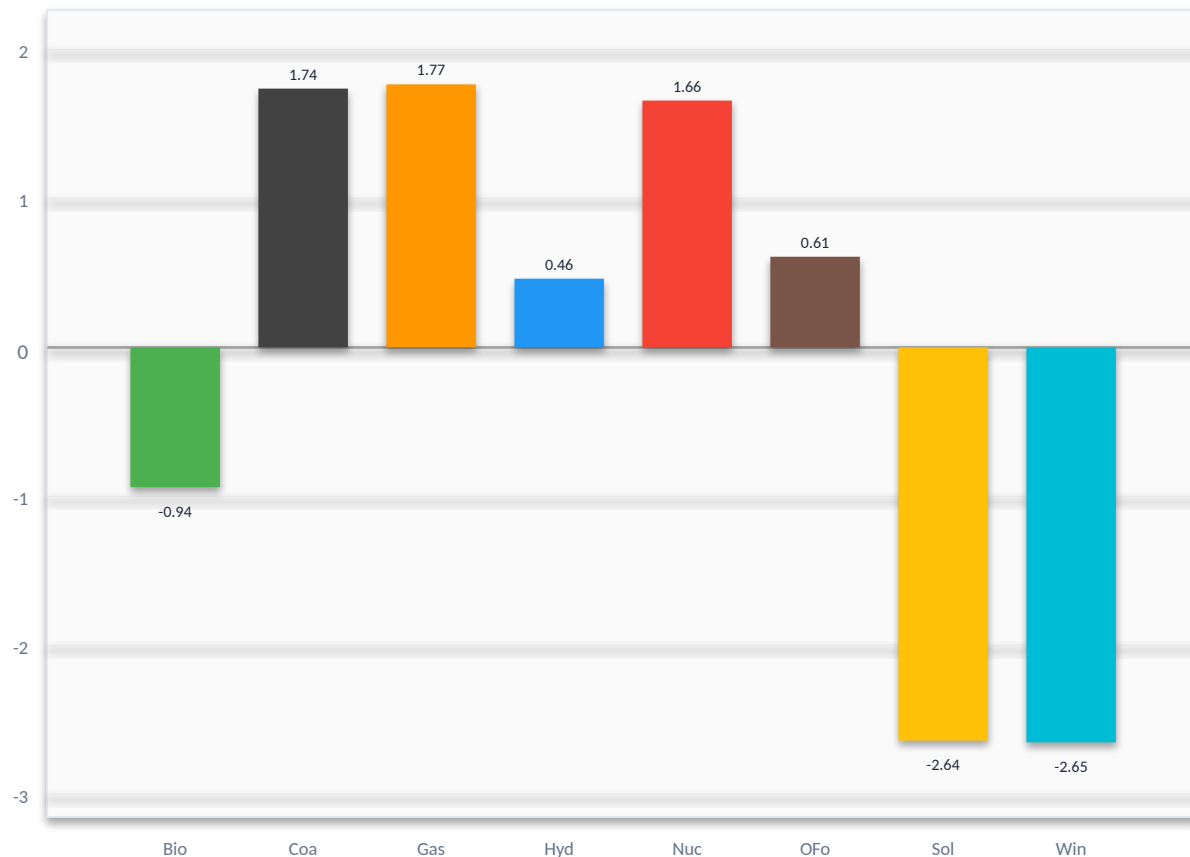
$p = 0.1168$

Dominant: Gas

XZ Face — 2010

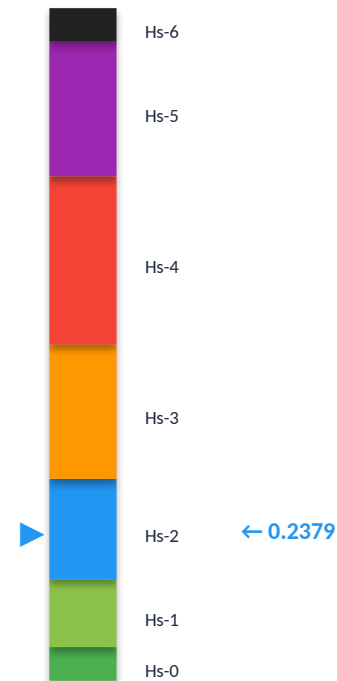
$H_s = 0.2379$ · Hs-2 STRUCTURED

$d_A = 4.93$ · Dom: Gas



H* Direct · Japan EMBER · 2010 · XZ Face · CLR zero-sum verified

Higgins Scale



$d_A = 4.935$

$N_{eff} = 4.88 / 8$

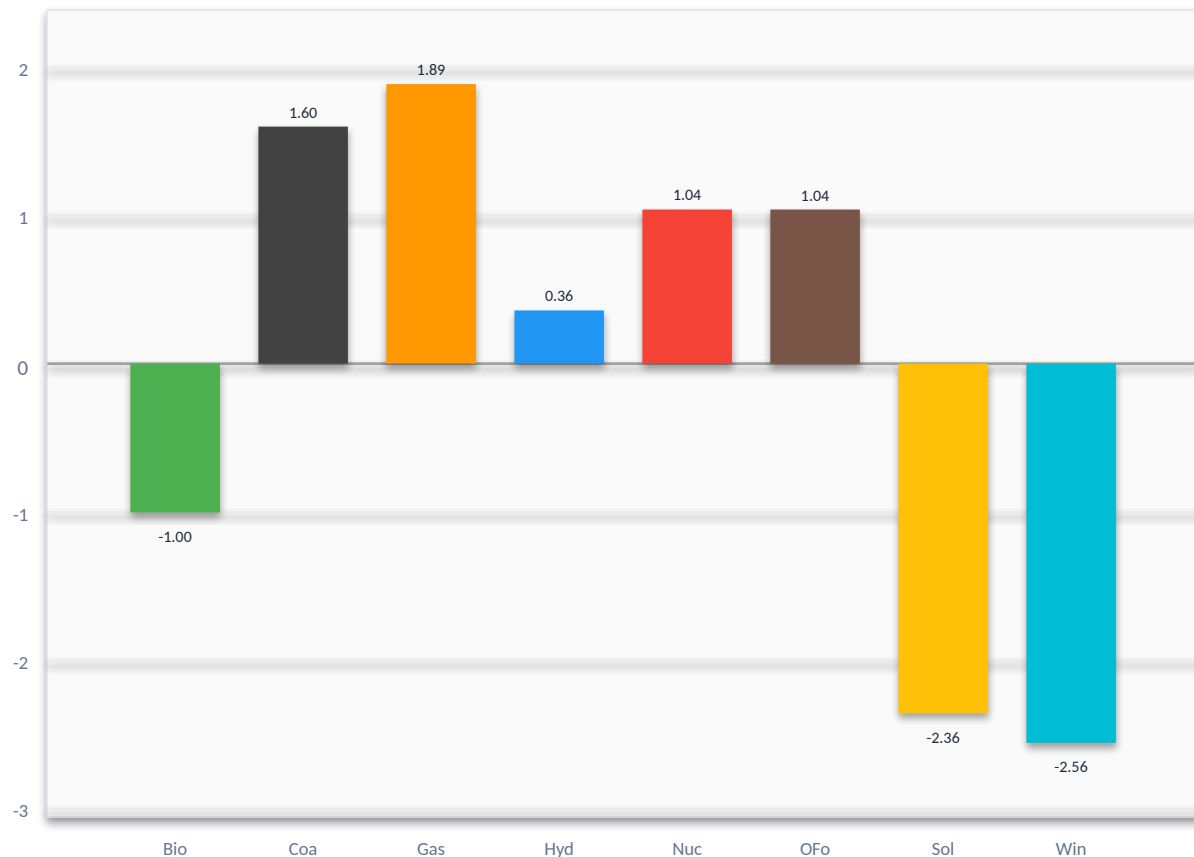
$p = 0.1741$

Dominant: Gas

XZ Face — 2011

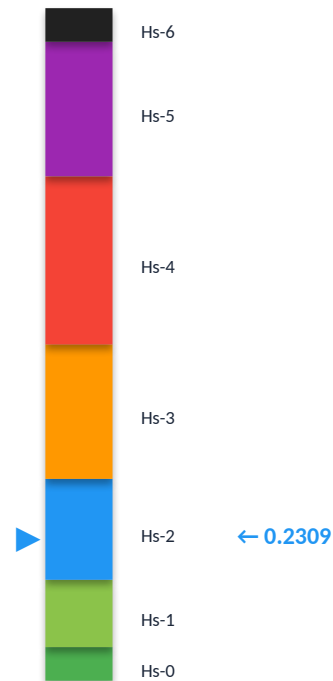
$H_s = 0.2309$ · Hs-2 STRUCTURED

$d_A = 4.64$ · Dom: Gas



H* Direct · Japan EMBER · 2011 · XZ Face · CLR zero-sum verified

Higgins Scale

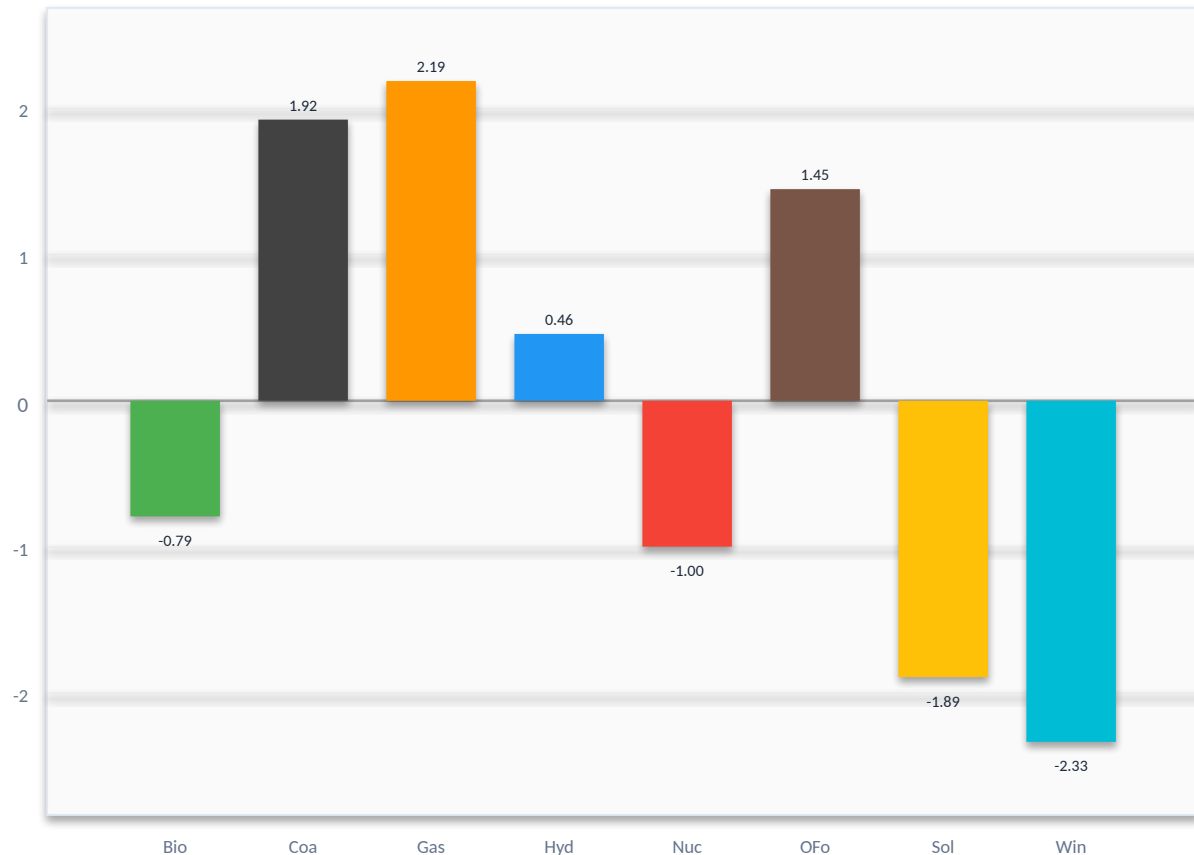


$d_A = 4.640$

$N_{eff} = 4.95 / 8$

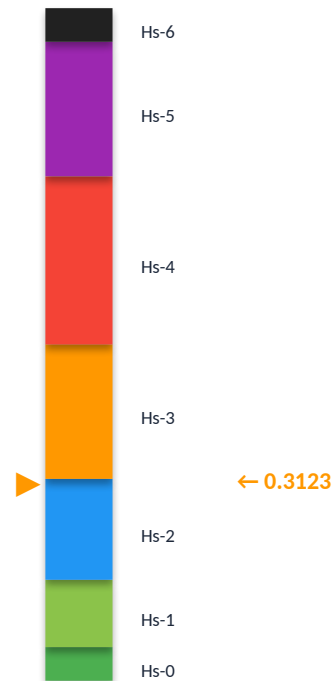
$p = 0.1403$

Dominant: Gas



H* Direct · Japan EMBER · 2012 · XZ Face · CLR zero-sum verified

Higgins Scale

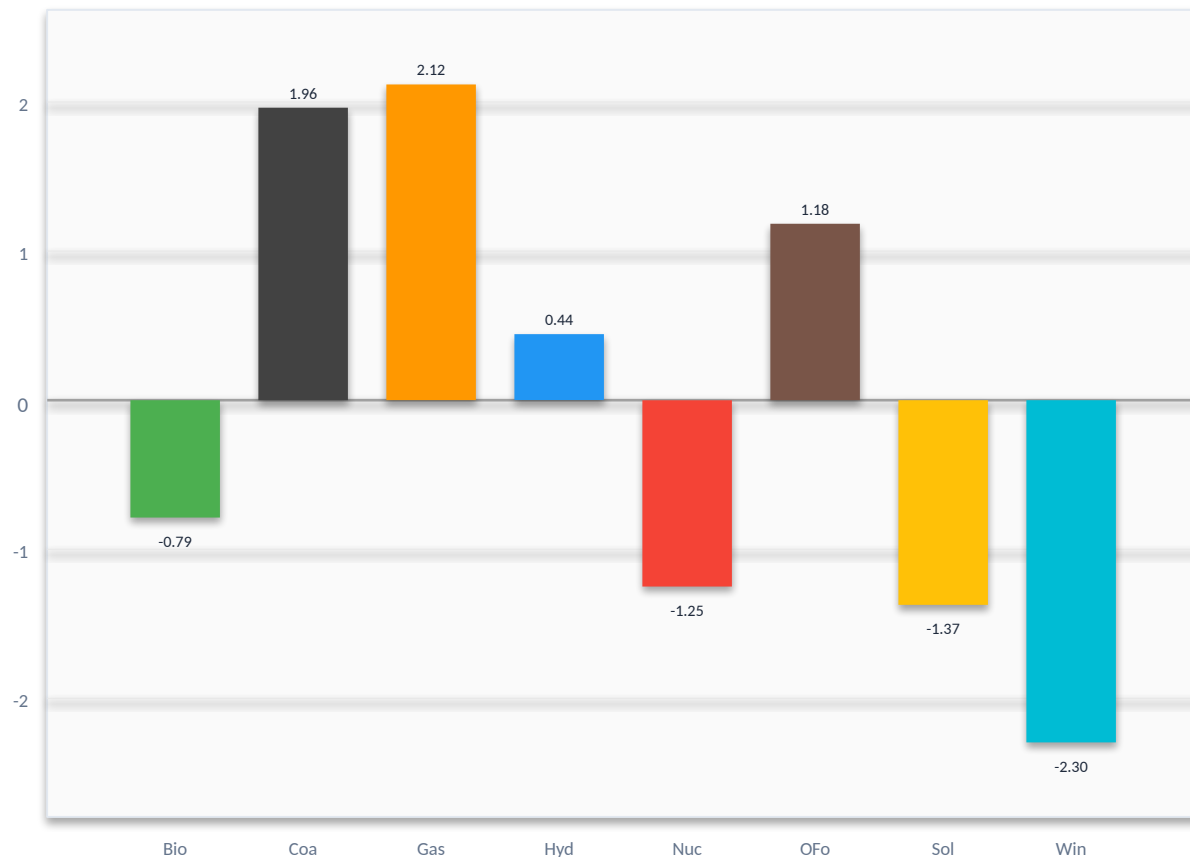


$d_A = 4.631$

$N_{eff} = 4.18 / 8$

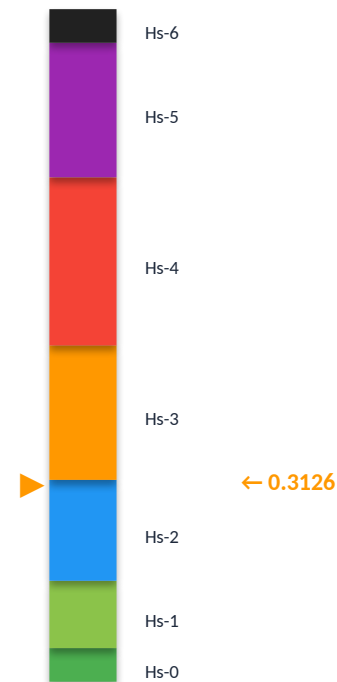
$\rho = 0.1956$

Dominant: Gas



H* Direct · Japan EMBER · 2013 · XZ Face · CLR zero-sum verified

Higgins Scale

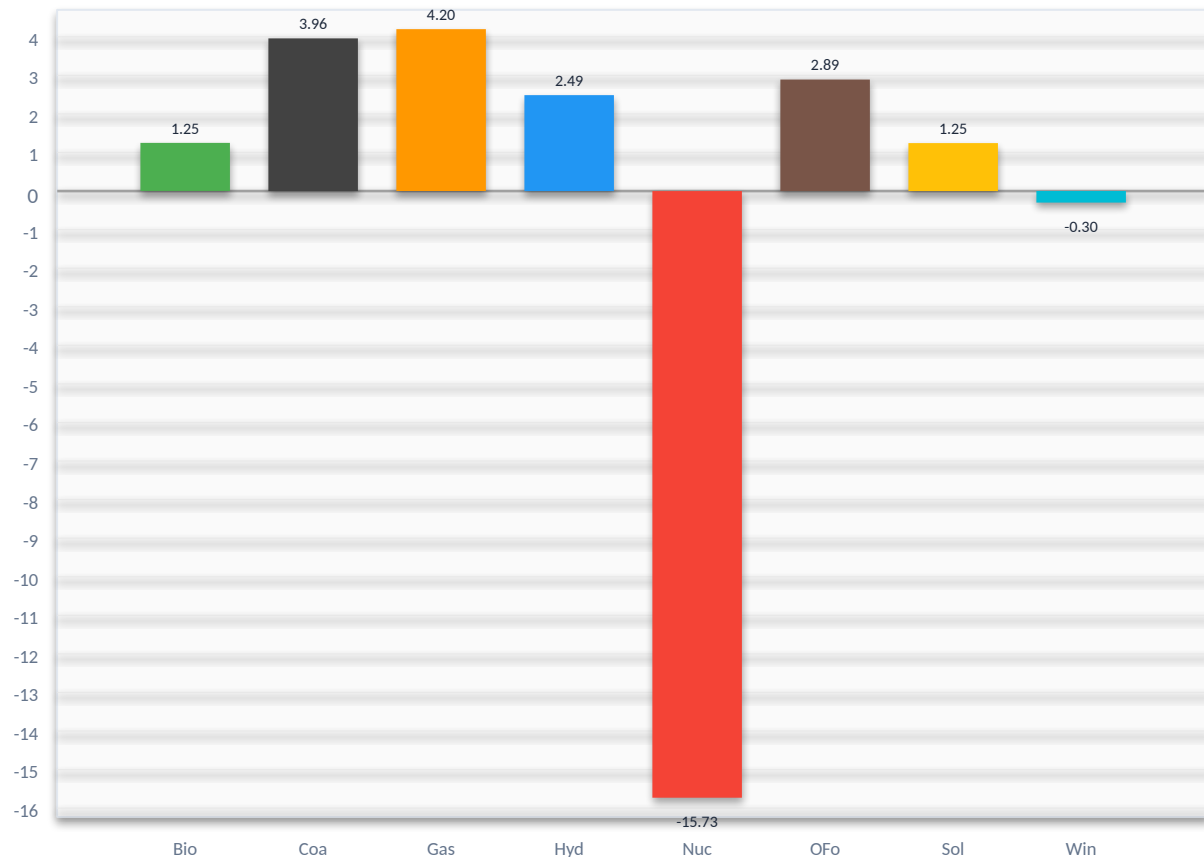


d_A = 4.390

N_eff = 4.18 / 8

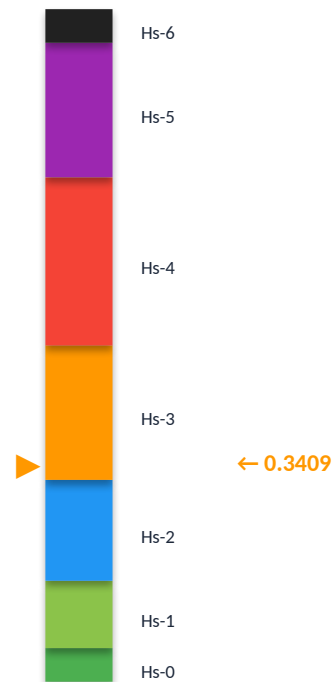
p = 0.1928

Dominant: Gas



H* Direct · Japan EMBER · 2014 · XZ Face · CLR zero-sum verified

Higgins Scale

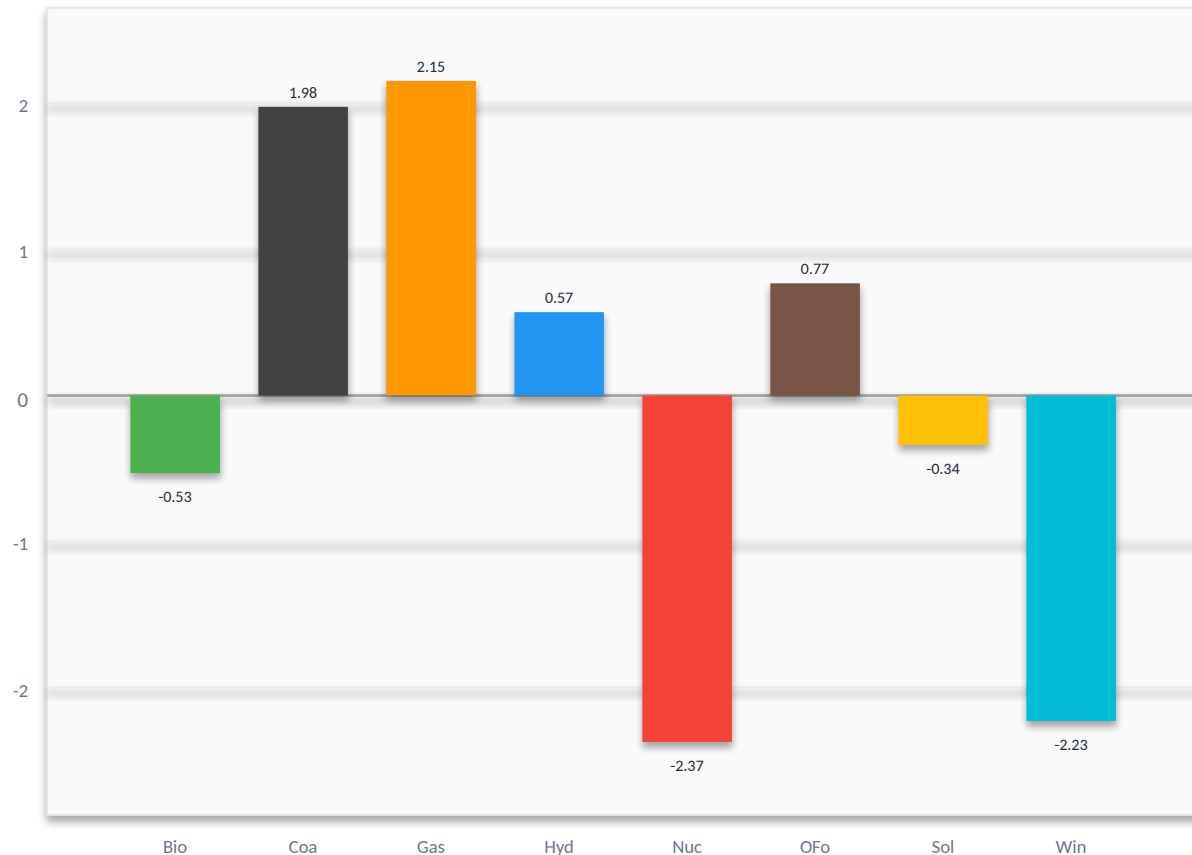


$d_A = 17.277$

$N_{eff} = 3.94 / 8$

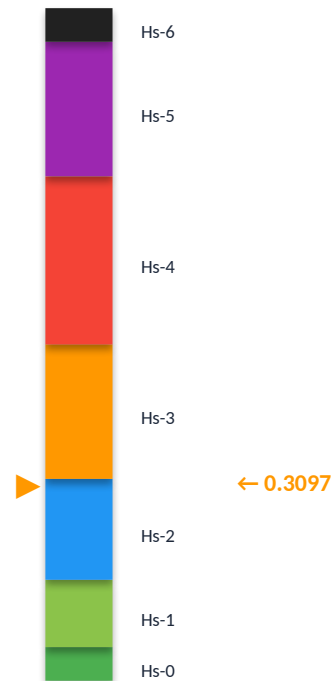
$p = 0.0192$

Dominant: Gas



H* Direct · Japan EMBER · 2015 · XZ Face · CLR zero-sum verified

Higgins Scale

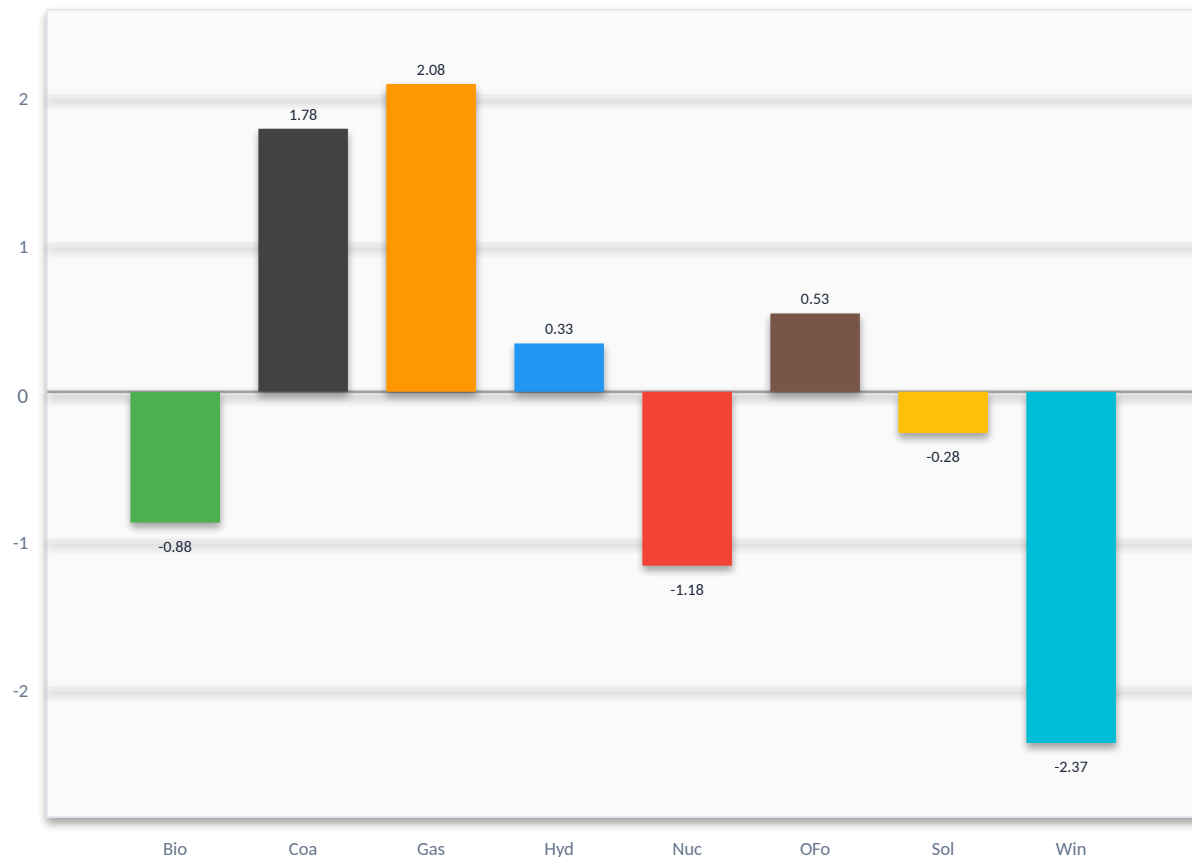


$d_A = 4.522$

$N_{eff} = 4.20 / 8$

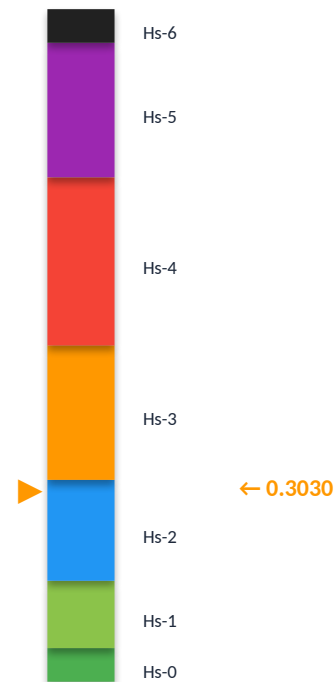
$\rho = 0.1427$

Dominant: Gas



H* Direct · Japan EMBER · 2016 · XZ Face · CLR zero-sum verified

Higgins Scale

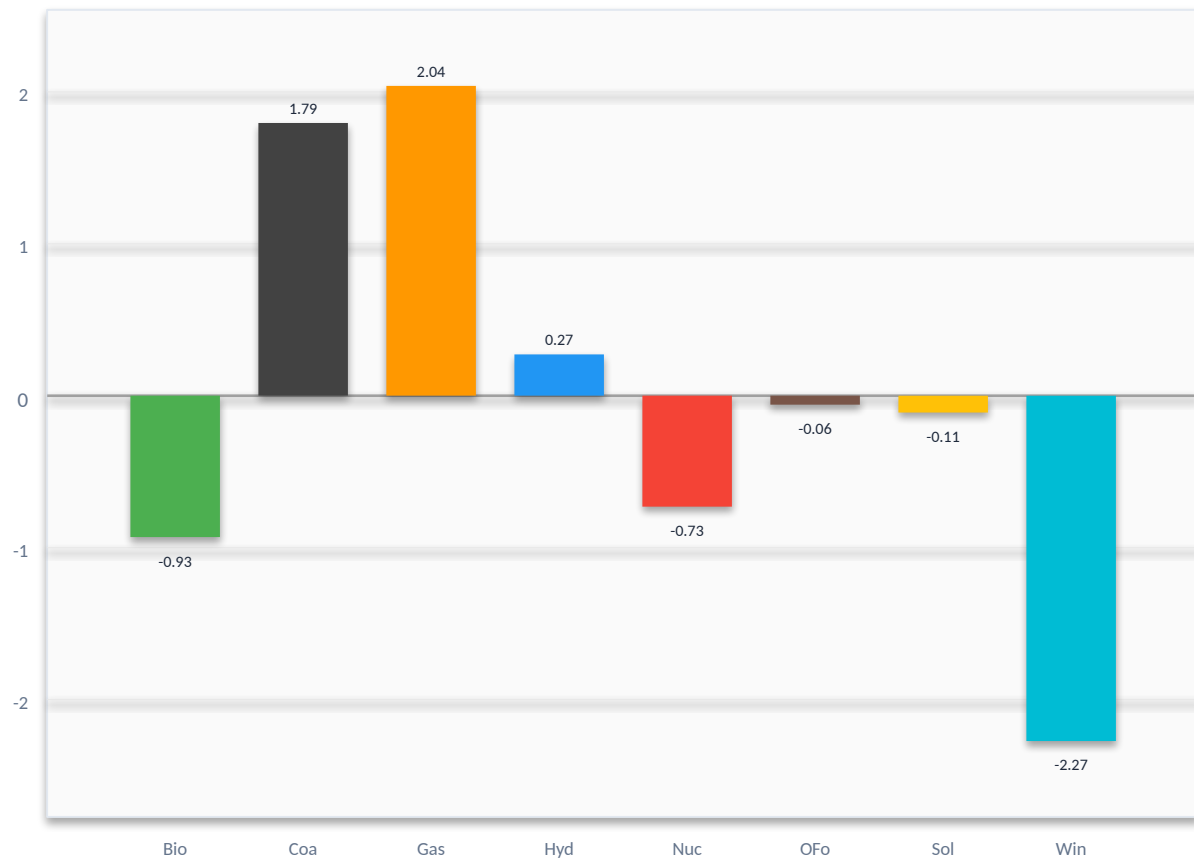


$d_A = 3.967$

$N_{eff} = 4.26 / 8$

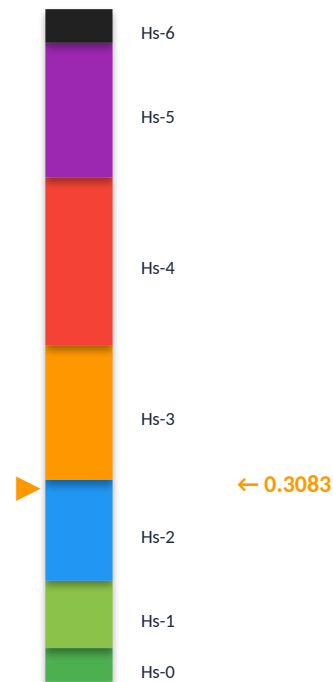
$\rho = 0.1177$

Dominant: Gas

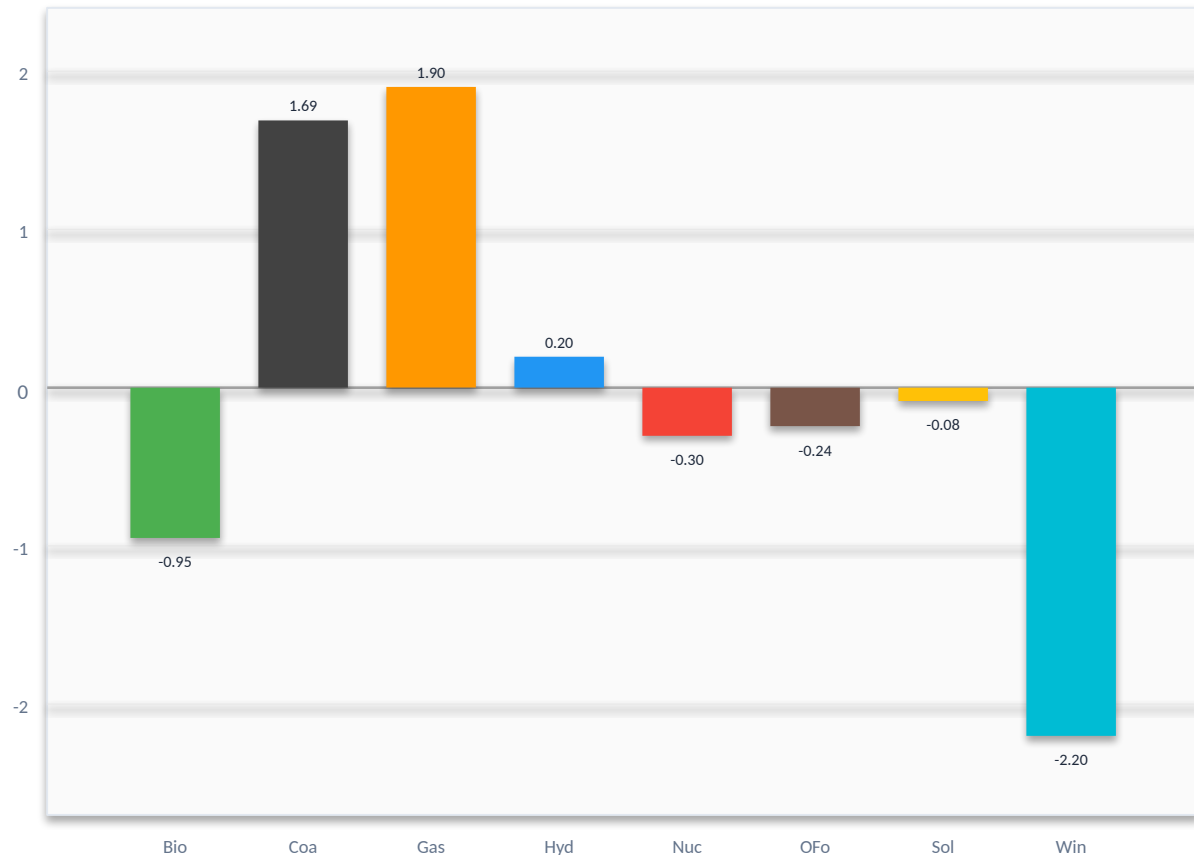


H* Direct · Japan EMBER · 2017 · XZ Face · CLR zero-sum verified

Higgins Scale

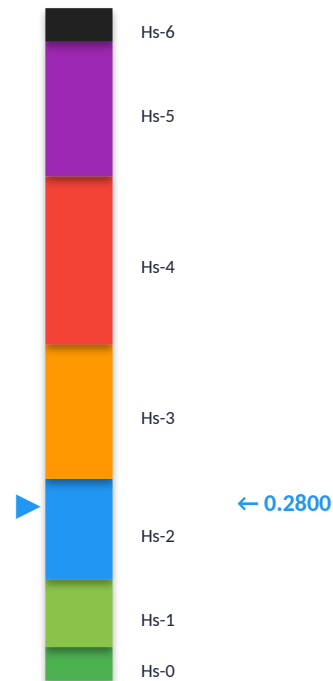
 $d_A = 3.747$ $N_{eff} = 4.21 / 8$ $p = 0.0263$

Dominant: Gas



H* Direct · Japan EMBER · 2018 · XZ Face · CLR zero-sum verified

Higgins Scale

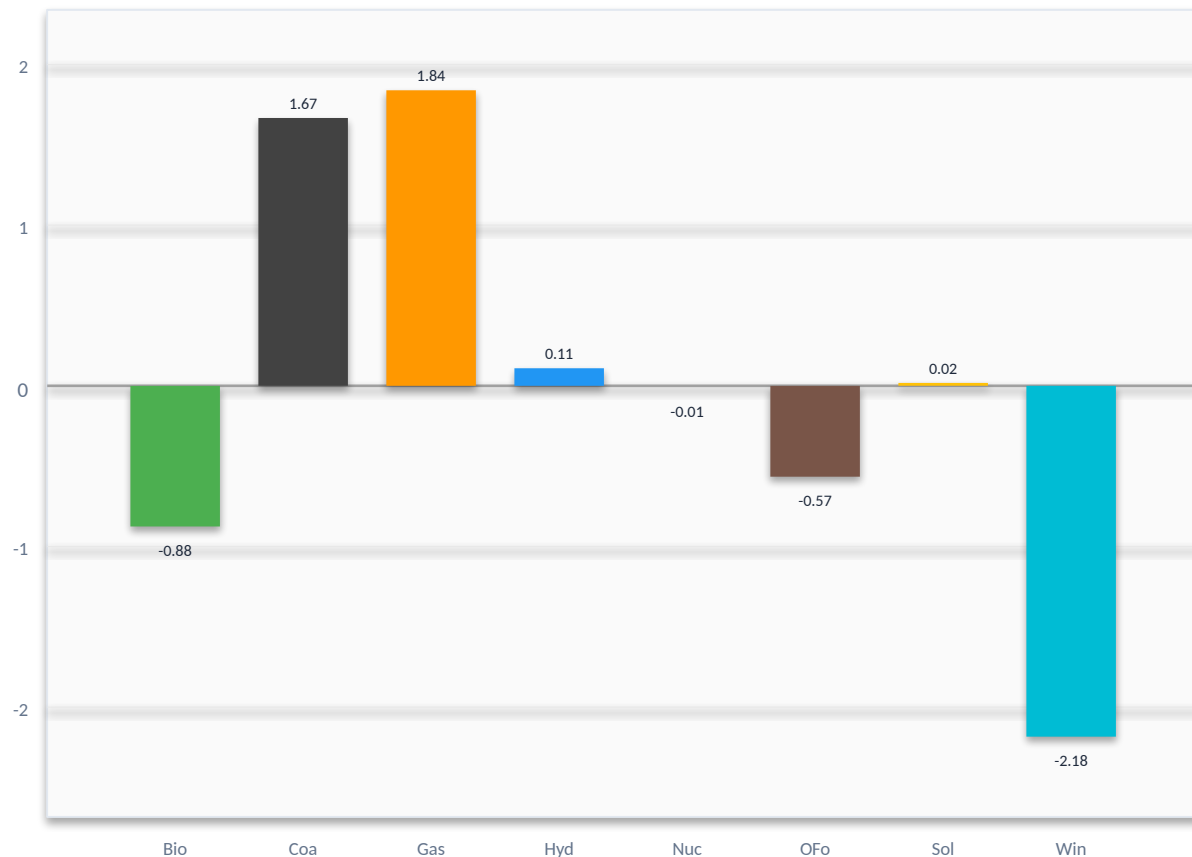


d_A = 3.523

N_{eff} = 4.47 / 8

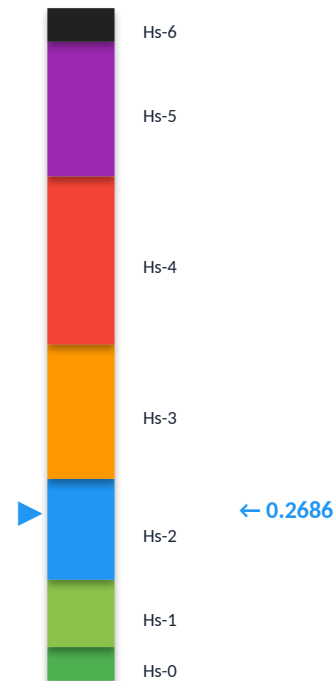
ρ = 0.0384

Dominant: Gas

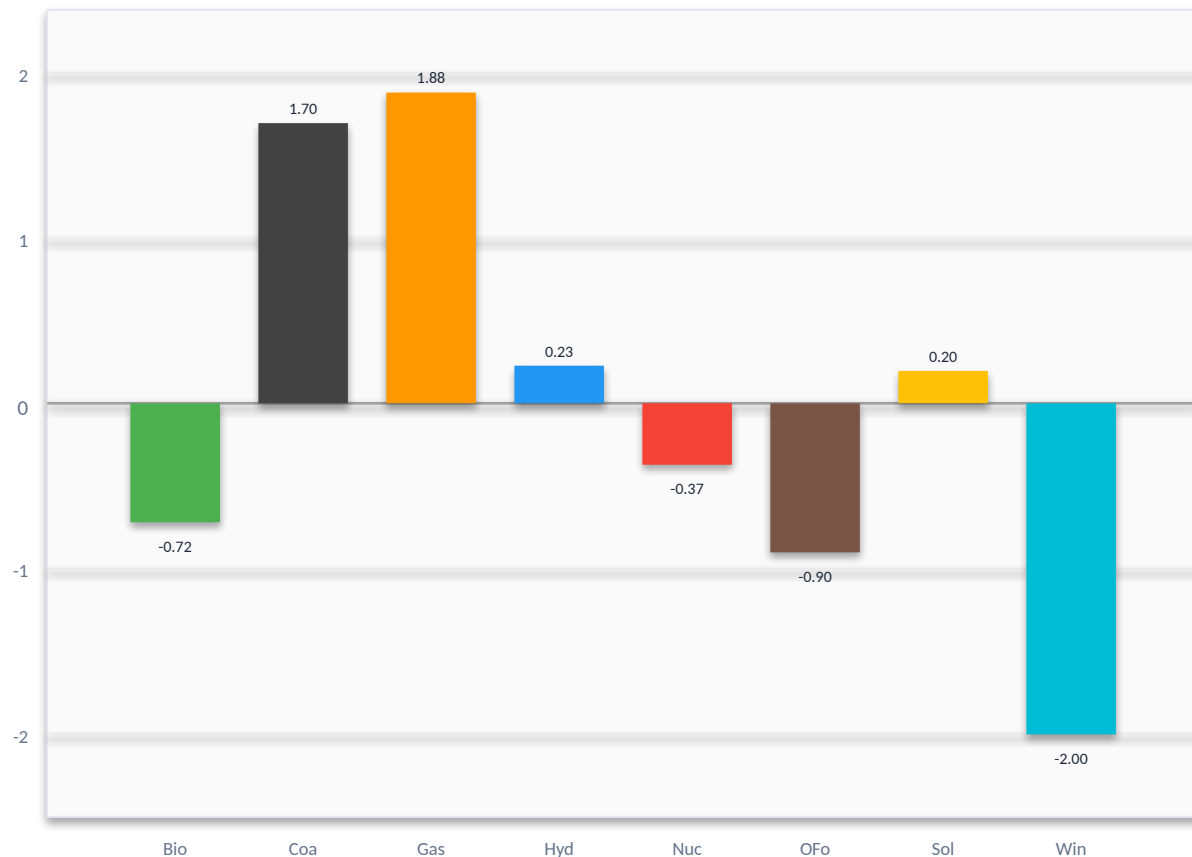


H* Direct · Japan EMBER · 2019 · XZ Face · CLR zero-sum verified

Higgins Scale

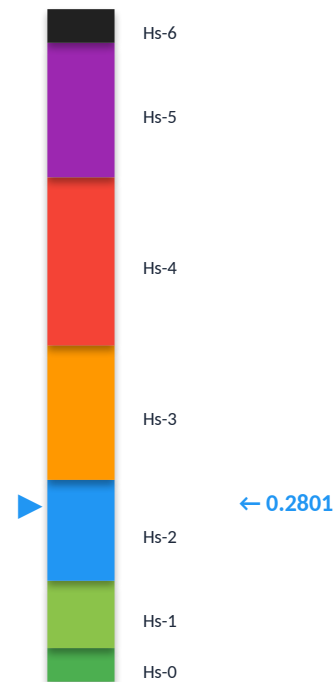
 $d_A = 3.468$ $N_{eff} = 4.58 / 8$ $p = 0.0042$

Dominant: Gas

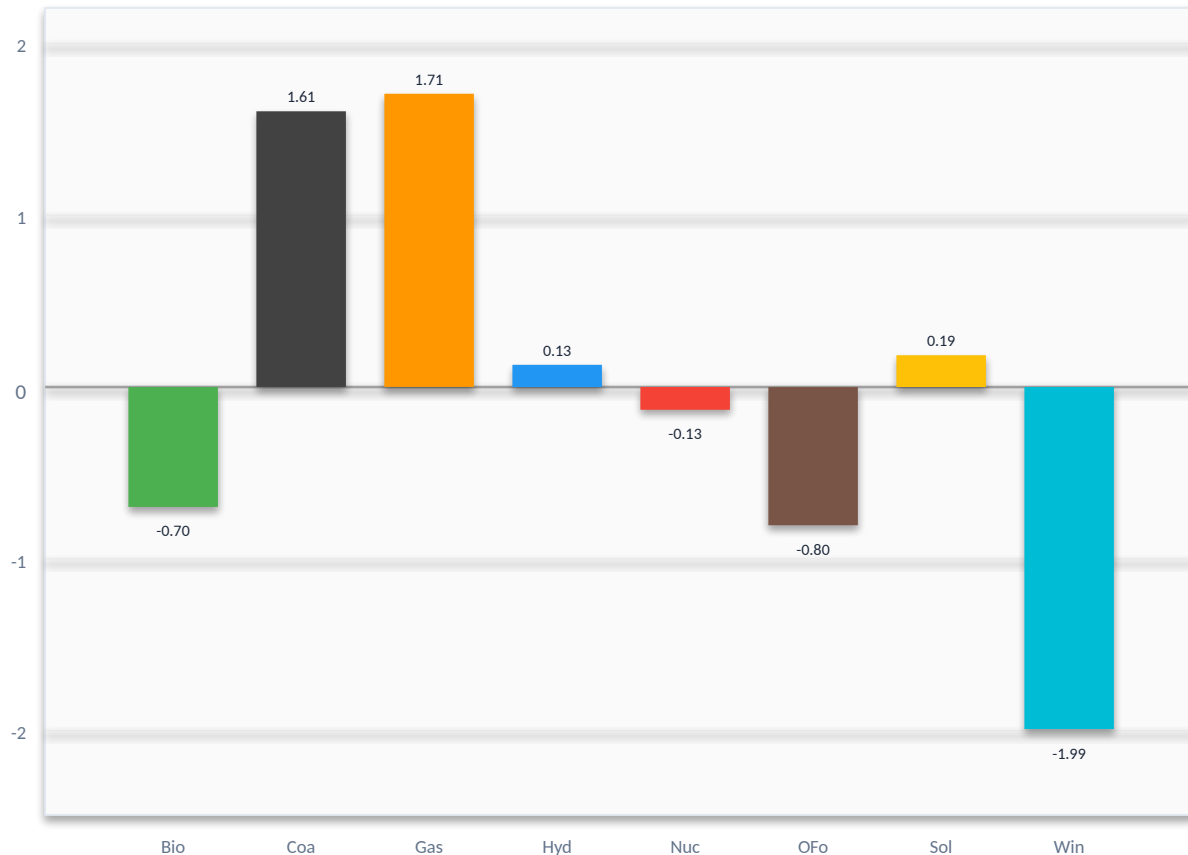


H* Direct · Japan EMBER · 2020 · XZ Face · CLR zero-sum verified

Higgins Scale

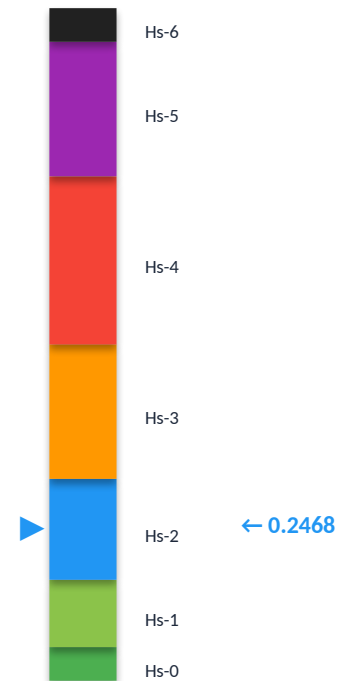
 $d_A = 3.463$ $N_{eff} = 4.47 / 8$ $p = 0.0977$

Dominant: Gas



H* Direct · Japan EMBER · 2021 · XZ Face · CLR zero-sum verified

Higgins Scale

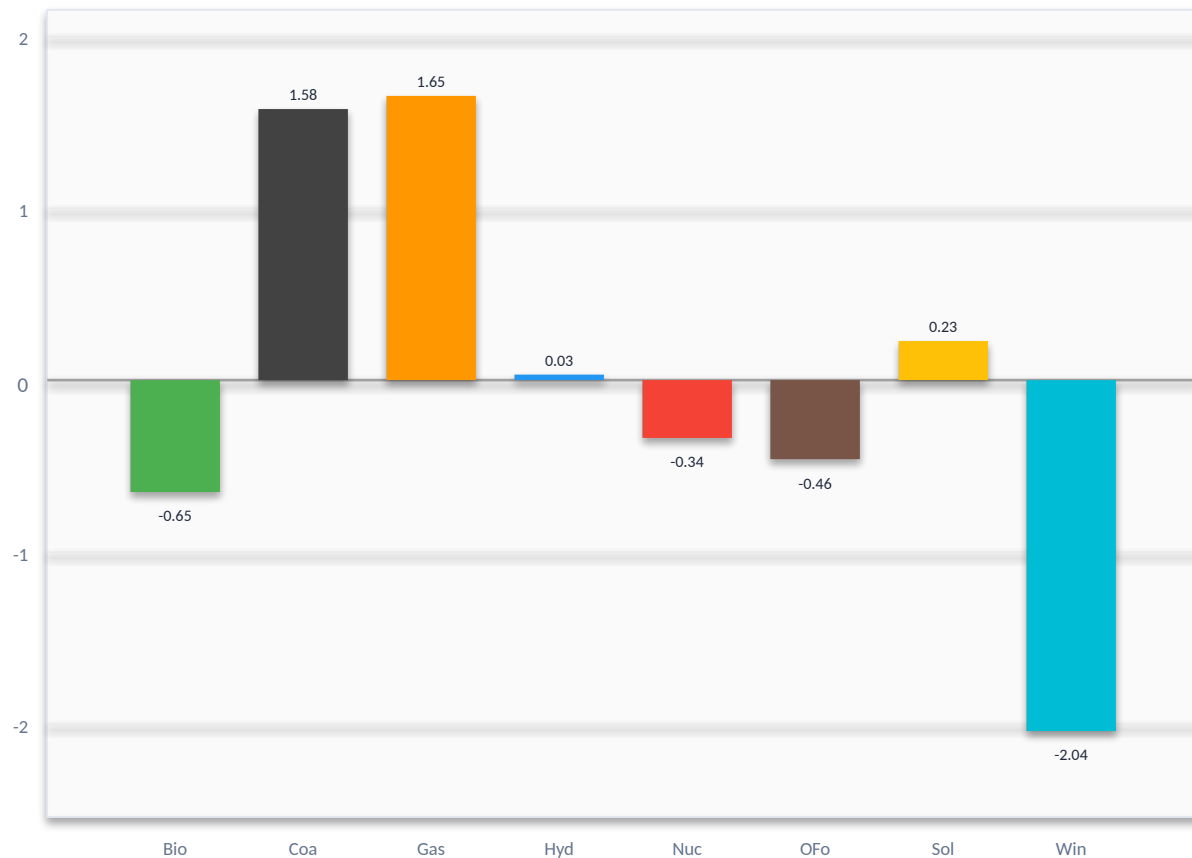


$d_A = 3.266$

$N_{eff} = 4.79 / 8$

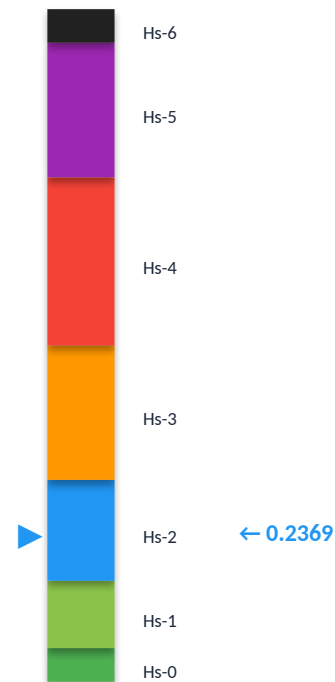
$p = 0.0652$

Dominant: Gas

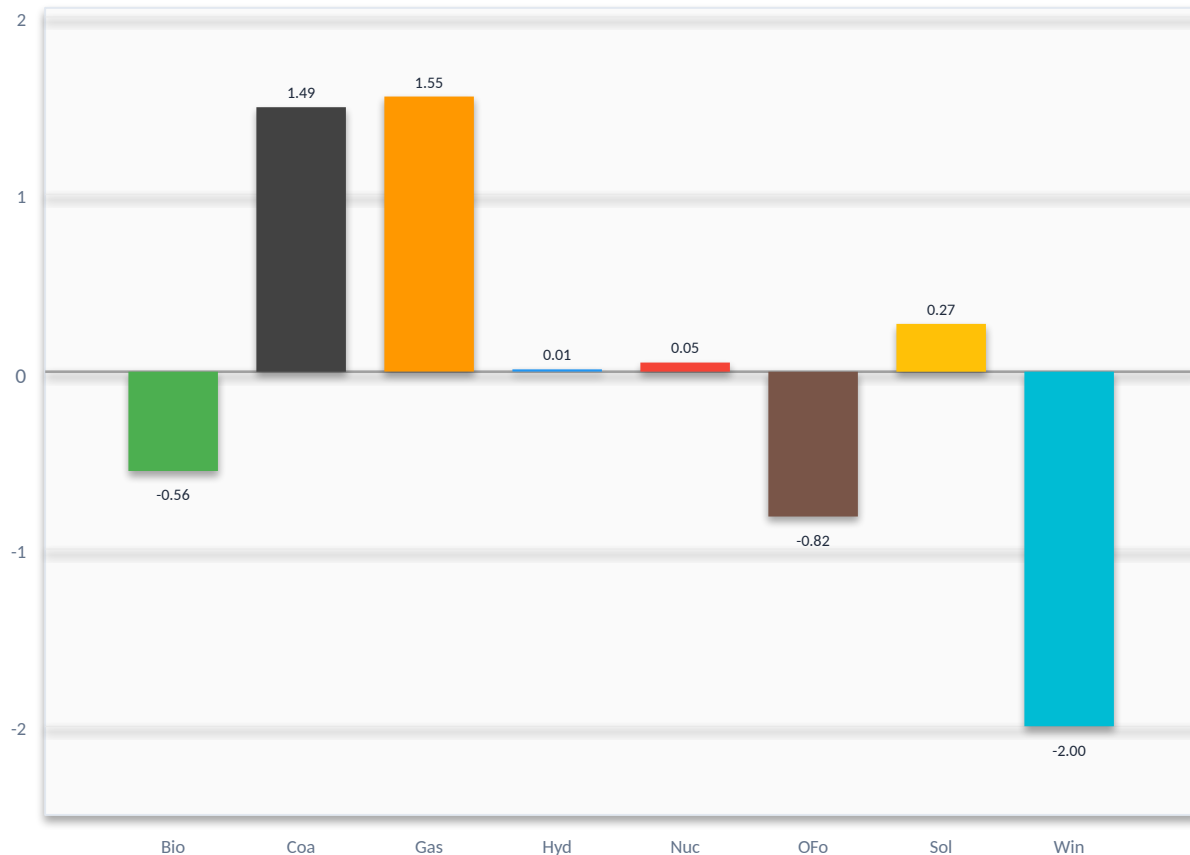


H* Direct · Japan EMBER · 2022 · XZ Face · CLR zero-sum verified

Higgins Scale

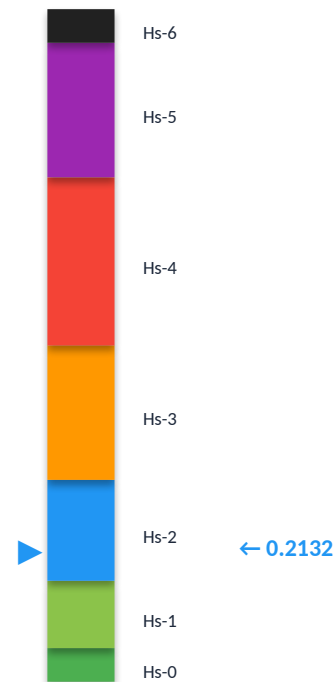
 $d_A = 3.192$ $N_{eff} = 4.89 / 8$ $p = 0.0156$

Dominant: Gas

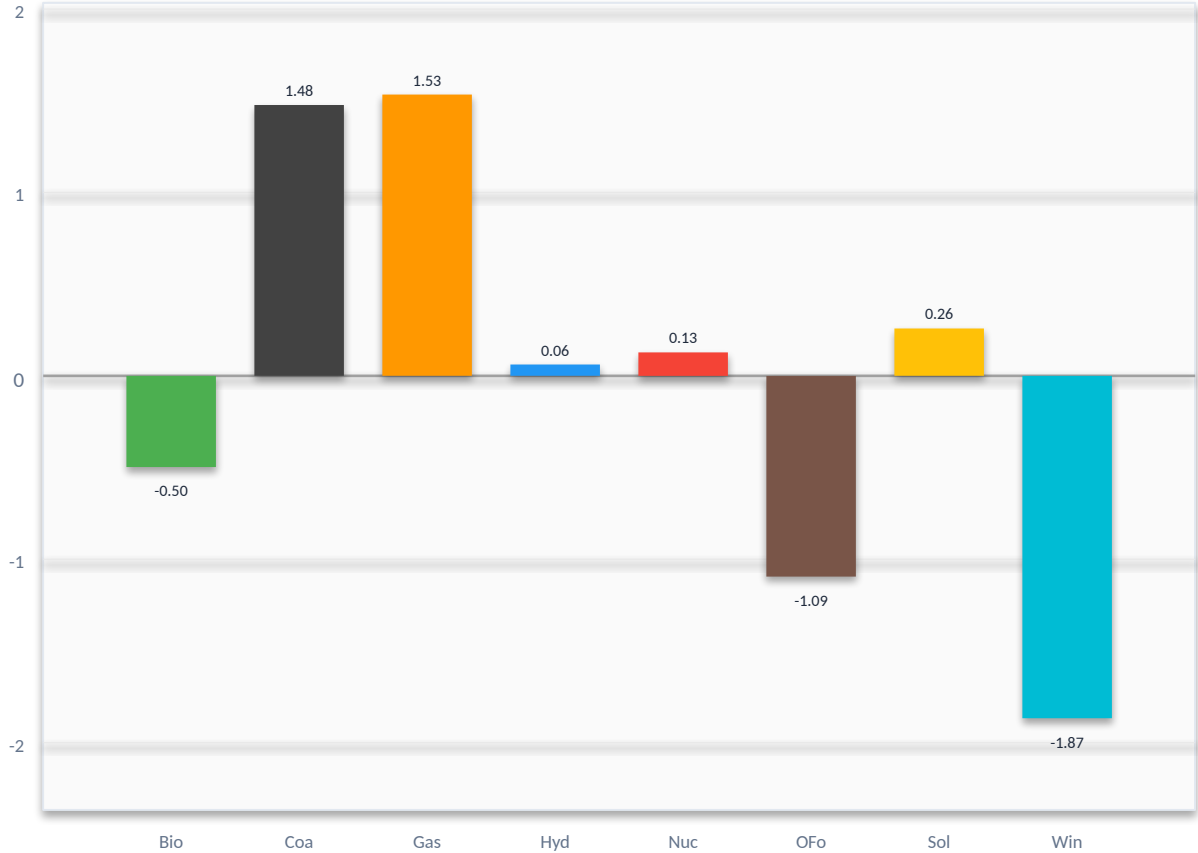


H* Direct · Japan EMBER · 2023 · XZ Face · CLR zero-sum verified

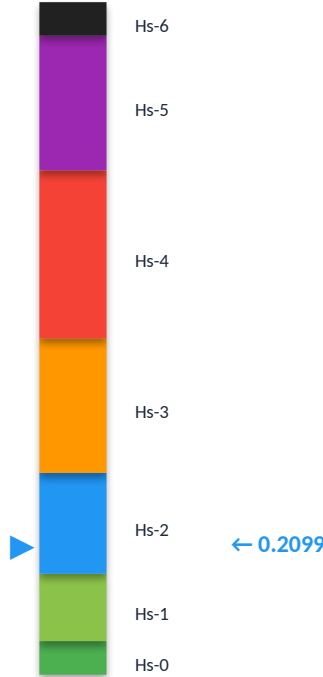
Higgins Scale

 $d_A = 3.112$ $N_{eff} = 5.13 / 8$ $p = 0.0065$

Dominant: Gas



Higgins Scale

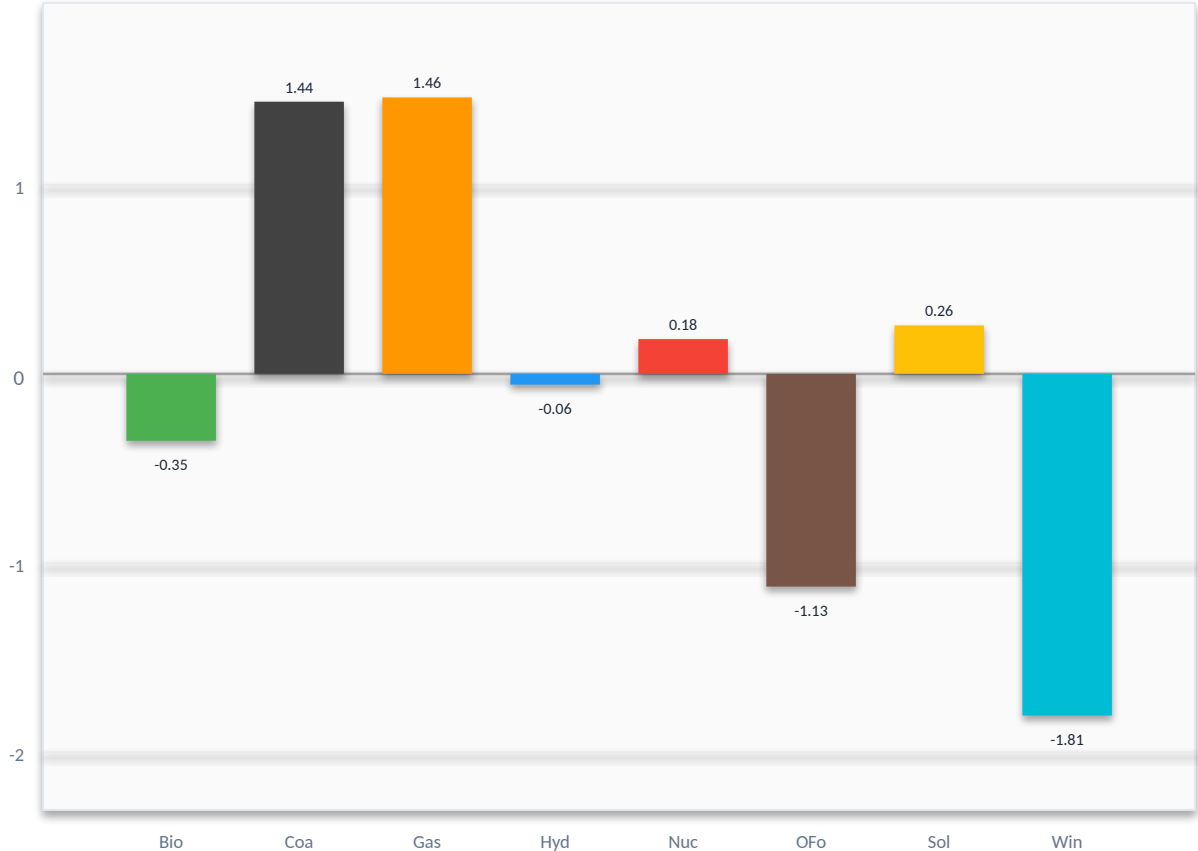


d_A = 3.091

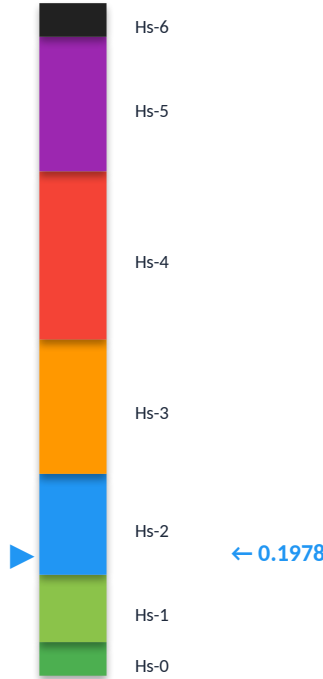
N_eff = 5.17 / 8

p = 0.0331

Dominant: Gas



Higgins Scale



d_A = 2.995

N_{eff} = 5.30 / 8

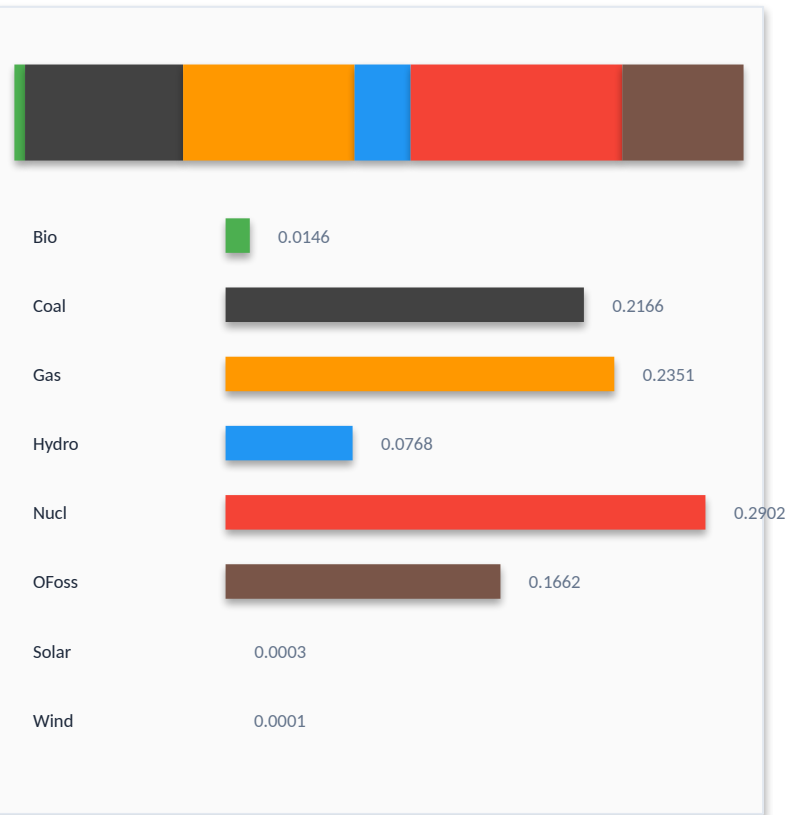
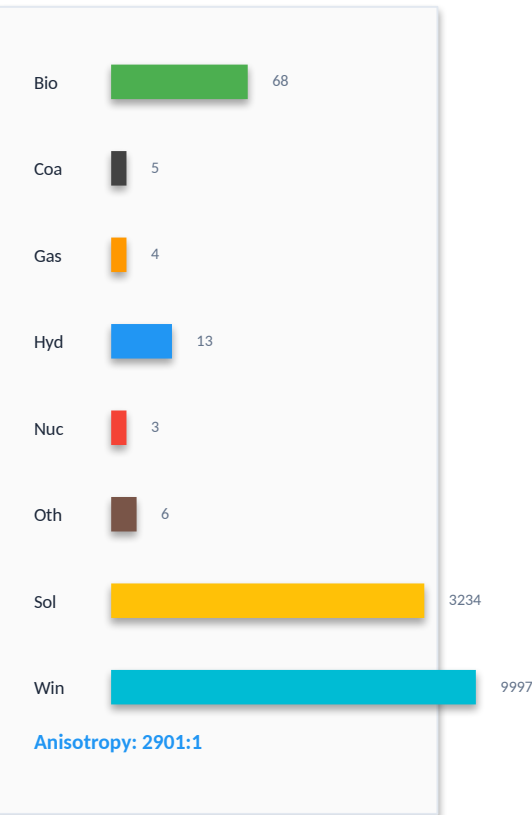
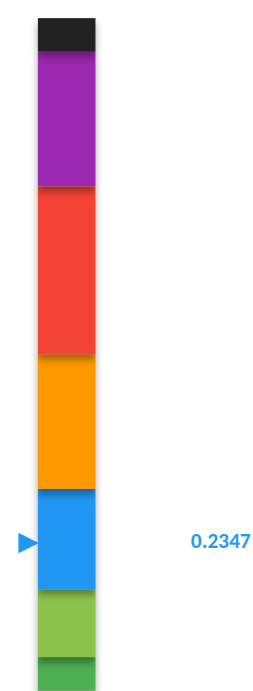
ρ = 0.0315

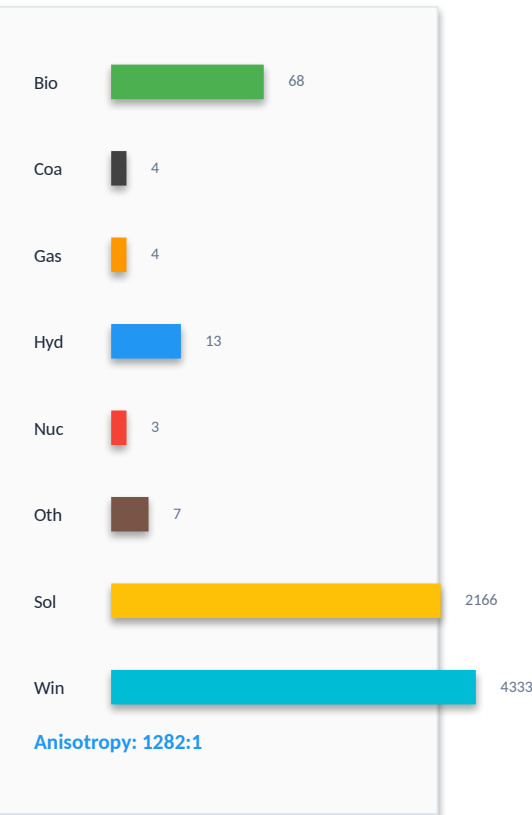
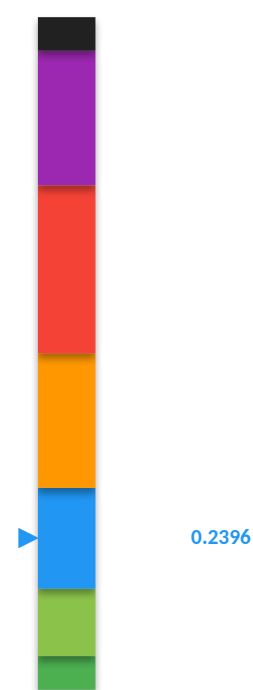
Dominant: Gas

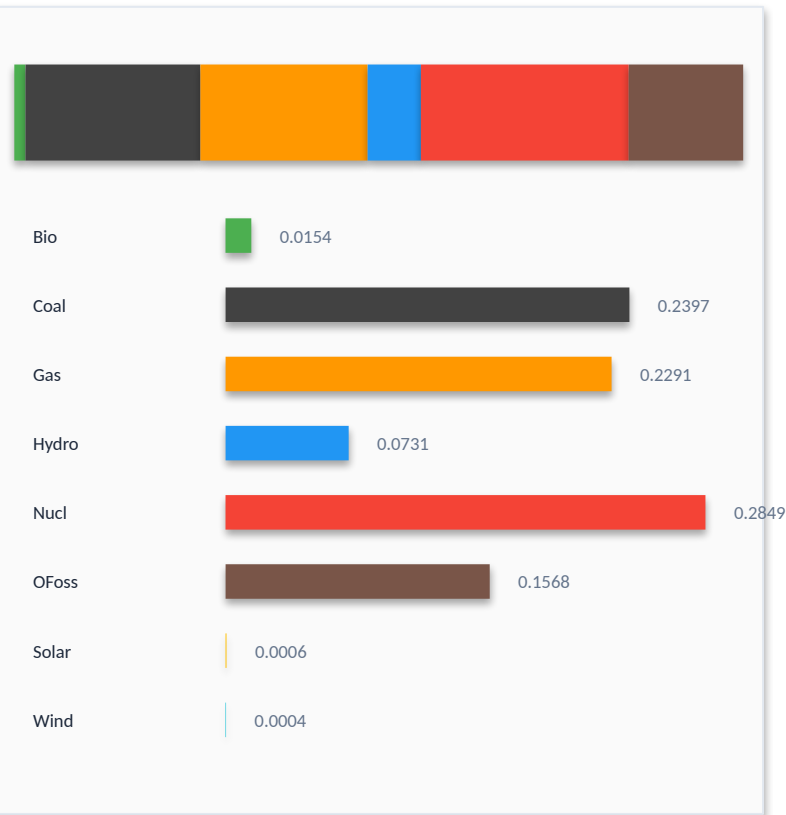
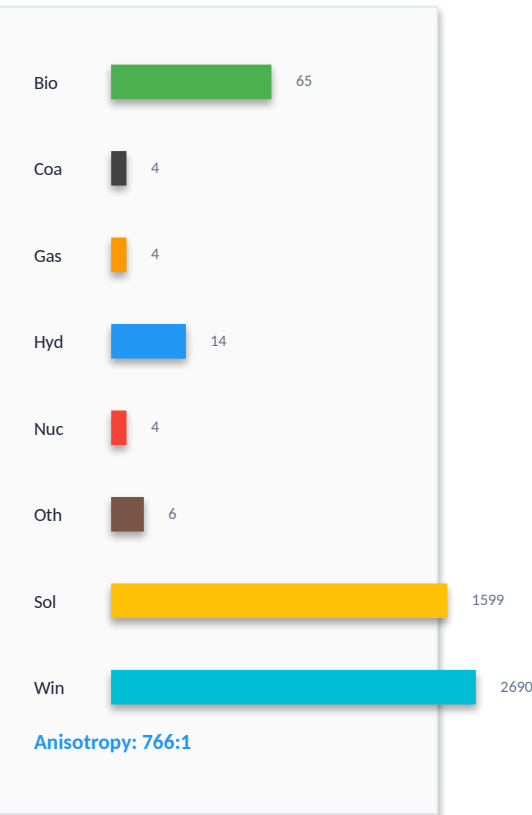
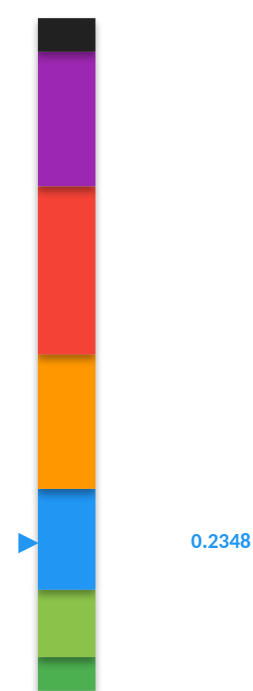
YZ Face — Side Elevation

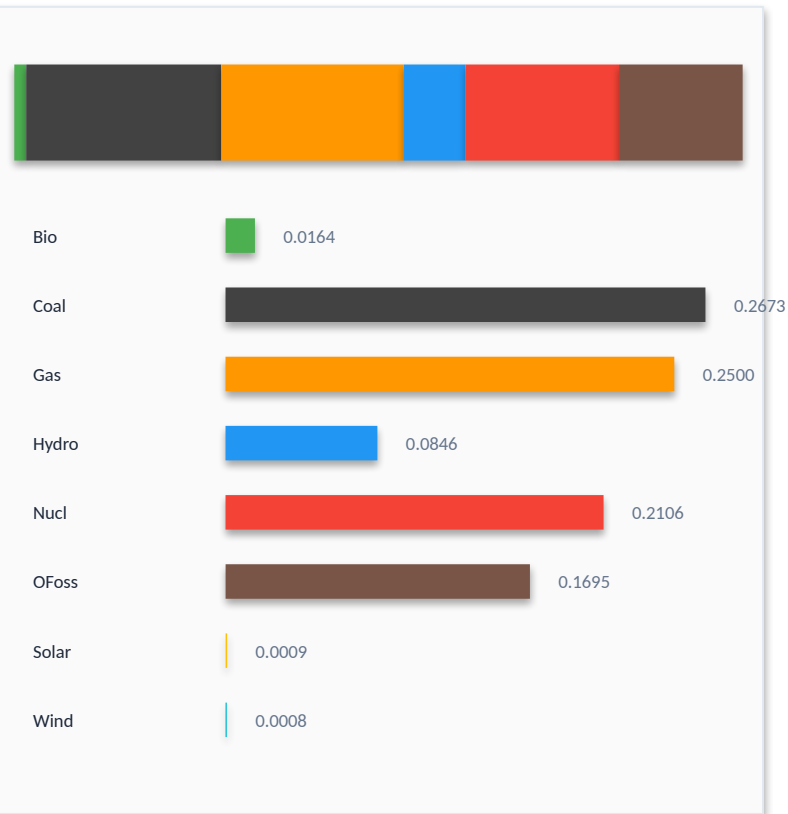
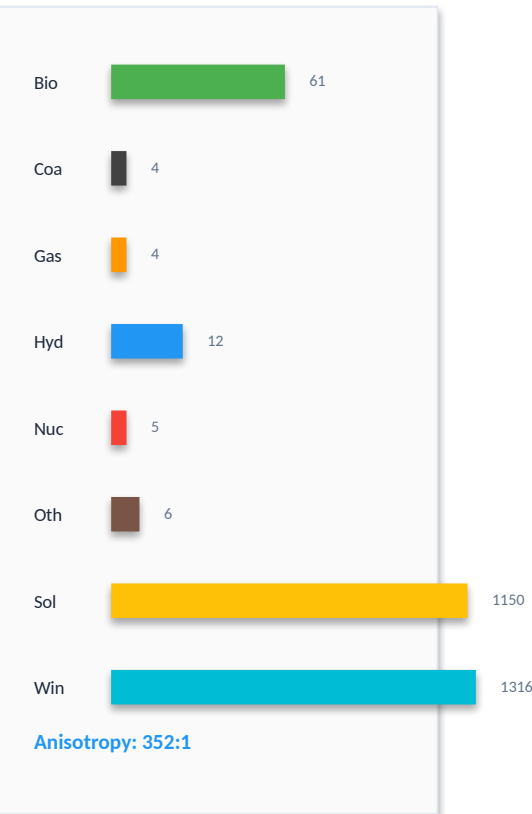
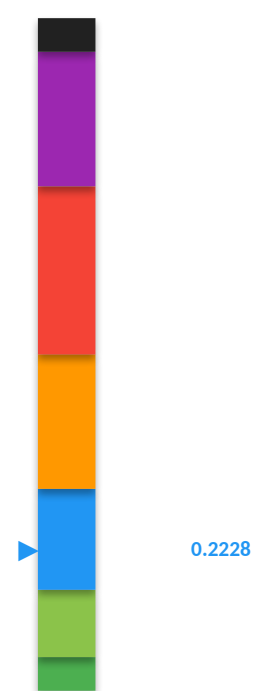
Compositional fractions · Metric tensor sensitivity · Higgins Scale

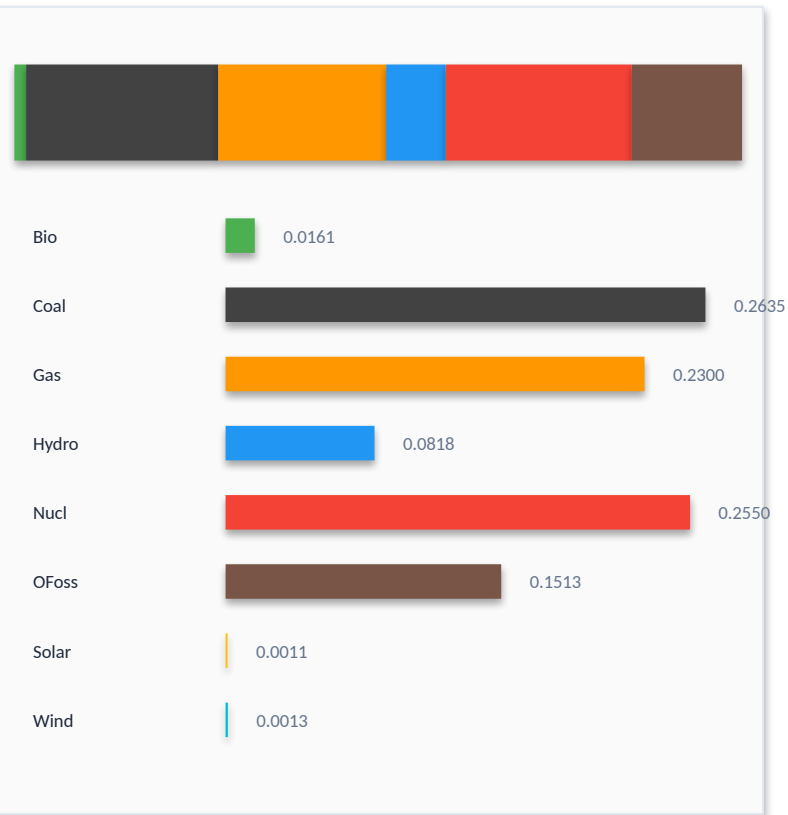
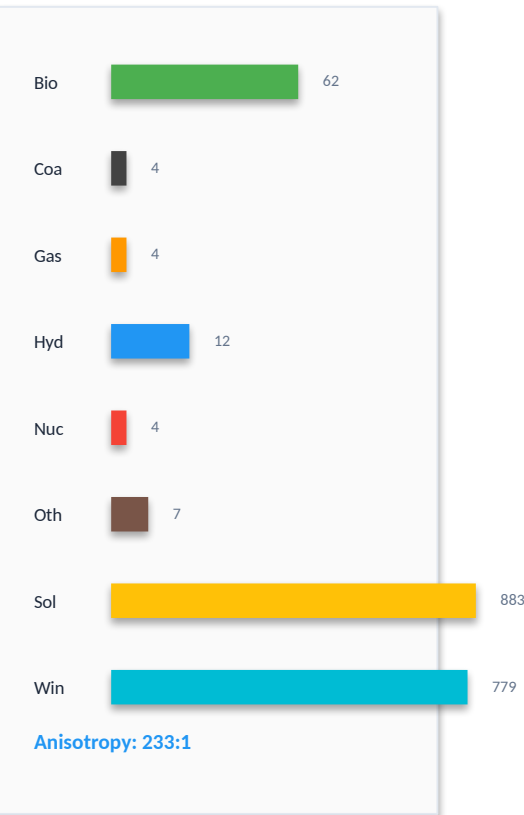
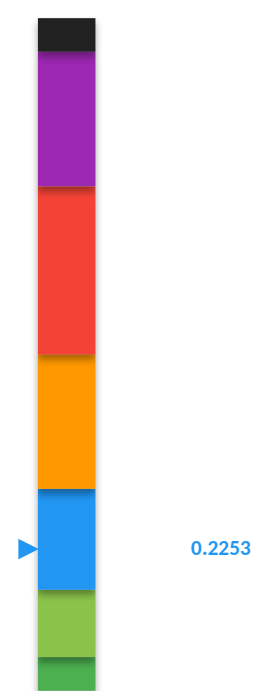
26 annual profiles · Each composition's simplex fractions and g_{jj} values

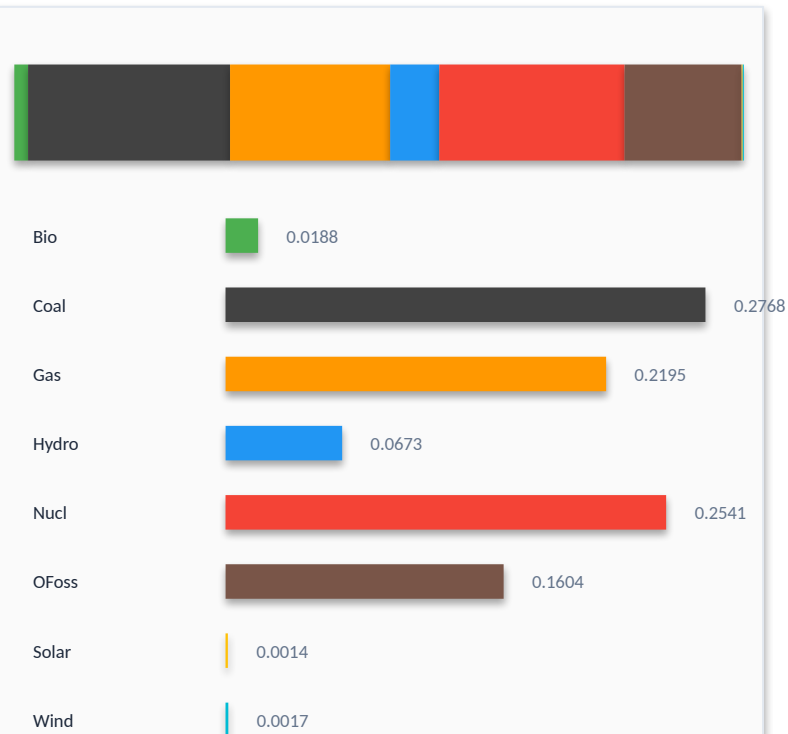
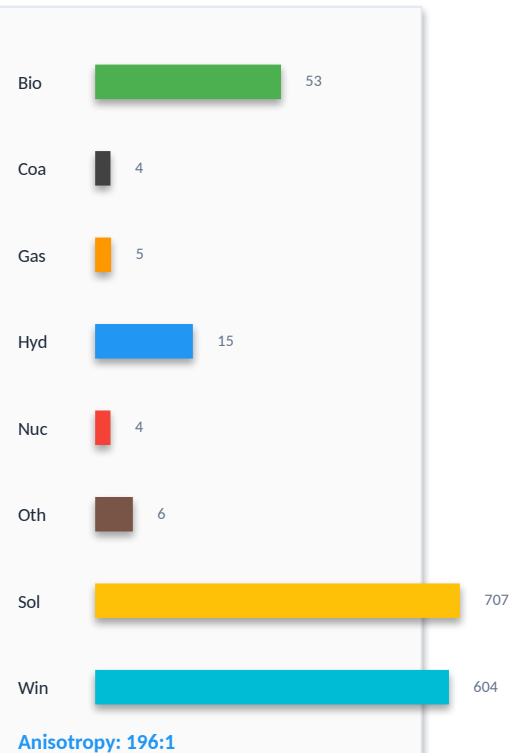
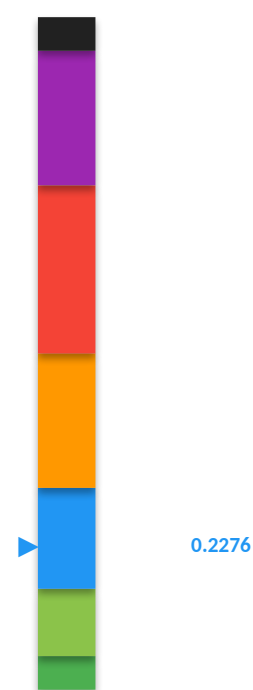
Simplex Fractions $\tilde{\mathbf{x}} = \mathbf{C}(\mathbf{x})$ Metric Tensor $\mathbf{g}_{jj} = 1/x_j$  H_s 

Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

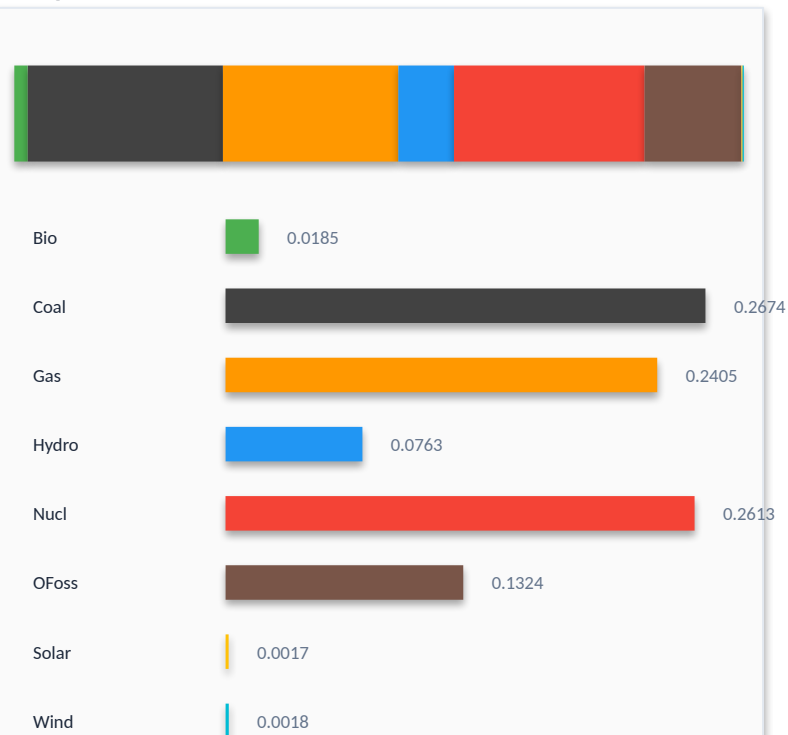
Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

Simplex Fractions $\tilde{\mathbf{x}} = \mathbf{C}(\mathbf{x})$ Metric Tensor $\mathbf{g}_{jj} = 1/x_j$  H_s 

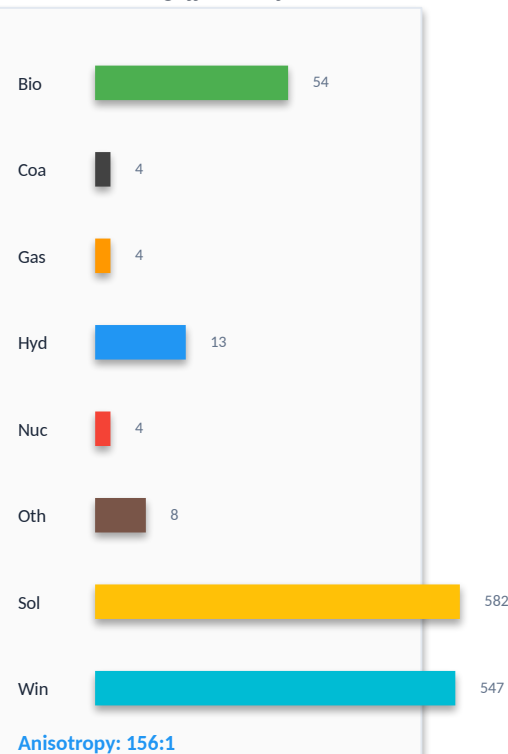
Simplex Fractions $\tilde{\mathbf{x}} = \mathbf{C}(\mathbf{x})$ Metric Tensor $\mathbf{g}_{jj} = 1/x_j$  H_s 

Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

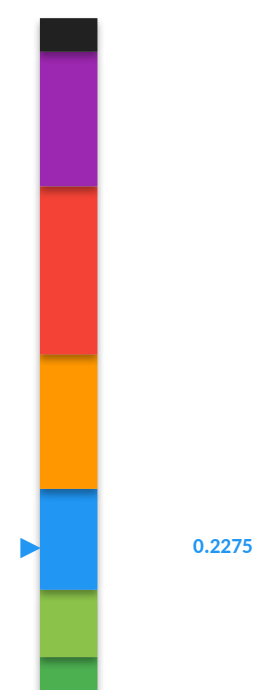
Simplex Fractions $\tilde{x} = C(x)$



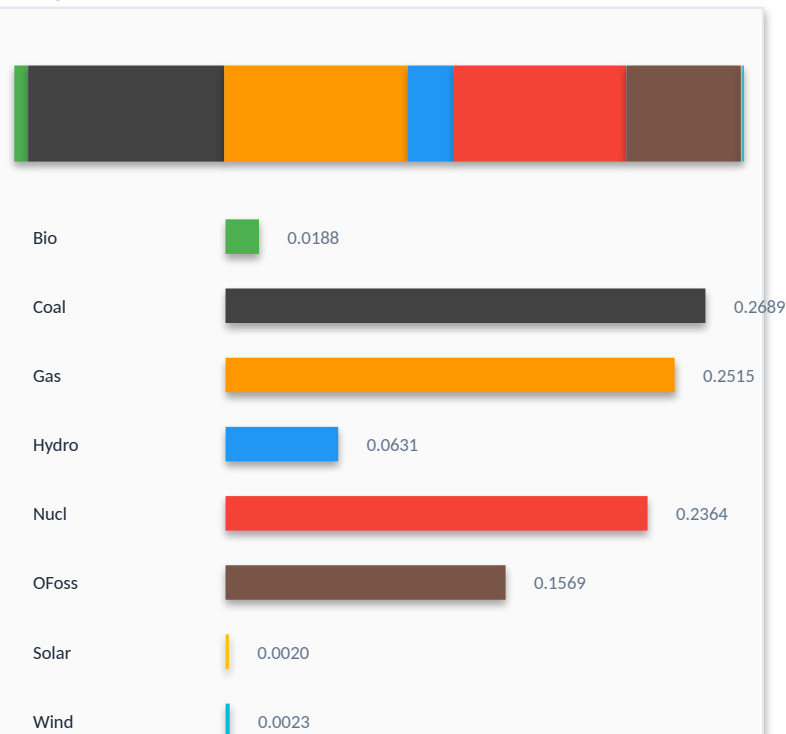
Metric Tensor $g_{jj} = 1/x_j$



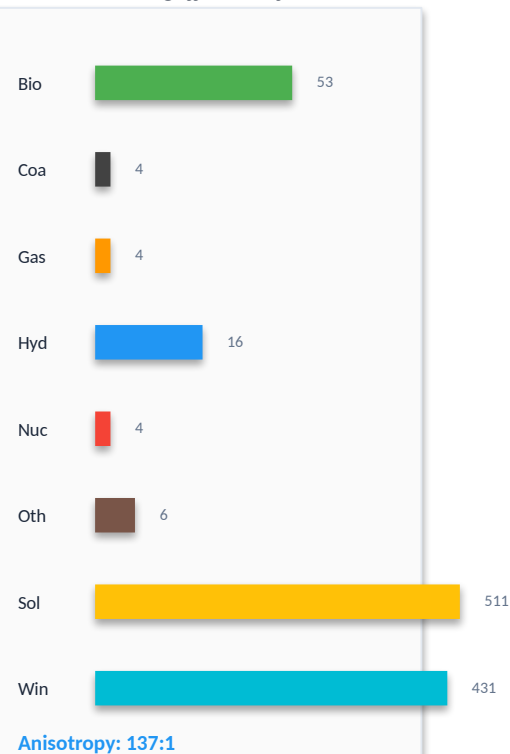
H_s



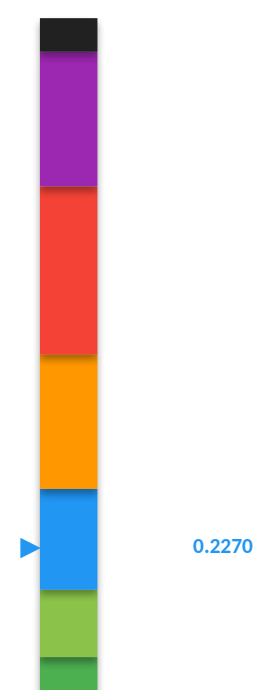
Simplex Fractions $\tilde{x} = C(x)$

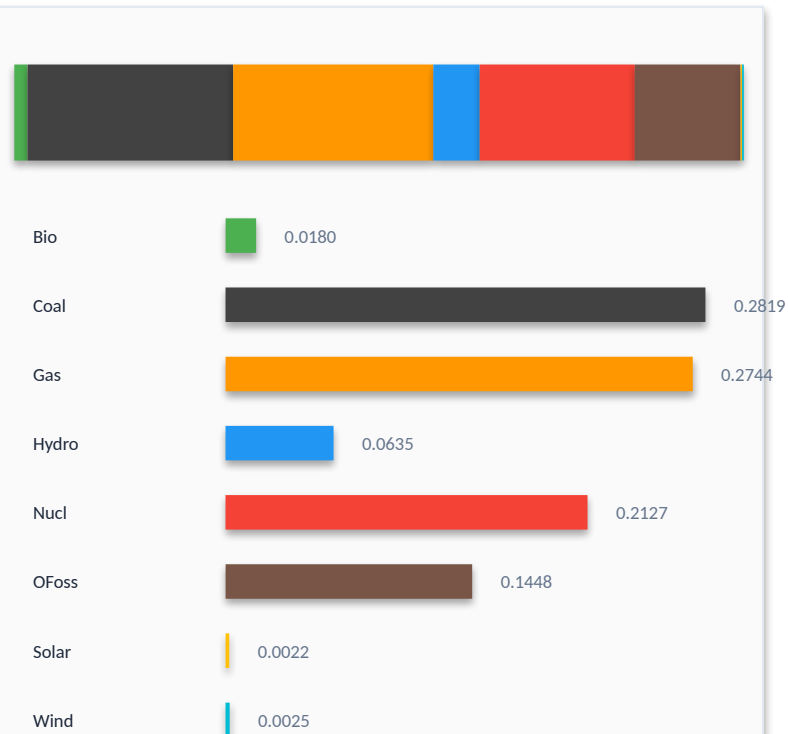
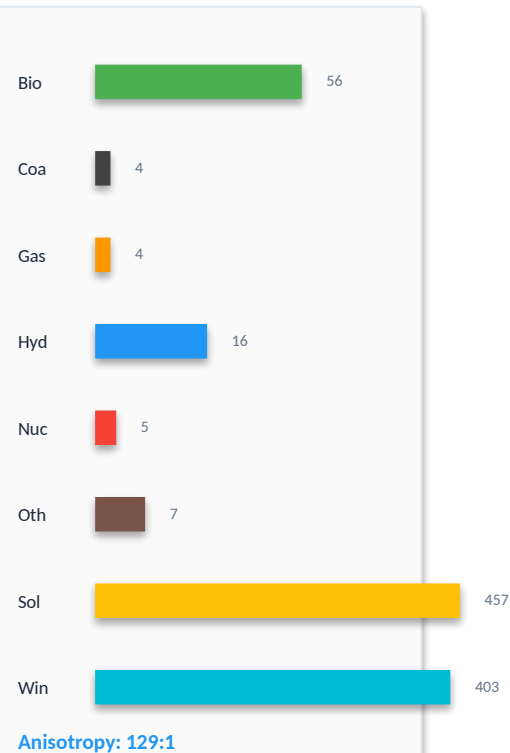
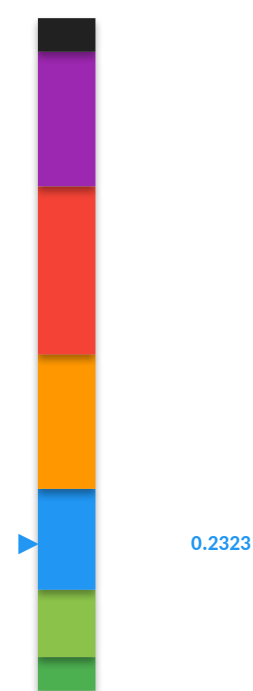


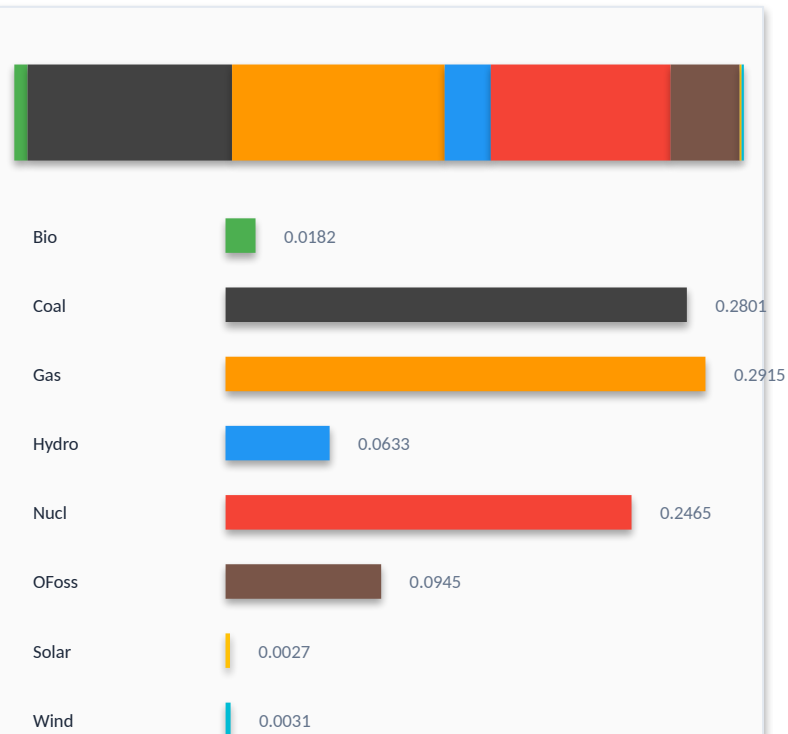
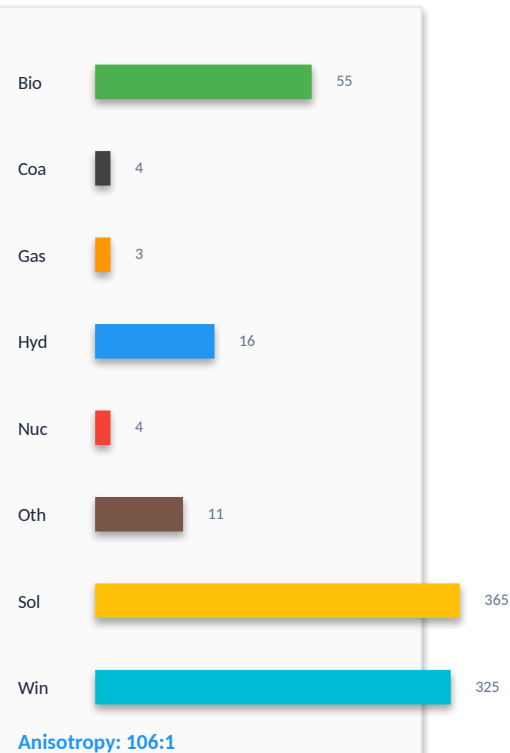
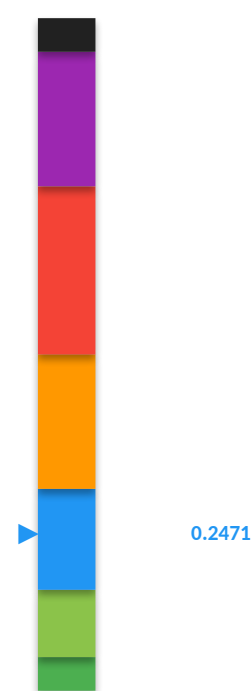
Metric Tensor $g_{jj} = 1/x_j$

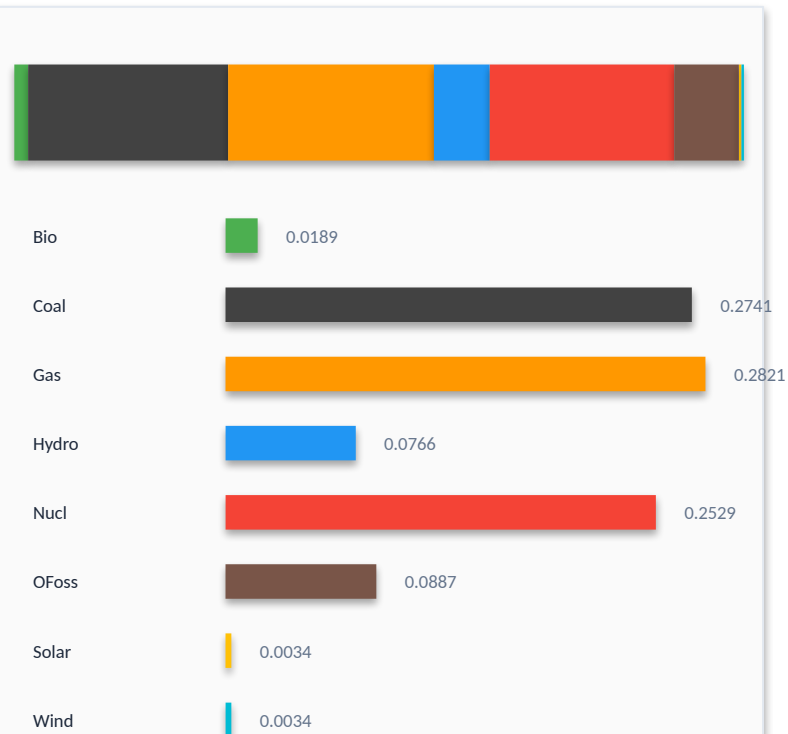
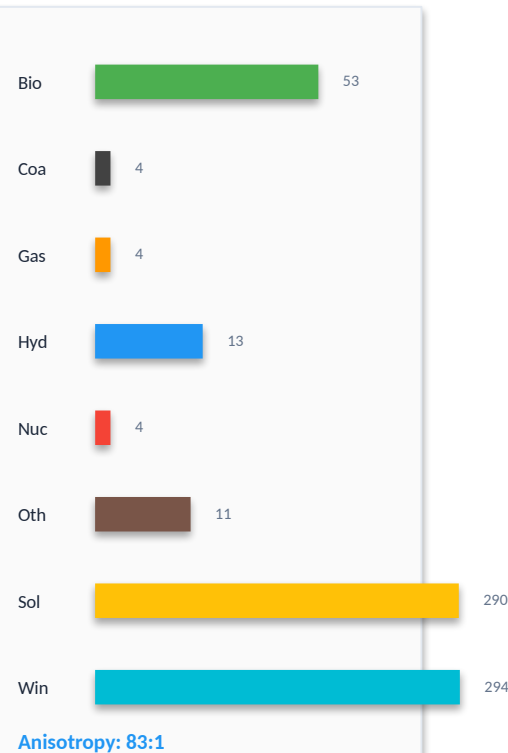
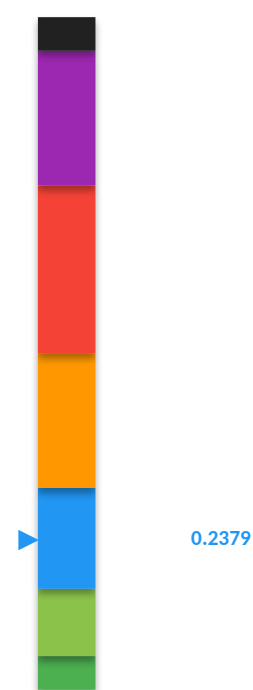


H_s

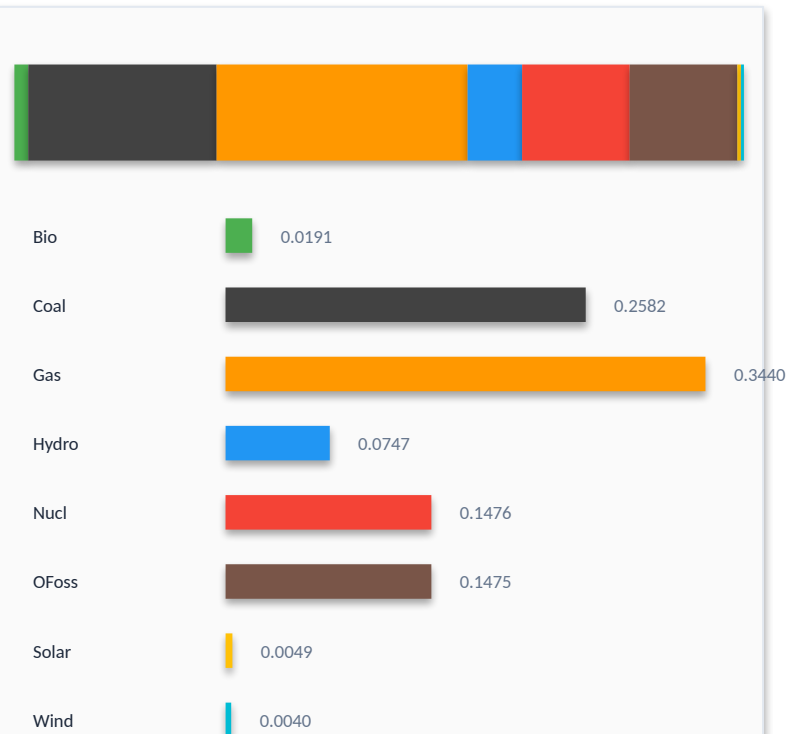


Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

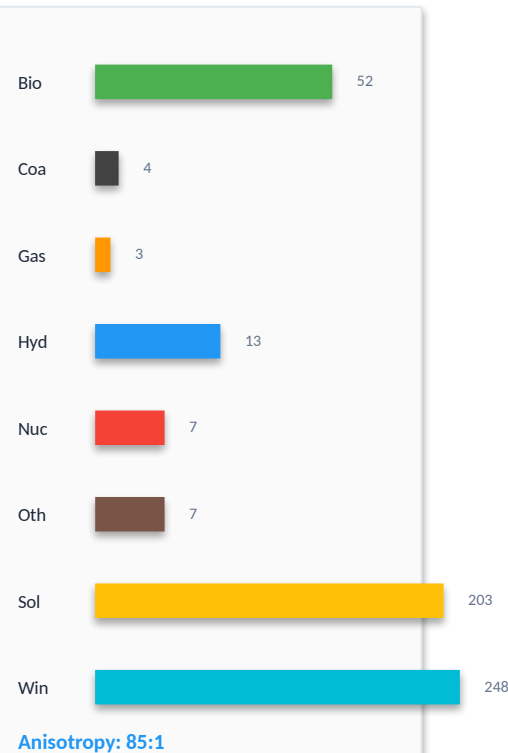
Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

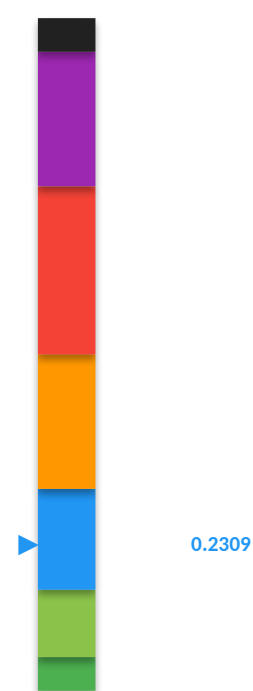
Simplex Fractions $\tilde{x} = C(x)$



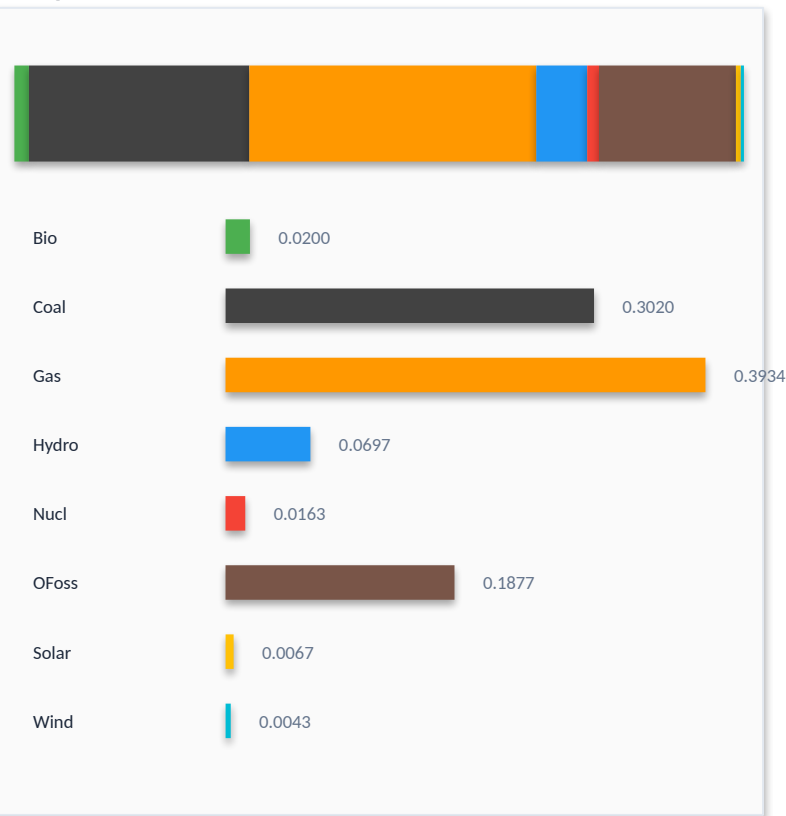
Metric Tensor $g_{jj} = 1/x_j$



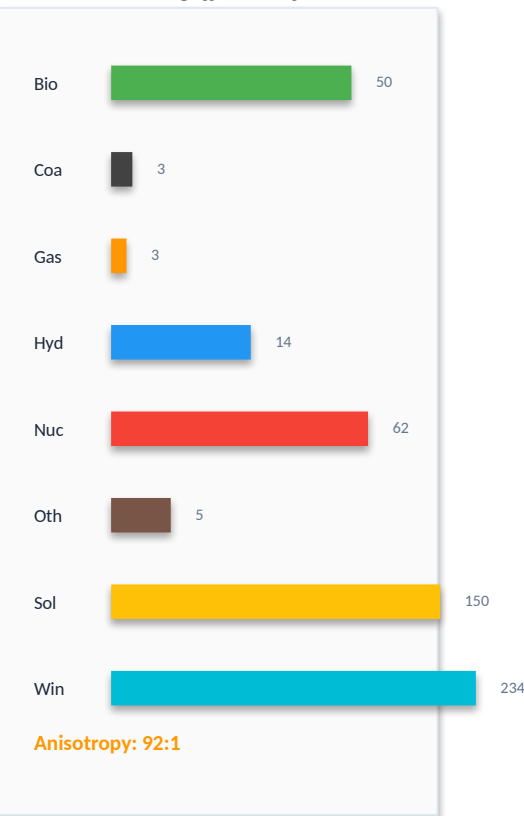
H_s



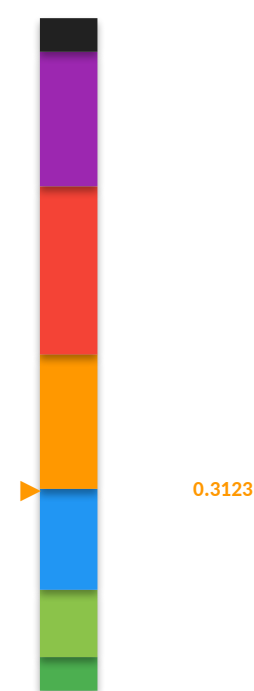
Simplex Fractions $\tilde{x} = C(x)$



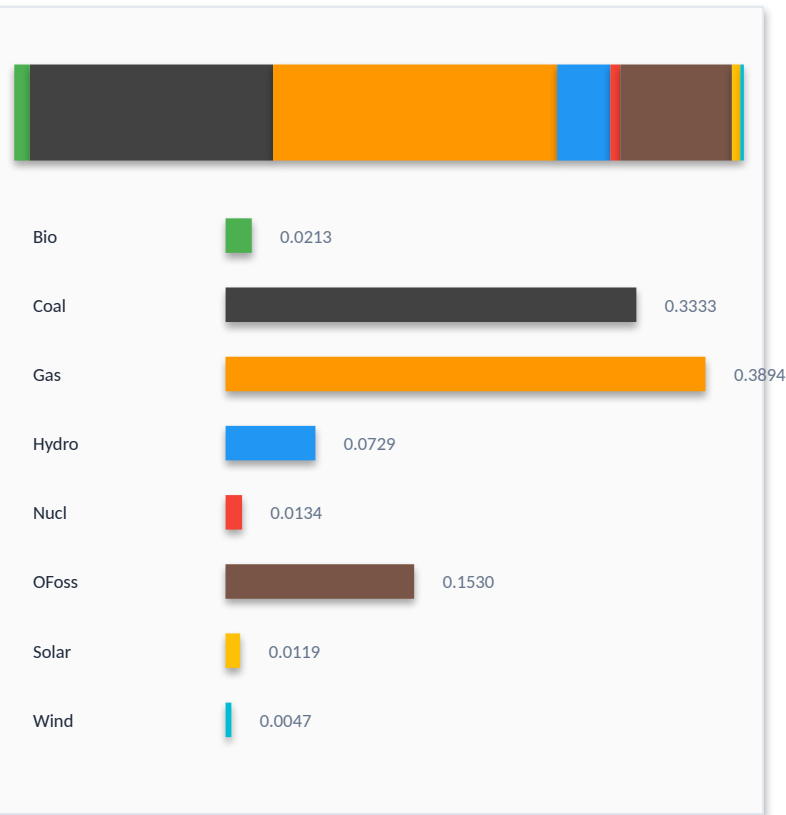
Metric Tensor $g_{jj} = 1/x_j$



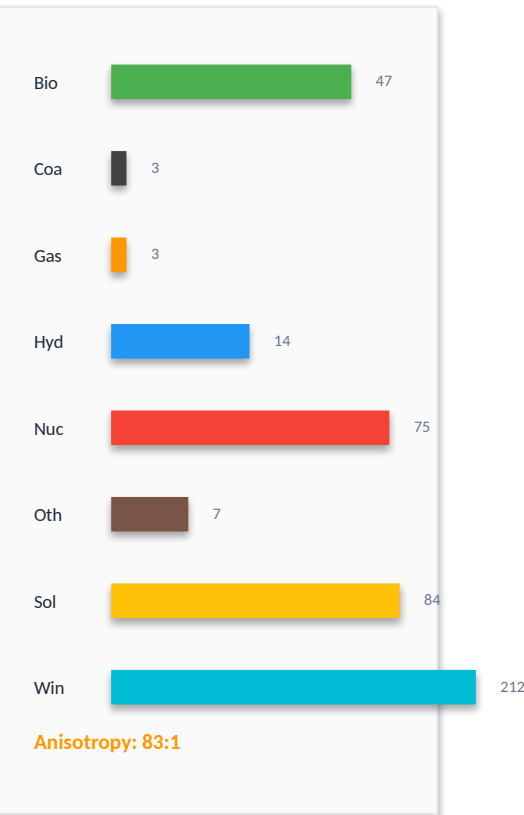
H_s



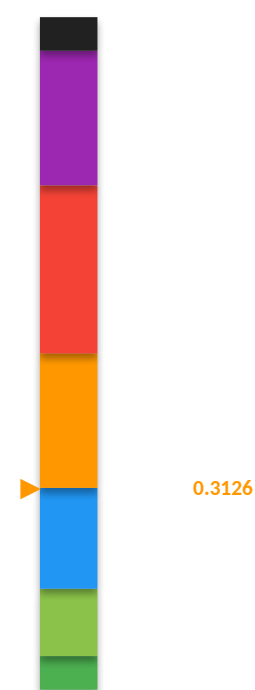
Simplex Fractions $\tilde{x} = C(x)$

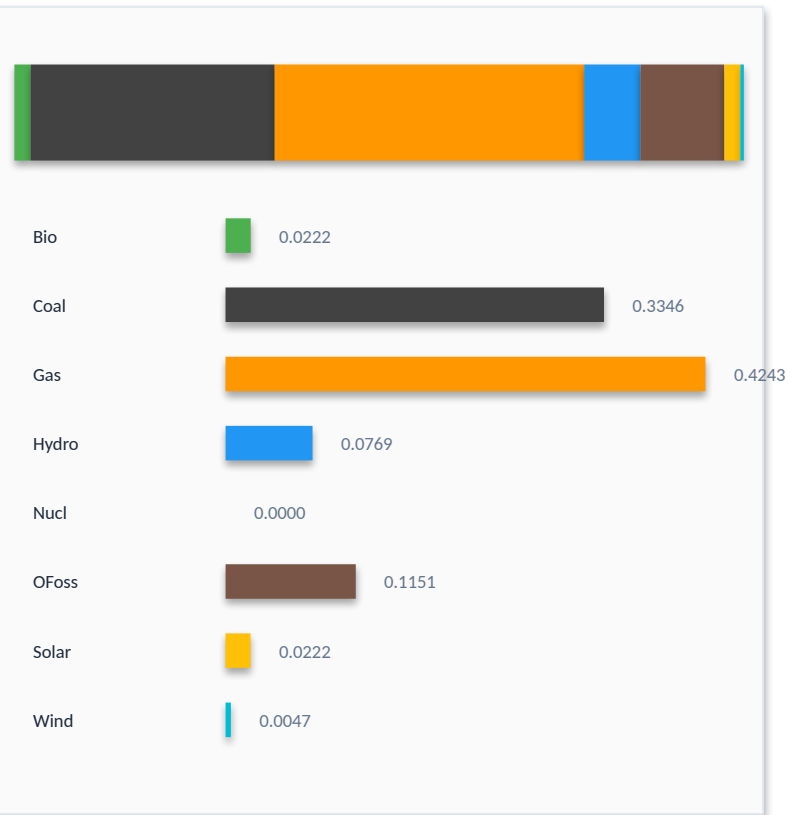
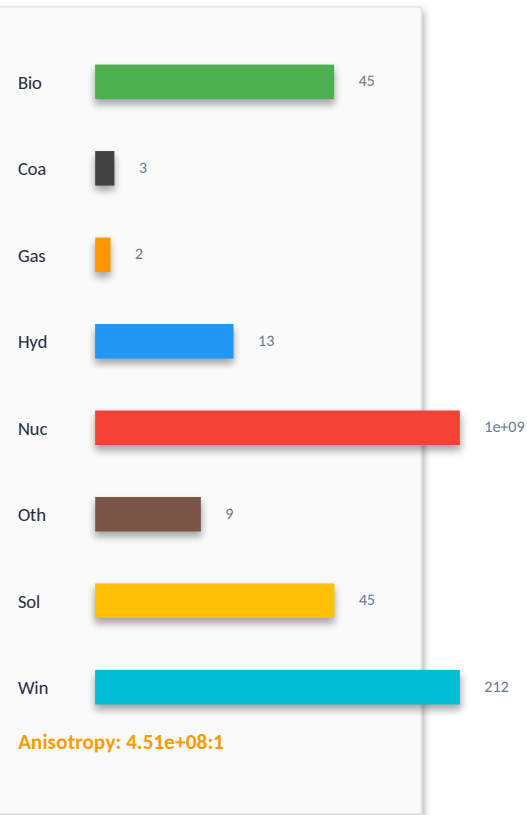
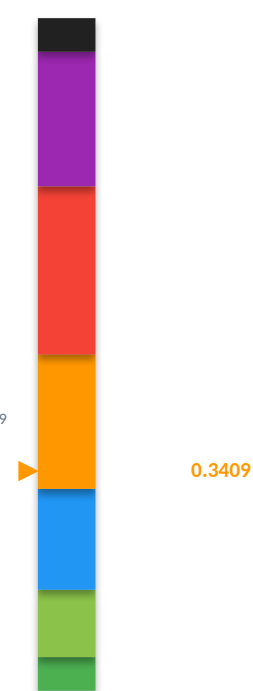


Metric Tensor $g_{jj} = 1/x_j$



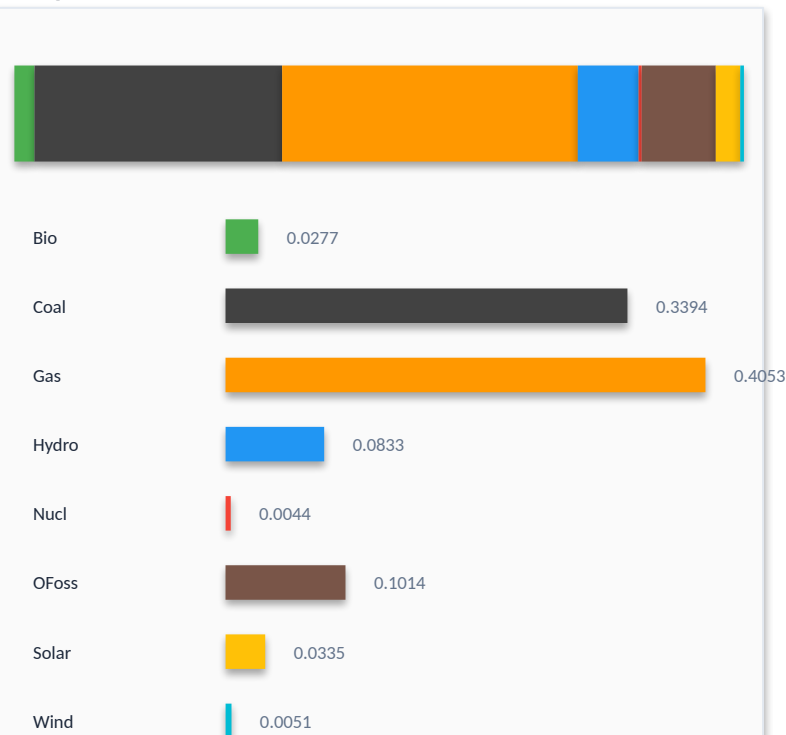
H_s



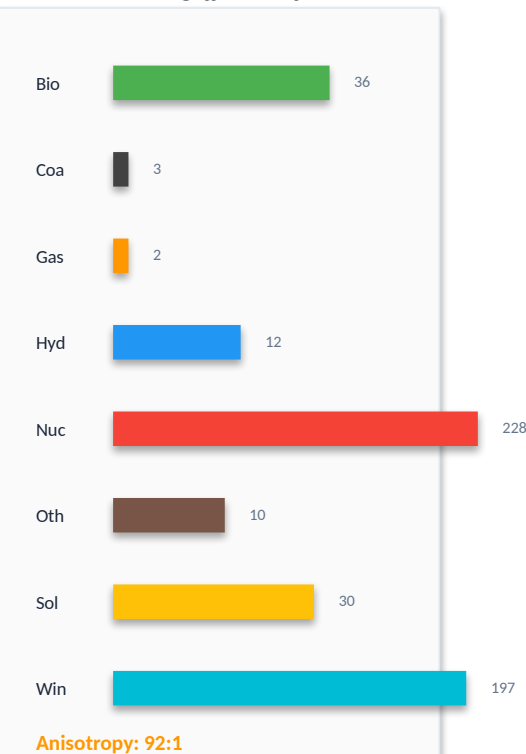
Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

Anisotropy: 4.51e+08:1

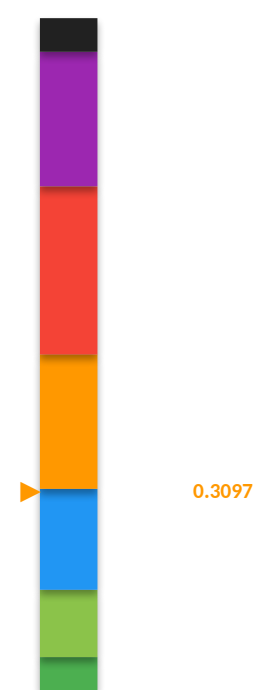
Simplex Fractions $\tilde{X} = C(x)$



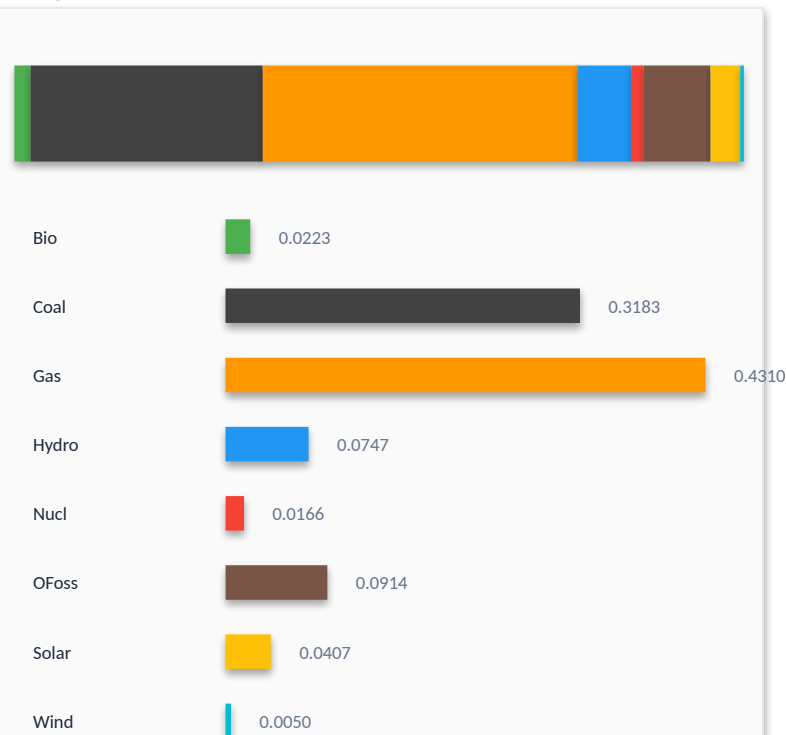
Metric Tensor $g_{jj} = 1/x_j$



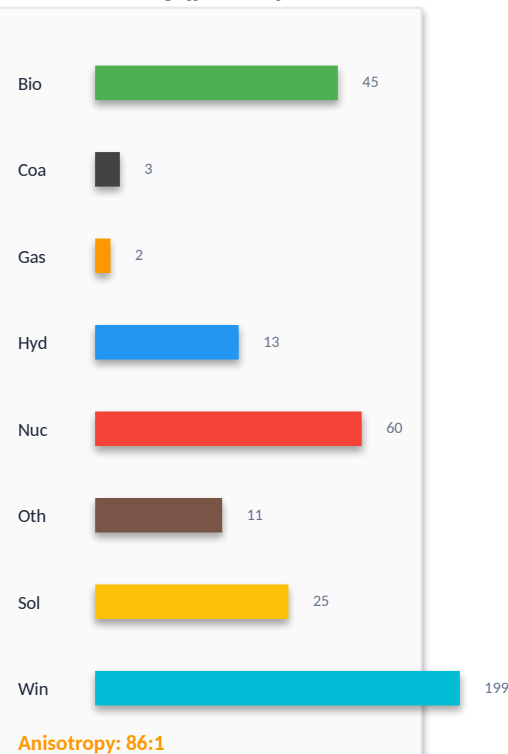
H_s



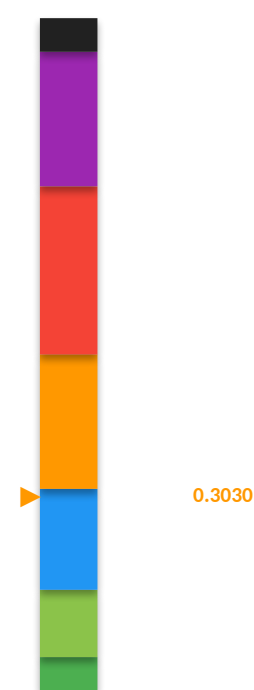
Simplex Fractions $\tilde{x} = C(x)$

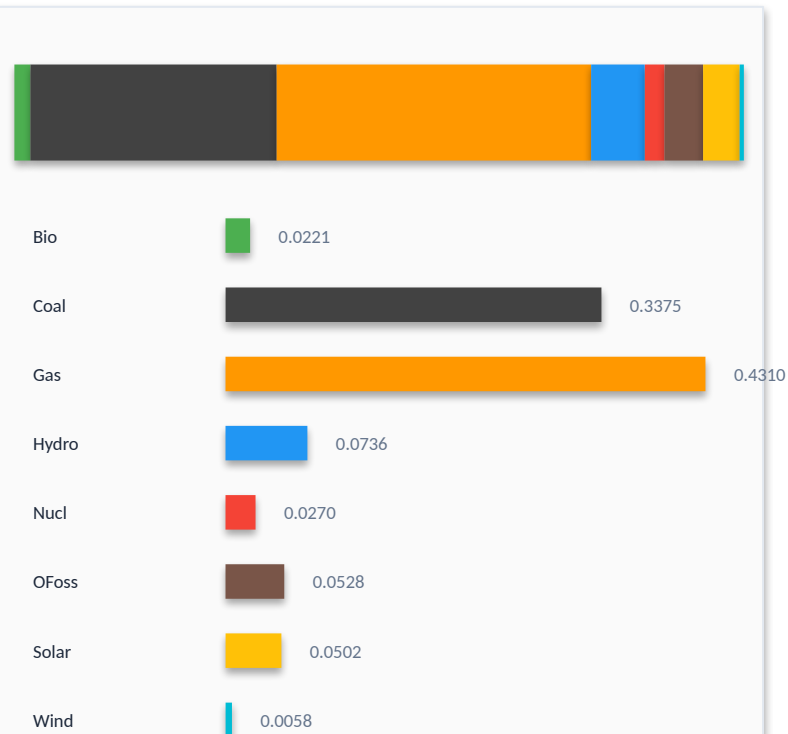
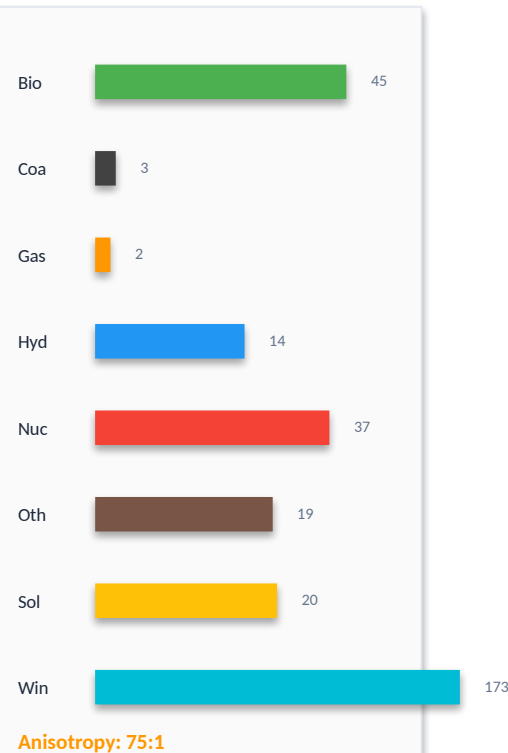
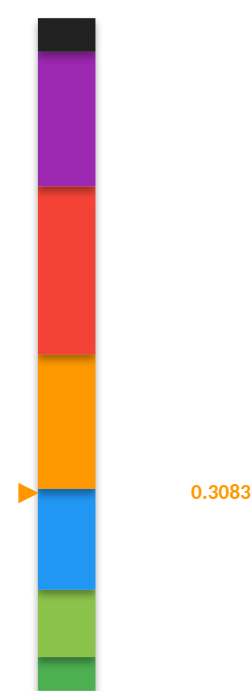


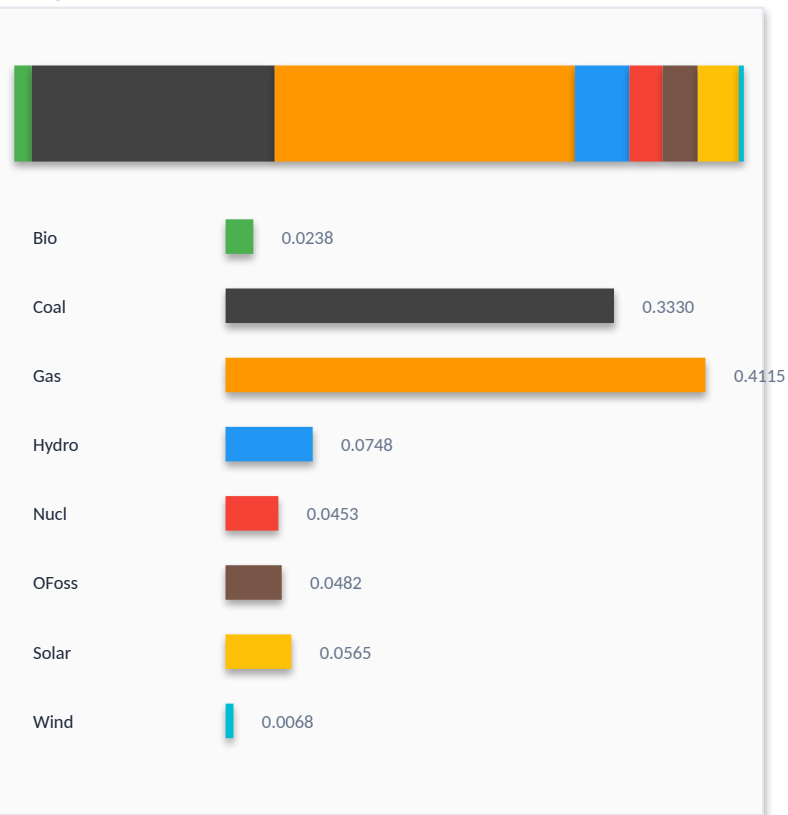
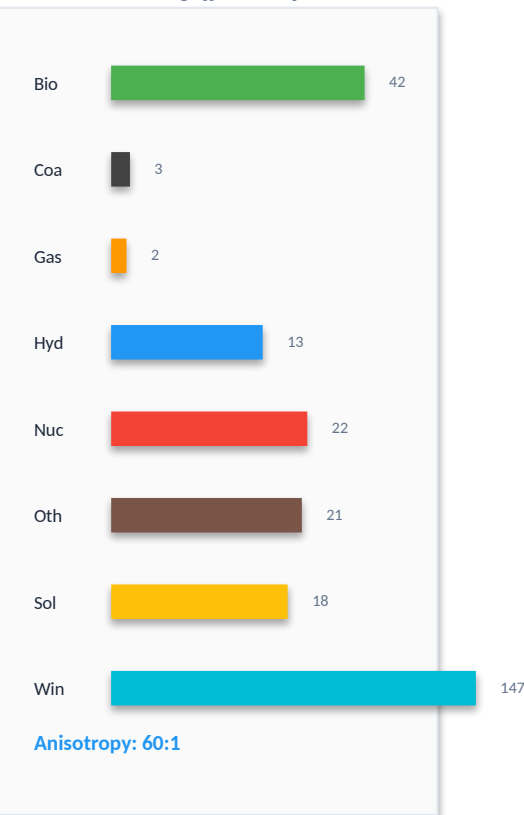
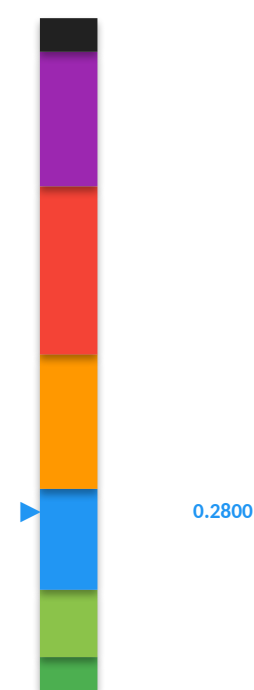
Metric Tensor $g_{jj} = 1/x_j$

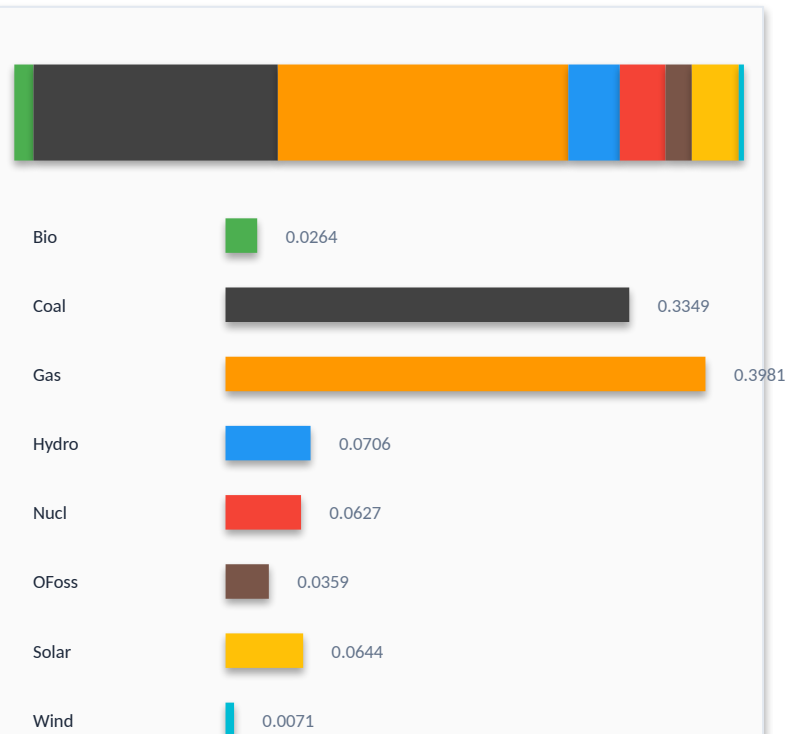
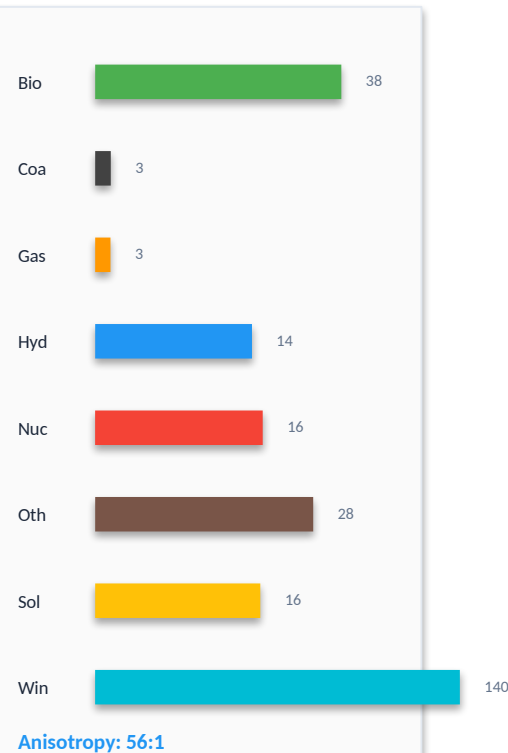
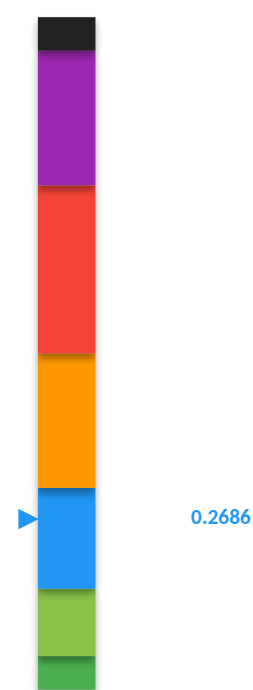


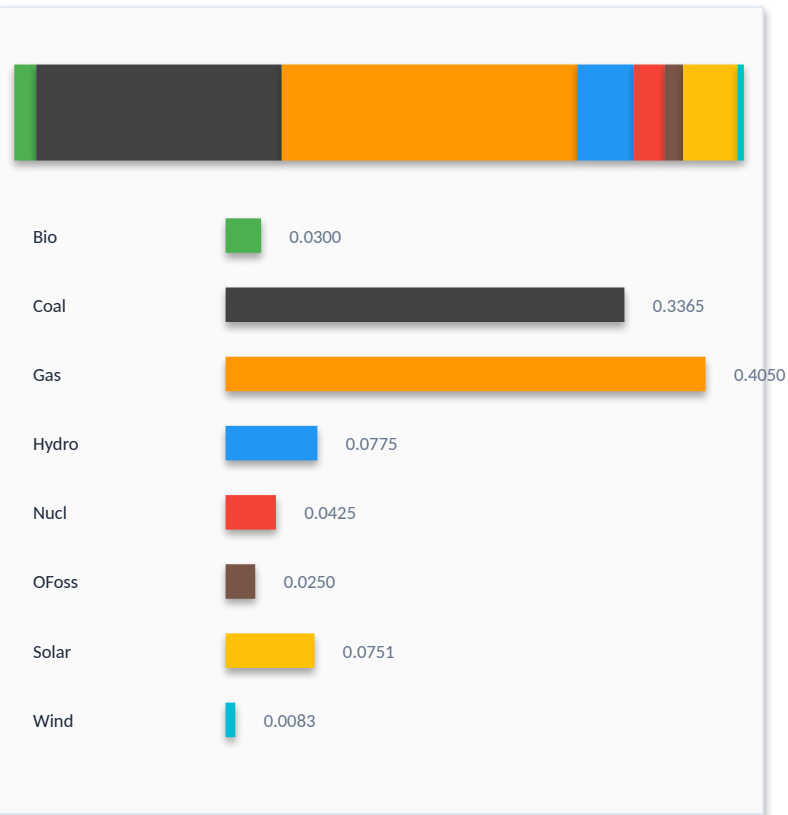
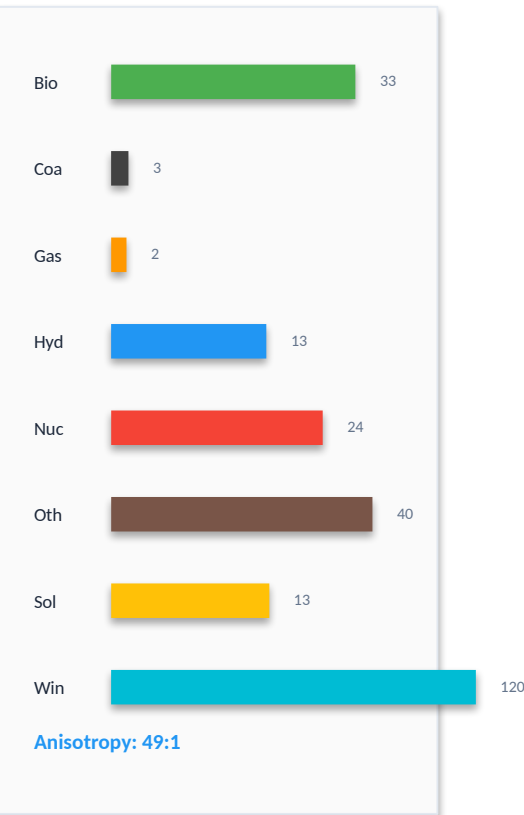
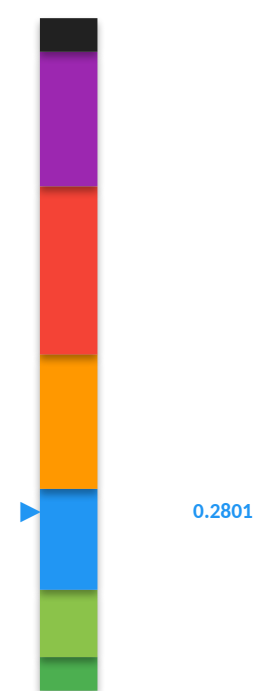
H_s

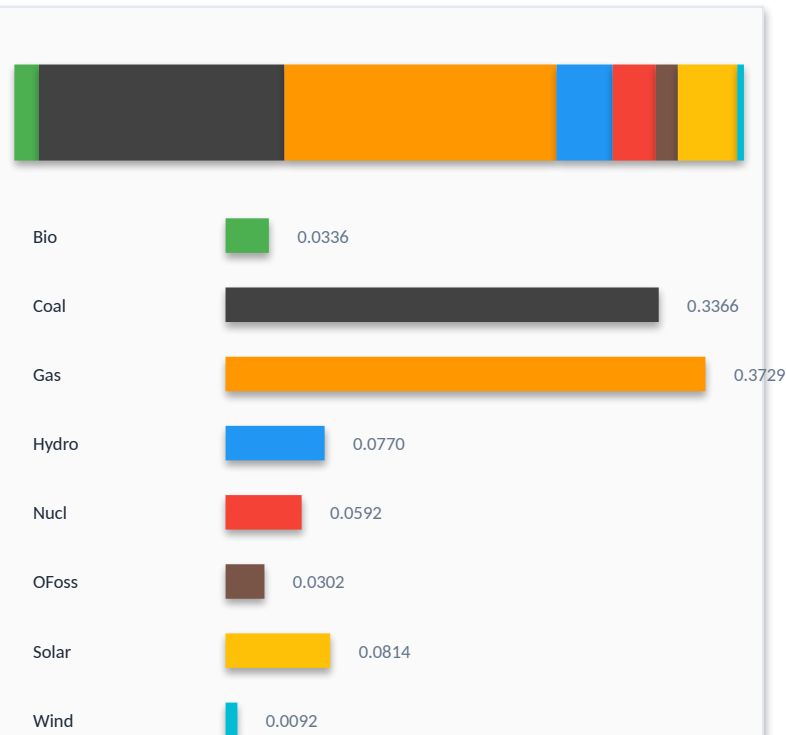
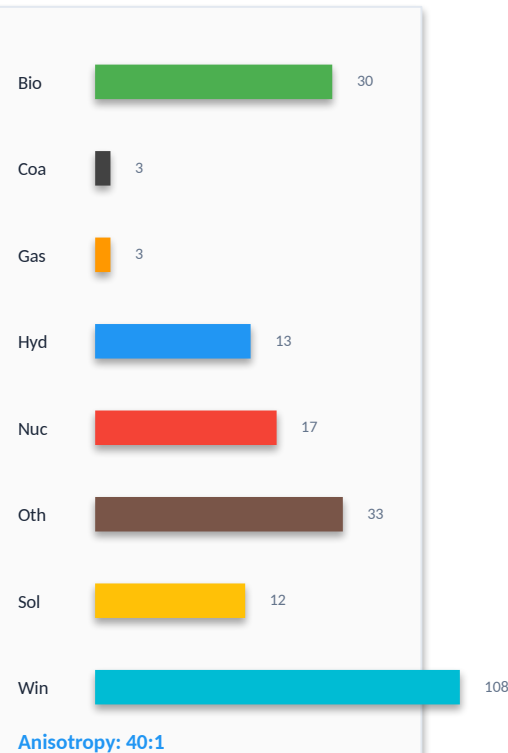
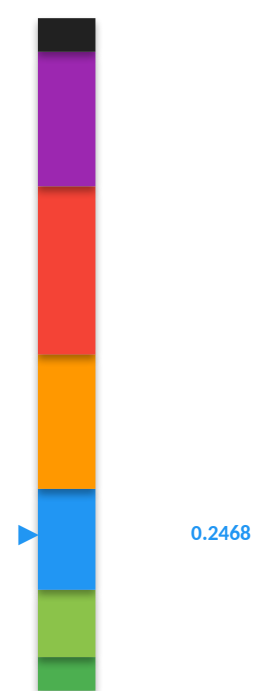


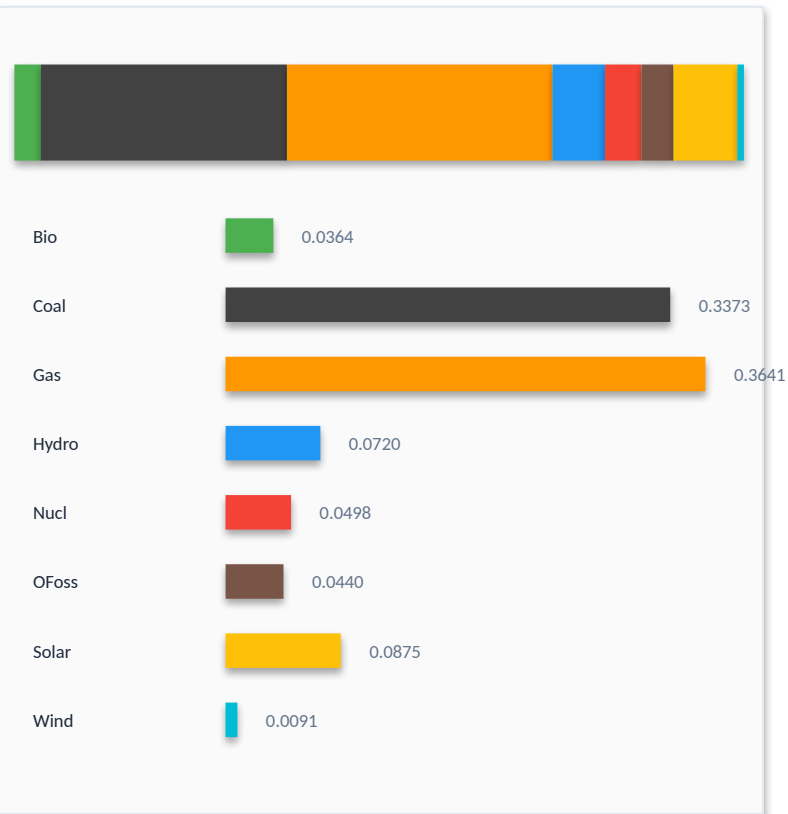
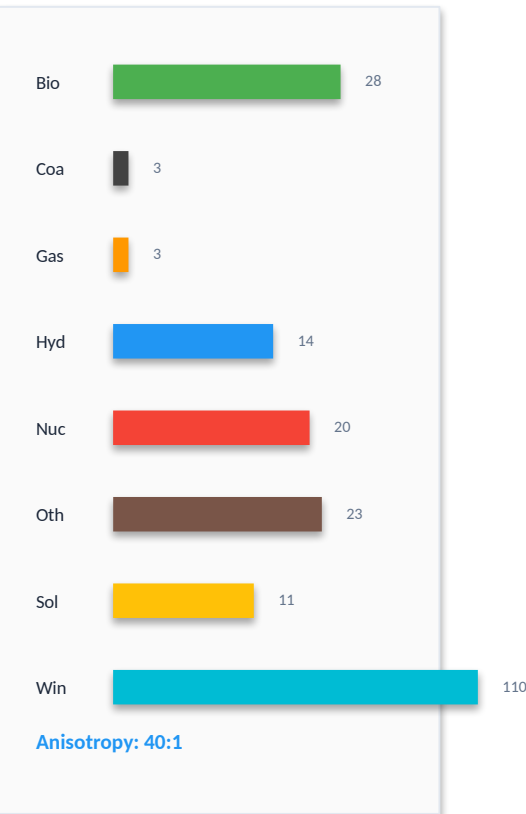
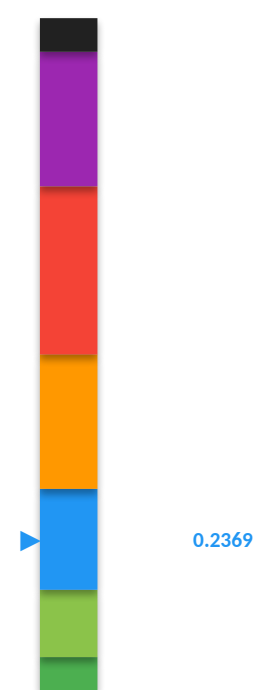
Simplex Fractions $\tilde{X} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

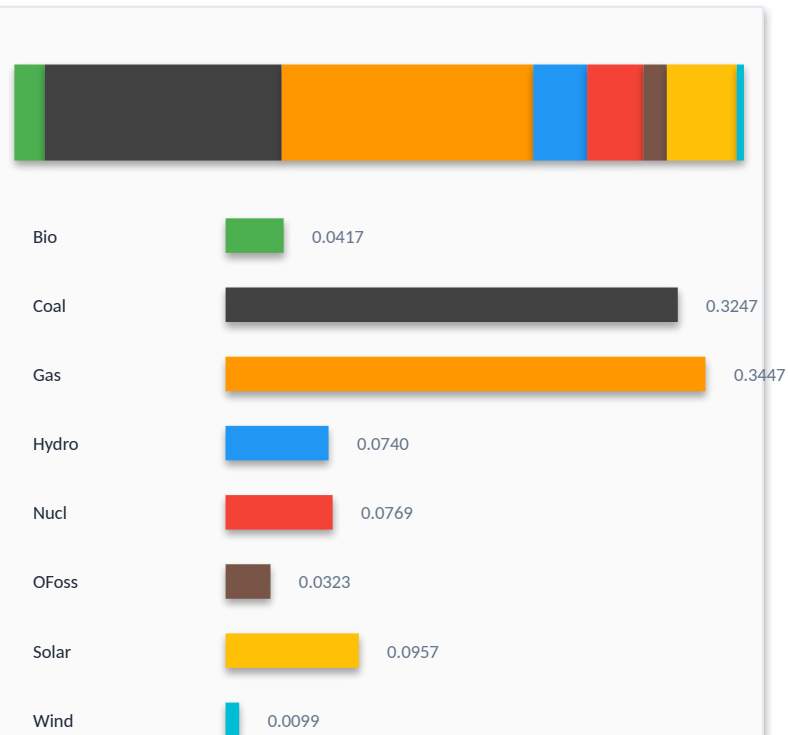
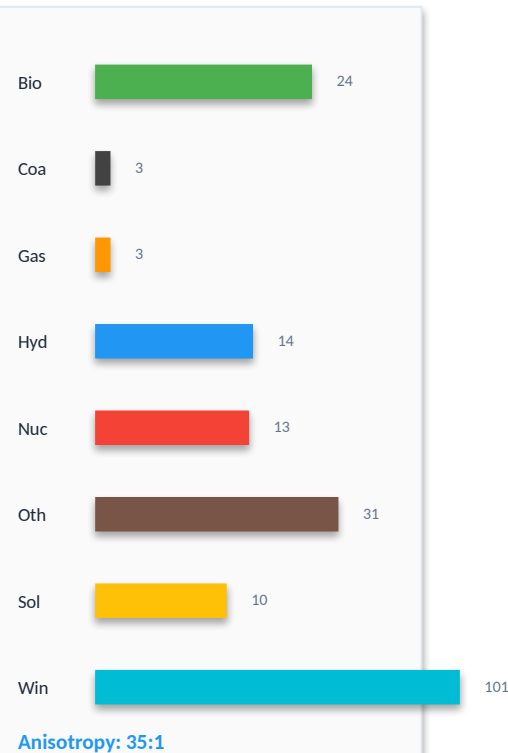
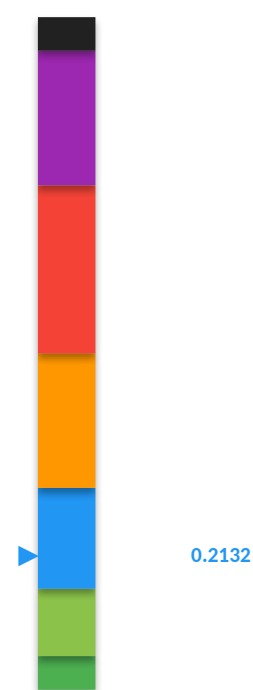
Simplex Fractions $\tilde{\mathbf{X}} = \mathbf{C}(\mathbf{x})$ Metric Tensor $\mathbf{g}_{jj} = 1/x_j$  H_s 

Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

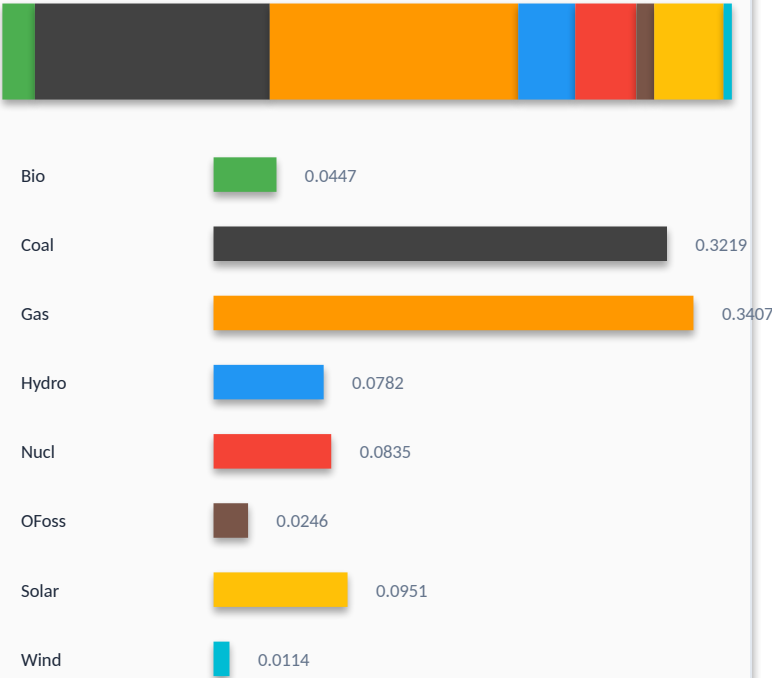
Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

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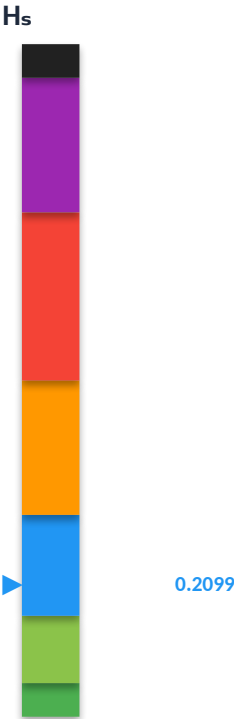
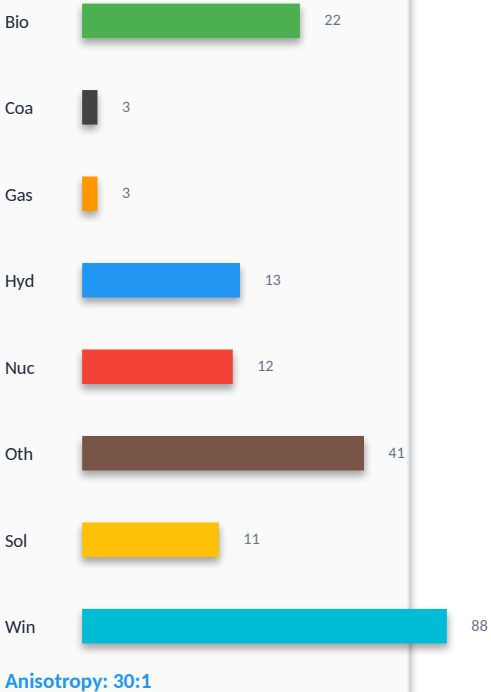
Simplex Fractions $\tilde{\mathbf{x}} = \mathbf{C}(\mathbf{x})$ Metric Tensor $\mathbf{g}_{jj} = 1/x_j$  H_s 

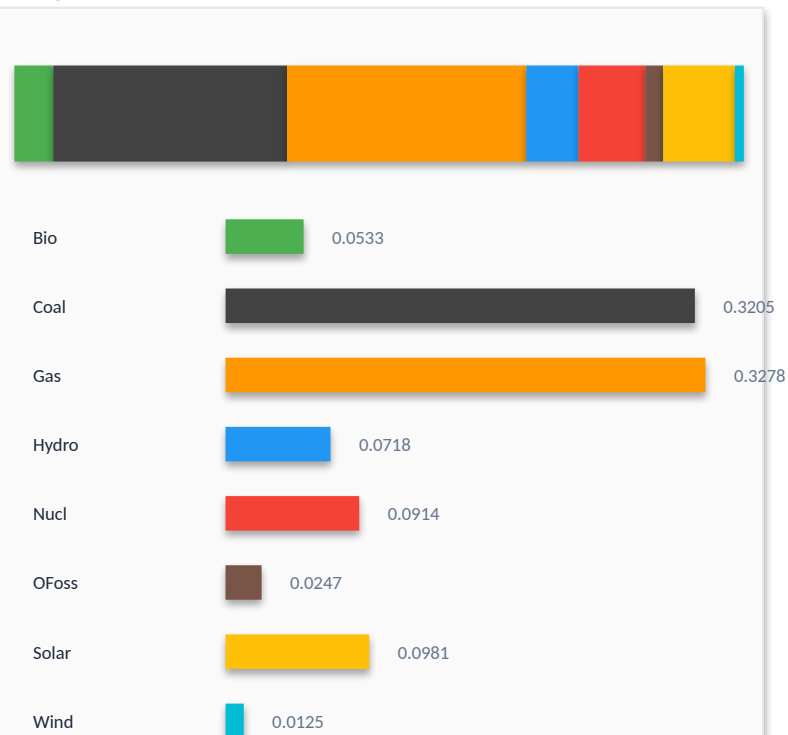
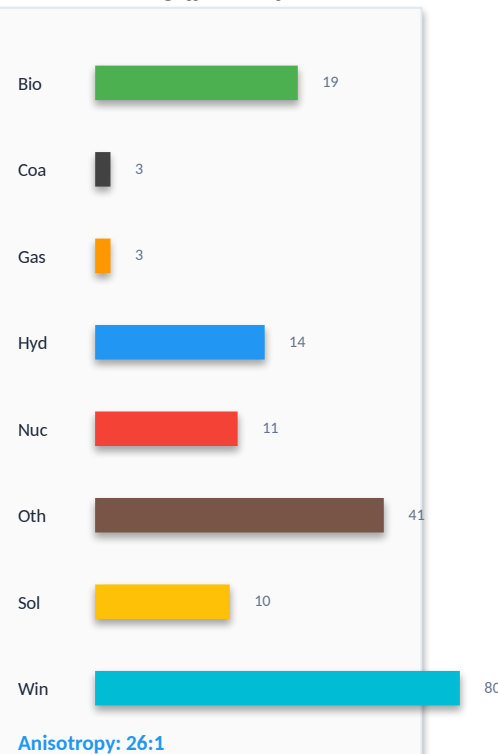
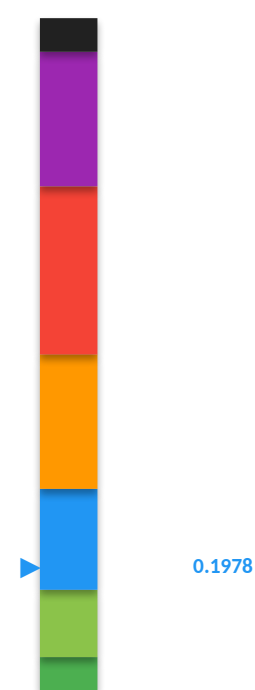
Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

Simplex Fractions $\tilde{X} = C(x)$

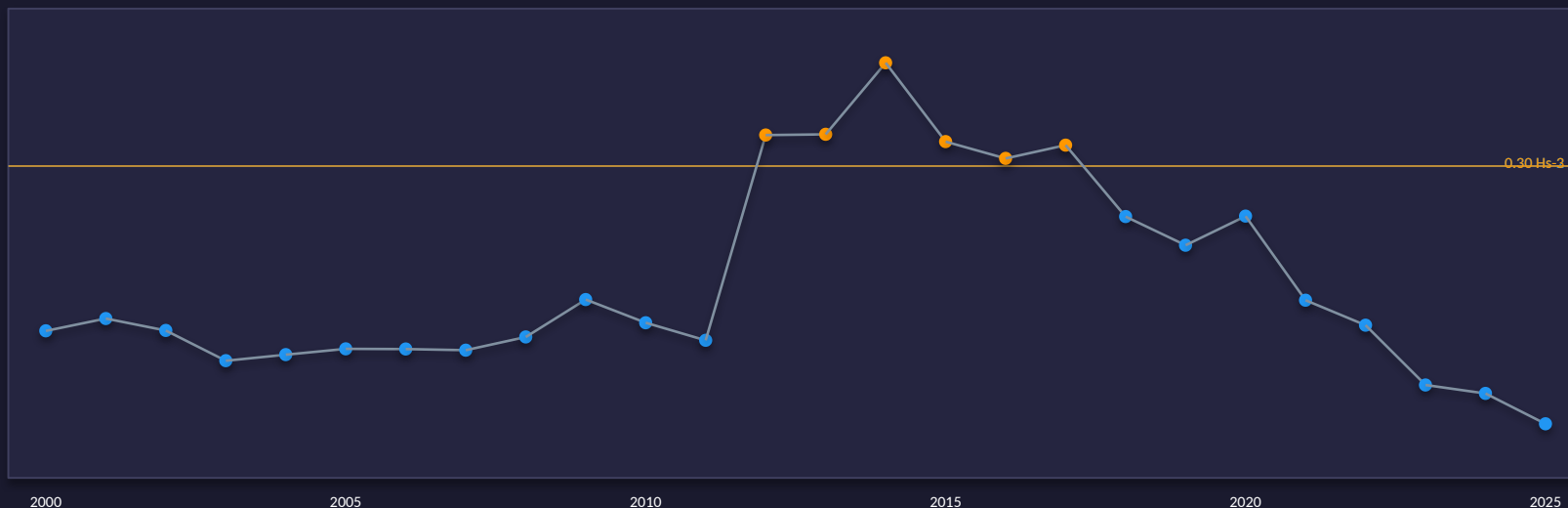


Metric Tensor $g_{jj} = 1/x_j$



Simplex Fractions $\tilde{x} = C(x)$ Metric Tensor $g_{jj} = 1/x_j$  H_s 

Japan EMBER — 26-Year Trajectory Summary



H_s range: 0.1978 – 0.3409 · Ring transitions: Hs-2 → Hs-3 → Hs-2 · d_A range: 2.99 – 8.40

2014 anomaly: Nuclear → 0, d_A = 17.28, anisotropy 450,880,000:1 · Post-Fukushima Gas dominance → gradual diversification (H_s declining toward 0.20)